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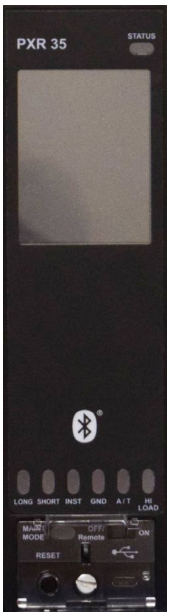
Power Xpert® Release trip unit for Magnum PXR and Power Defense SB circuit breakers PXR 20/25/35 user manual



PXR 20



PXR 25



PXR 35



Powering Business Worldwide

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⚠ WARNING

Dangerous electrical voltage!

Do not attempt to install or perform maintenance on equipment while it is energized. Death or severe injury can result from contact with energized equipment. Always verify that no voltage is present before proceeding. Always follow safety procedures. Eaton is not liable for misapplication or misinstallation of its products.

Observe all recommendations, notes, cautions, and warnings related to the safety of personnel and equipment. Observe and comply with all general and local health and safety laws, codes, and procedures.

⚠ WARNING

Do not attempt to install, test, or perform maintenance on equipment while it is energized. Death or severe personal injury can result from contact with energized equipment. De-energize the circuit and disconnect the circuit breaker before performing maintenance or tests. Any tripping operation will cause disruption of service and possible personal injury, resulting in the unnecessary switching of connected equipment. Testing a circuit breaker while it is in-service and carrying load current is not recommended. Testing of a circuit breaker that results in the tripping of the circuit breaker should be done only with the circuit breaker in the test or disconnected cell positions or while the circuit breaker is on a test bench.

⚠ WARNING

Electrical shock or burn injury can occur when working on power systems. Always turn off the main power that is supplying the circuit breaker before conducting tests. Test out of the cell, if possible.

⚠ CAUTION

Exercise care when replacing the battery to ensure that it is installed correctly. Accidentally installing the battery in the reverse direction will not harm the trip unit nor the battery, but will defeat the function of the battery.

⚠ IMPORTANT

Auxiliary power is not required to provide current protection features. Protection is active well below any overload. The trip unit begins to power-up at very low levels of current at approximately 40 A.

SUPPORT

Eaton provides technical support for the Magnum PXR and Power Defense SB circuit breakers and trip units. You can contact us via several methods. This link: www.eaton.com/eatoncare will take you to the on-line portal where you can contact Technical Support via phone, e-mail, or view FAQs.

For support via phone, you can call EatonCare and navigate the menu by:

- Call 877-ETN-CARE (877-386-2273)
- Option 2 for the Technical Resource Center
- Option 2 for Circuit Protection Products
- Option 3 for Low-voltage and Medium-voltage Power Circuit Breakers

1. The Power Xpert Release trip unit

1.1 System overview

Power Xpert Release® (PXR) trip unit has features and flexibility that allow configuration for a wide variety of protection applications. The PXR is available in multiple models for the Magnum PXR and Power Defense SB (PD-SB) circuit breaker family. The models range from simple current protection devices through more complex protection, metering, and communication functionality. Communication options support integration into a variety of systems to monitor performance. Advanced metering of currents, voltages, power, and energy allows evaluation of real time energy use.

The PXR trip unit, along with sensors and a trip actuator, provides the protective function. The PXR analyzes signals from the sensors; if current level and time delay settings are exceeded then the PXR trip unit will send a signal and trip the circuit breaker. The automatic overload and short circuit tripping characteristics for a specific circuit breaker are determined by the current rating and user selected protection settings.

The internal current sensors consist of two coils; as current begins to flow through the circuit breaker, the iron core coil generates the energy which powers the trip unit. At the same time, the air core coil provides signals which are processed to determine the magnitude of current the breaker is carrying. Auxiliary control power is not required for current protection functions.

The mechanical action required to initiate tripping of the circuit breaker is provided by a trip actuator, an integral part of the circuit breaker mechanism, which also includes a charging lever and mechanical trip button. The trip actuator is automatically reset by the mechanism.

The master connection diagrams show how certain functions of the trip unit are connected to external circuits. All wiring is landed on the secondary connection system directly above the circuit breaker. See Section 8 - Documentation Cross Reference for the low voltage power circuit breakers user manual and other related documents.

The PXR trip units are listed by Underwriters Laboratories Inc. (UL) for use in Magnum PXR and PD-SB circuit breakers. For more information on Magnum PXR low voltage power breakers, please visit: www.eaton.com/magnumpxr. For more information on PD-SB low voltage power breakers please visit www.eaton.com/powerdefensesb.

Power Xpert Protection Manager (PXPM) the windows based software for the PXR trip units provides cost savings and advanced functionality. This intuitive user interface allows for simple trip unit configuration, testing, and real time reporting of operating conditions, including current, voltage, power, and energy. The software also provides waveform captures, post event logs, as well as the ability to check critical health performance metrics.

Each PXR trip unit has embedded secondary injection and testing features which run using PXPM software on your personal computer when connected via a USB cable. These unique features offer savings through labor reduction and avoid the need for expensive proprietary testing kits. PXPM's extensive reporting features allow saving configuration files and generating reports in PDF format for sharing, as well as off-line configuration files for use in restoring or replicating trip unit configuration. Refer to Section 5 - Power Xpert Protection Manager software for details.

For more information on Power Xpert Protection Manager (PXPM) software, please visit: www.eaton.com/pxpm.

Power Distribution Monitoring and Control provides a user-friendly interface for interaction with multiple PXR trip units, legacy trip units, meters, starters, and drives. Installed in your switchboards, switchgear, and motor control centers, it is the engine that produces the user interface. You can view the screens locally using an HMI mounted in or near the switchgear or remotely by using a standard web browser.

The Power Distribution Monitoring and Control Enhanced features allow you to safely monitor and operate the Magnum circuit breaker with PXR trip units, as well as enable Arc Flash Reduction Maintenance System™ (ARMS or Maintenance Mode) remotely, increasing safety for all personnel.

For more information on Power Distribution Monitoring and Control, please visit: www.eaton.com/PDMonitoringControl.

Table 1. Trip unit catalog numbers and descriptions.

Trip unit	Trip unit catalog number	Breaker catalog digit 13, 14	Protection	ARMS	Communication	Metering
PXR 20	PXR20VN30LNNC	2A	LSI	N	Adapter module (CAM)	Current
	PXR20VN30LGNC	2C	LSIG	N	Adapter module (CAM)	Current
	PXR20VN30LNAM	2H	LSI	Y	Modbus RTU & CAM	Current
	PXR20VN30LGAM	2F	LSIG	Y	Modbus RTU & CAM	Current
PXR 25	PXR25VN30LNAM	2Q	LSI	Y	Modbus RTU & CAM	Current, voltage, power, energy
	PXR25VN30LGAM	2S	LSIG	Y	Modbus RTU & CAM	Current, voltage, power, energy
PXR 35	PXR35VN30LDAE	3A	LSI	Y	Modbus RTU & Ethernet	Current, voltage, power, energy
	PXR35VN30LGAE	3B	LSIG	Y	Modbus RTU & Ethernet	Current, voltage, power, energy
	PXR35VN30LDAB	3C	LSI	Y	Modbus RTU, Ethernet, & Bluetooth	Current, voltage, power, energy
	PXR35VN30LGAB	3D	LSIG	Y	Modbus RTU, Ethernet, & Bluetooth	Current, voltage, power, energy

Notes:
LSI = Long, short, and instantaneous protection are available.
LSIG = Long, short, instantaneous, and ground protection are available.
CAM = Communication adapter module, see Section 4.4 Communications adapter modules (CAMs).
Ethernet = Web UI, Modbus TCP, and IEC 61850 GOOSE Messaging

1.2 Catalog numbering and styles

This user manual applies to the trip unit catalog numbers shown in Table 1. Firmware revisions (as of the date of this manual) include 3.00, 3.01, and 3.02 for the PXR 20/25, 1.0.0 and 2.0.0 for the PXR 35. These revisions of firmware are used with Magnum PXR and PD-SB low voltage circuit breakers.

1.3 Primary protection features

The Time-Current Curves (TCC) for the PXR trip units when used in Magnum PXR and PD-SB circuit breakers are found in document TD013172EN. All protection settings shall be made by following the recommendations of the specifying engineer in charge of the installation.

Table 2. Primary protection features.

Protection			PXR 20	PXR 25	PXR 35
Overload protection (long delay) (L) ANSI 51P	Long delay slope		I^2t	I^2t, I^4t, It, $I^{0.5}t$	ANSI: I^2t, I^4t, It, $I^{0.5}t$ IEEE: Moderate, Extreme, Very Extreme IEC: A, B, C
	Long delay pickup	x (I_n)	0.4, 0.5, 0.6, 0.7, 0.75, 0.8, 0.9, 0.95, 0.98, 1.0	0.4 to 1.0 in steps of 0.01	0.4 to 1.0 in steps of 0.01
	Long delay time Setting	Seconds	0.5, 1, 2, 4, 7, 10, 12, 15, 20, 24	0.5 to 24① in steps of 0.1	0.5 to 24① in steps of 0.1
	Long delay thermal memory		Enable/disable	Enable/disable	Enable/disable
	Thermal memory alarm	% x Long Delay Trip Time			Off, 50 to 100 in steps of 5
	High load alarm 1④	% x (I_r)	OFF or 50% to < high load alarm 2 in steps of 1.0%, maximum value is 120% if high load alarm 2 is OFF		
	High load alarm 2④	% x (I_r)	OFF or > high load alarm 1 to 120% steps of 1.0%, minimum value is 50% if high load alarm 1 is OFF		
	High load alarm time delay				1 to 60 seconds in steps of 1 second
Short circuit protection (S) ANSI 50TD ZSI: ANSI 68	Short delay slope		Flat, I^2t	Flat, I^2t	Flat, I^2t
	Short delay pickup	x (I_r)	1.5, 2, 2.5, 3, 4, 5, 6, 7, 8, 10	1.5 to 10 in steps of 0.1	1.5 to 14 in steps of 0.1
	Short delay time I^2t @ 8 x (I_r)	Seconds	0.1, 0.3, 0.4, 0.5	0.05 to 0.5 in steps of 0.01	0.05 to 0.5 in steps of 0.01
	Short delay time flat	Seconds	0.05, 0.1, 0.2, 0.3, 0.4, 0.5	0.05 to 0.5 in steps of 0.01	0.05 to 0.5 in steps of 0.01
	Short delay zone interlock		Enable/disable		
Neutral protection ANSI 50N ANSI 51N	Neutral protection ratio	% x (I_r)	OFF, 60, 100		
Instantaneous protection (I) ANSI 50P	Instantaneous pickup	x (I_n)	OFF, 2, 4, 5, 6, 7, 8, 10, 12, 15	OFF, 2 to 15 in steps of 0.1	OFF, 2 to 15 in steps of 0.1
	Making current release level determined by frame	A	Y		
Ground (earth) fault protection (G) ANSI 51G ZSI: ANSI 68	Ground fault sensing		Residual, source ground, or zero sequence		
	Ground fault pickup② - trip (I_g)	x (I_n)	OFF, 0.2, 0.4, 0.6, 0.8, 1.0	OFF, 0.2 to 1.0 in steps of 0.01	OFF, 0.2 to 1.0 in steps of 0.01
	Ground fault pickup② - alarm	x (I_n)	OFF, 0.2, 0.4, 0.6, 1.0	OFF, 0.2 to 1.0 in steps of 0.01	OFF, 0.2 to 1.0 in steps of 0.01
	Ground fault delay I^2t @ 0.625 x (I_n)	Seconds	0.1, 0.2, 0.3, 0.5, 1.0	0.1 to 1 in steps of 0.01	0.1 to 1 in steps of 0.01
	Ground fault delay flat	Seconds	0.05, 0.2, 0.3, 0.5, 1.0	0.05 to 1 in steps of 0.01	0.05 to 1 in steps of 0.01
	Ground fault zone interlock		Enable/disable		
	Ground fault thermal memory		Enable/disable		
	Ground fault pre-alarm	%	50 to 100 of I_g in steps of 1.0		
Maintenance mode protection (ARMS) (A) ANSI 50	Switch positions		ON or remote		
	Maintenance mode pickup③	x (I_n)	1.5, 2.5, 4.0, 6.0, 8.0, 10.0		
	Local indication of mode		Ring LED (blue) around switch	Ring LED (blue) around switch	Blue LED
	Relay contact for remote indication of mode		Y		

Notes:
① If I^4t, It, $I^{0.5}T$ slope is selected not all times are available, consult time-current curves. Step sizes are slope dependent.
② Ground fault pickup is limited to a maximum of 1200 A for ANSI/UL/CSA versions to comply NEC® standards. Licensable 1200 A bypass can be purchased via PXPM for PXR 35. The multiplier (I_n) is shown for residual sensing and I_n 1200 < A. The multiplier will be different for other sensing methods or $I_n \geq 1200$ A. See Section 3.6 - Ground fault settings for a complete description of the settings.
③ Maintenance mode pickup maximum setting may be limited based on rating I_n.
④ For PXR 35 high load alarm 1 or high load alarm 2 can be set to trigger an extended capture instead of alarm when the threshold is exceeded. The other high load alarm can be set to off or alarm.

1.4 Enhanced protection features

Both the PXR 25 and PXR 35 include protection options in addition to the primary protection features. Control power (AUX) to the trip unit is recommended when using the enhanced protection features.

Table 3. PXR 25 enhanced protection features. ①

	Enhanced protection		Units	Settings
ANSI 59	Voltage protection	Over voltage pickup	Volts	180 to 720 in steps of 1
		Over voltage delay	Seconds	1 to 300 in steps of 1
ANSI 27		Under voltage pickup	Volts	60 to 670 in steps of 1
		Under voltage delay	Seconds	1 to 300 in steps of 1
ANSI 47		Voltage unbalance pickup	% volts	5 to 25 in steps of 1
		Voltage unbalance delay	Seconds	1 to 300 in steps of 1
ANSI 46	Current protection	Current unbalance pickup	% Amps	5 to 25 in steps of 1
		Current unbalance delay	Seconds	1 to 300 in steps of 1
ANSI 78V		Phase loss pickup	% amps	75
		Phase loss delay	Seconds	1 to 240 in steps of 1
ANSI 32R	Power protection	Reverse power pickup	kW	1 to 65,500 in steps of 1
		Reverse power delay	Seconds	1 to 300 in steps of 1

Note:
① Three modes of operation are selectable for these protection functions: OFF, Alarm, or Trip.

The PXR 35 Motor Protection, Power Protection, and Frequency Protection each have a global enable/disable setting. Each of the individual protection features can be edited even when the global setting is disabled, but the function will not be active until the associated group has been set to enabled.

Table 4. PXR 35 enhanced protection features.

	Enhanced protection		Units	Setting
ANSI 59	Motor protection ①	Over voltage pickup	% Volts	102 to 150 in steps of 0.1
		Over voltage delay	Seconds	0.05 to 300 in steps of 0.05
ANSI 27		Under voltage pickup	% Volts	50 to 98 in steps of 0.1
		Under voltage delay	Seconds	0.05 to 300 in steps of 0.05
ANSI 47		Voltage unbalance pickup	% Volts	2 to 90 in steps of 1
		Voltage unbalance delay	Seconds	1 to 300 in steps of 0.05
ANSI 46		Phase rotation (sequence) pickup	—	ABC or CBA
ANSI 46		Phase rotation (sequence) delay	Seconds	5
ANSI 46		Current unbalance pickup	% Amps	2 to 90 in steps of 1
		Current unbalance delay	Seconds	1 to 300 in steps of 0.05
ANSI 78V		Phase loss pickup	% Amps	75
		Phase loss delay	Seconds	1 to 240 in steps of 1
ANSI 32R		Reverse power pickup	kW	1 to 65,500 in steps of 1
		Reverse power delay	Seconds	0.05 to 300 in steps of 0.05
		Reverse reactive power pickup	kvar	1 to 65,500 in steps of 1
		Reverse reactive power delay	Seconds	1 to 300 in steps of 1

Table 4. PXR 35 enhanced protection features (continued).

	Enhanced protection		Units	Setting
ANSI 32P	Power protection ①	Real power pickup	kW	1 to 65,500 in steps of 1
		Real power delay	Seconds	1 to 300 in steps of 1
		Apparent power pickup	kVA	1 to 65,500 in steps of 1
		Apparent power delay	Seconds	1 to 300 in steps of 1
		Reactive power pickup	kvar	1 to 65,500 in steps of 1
		Reactive power delay	Seconds	1 to 300 in steps of 1
		Real power demand pickup	kW	1 to 65,500 in steps of 1
		Real power demand delay	Windows	1 to 10 in steps of 1
		Apparent power demand pickup	kVA	1 to 65,500 in steps of 1
		Apparent power demand delay	Windows	1 to 10 in steps of 1
ANSI 55		Power factor pickup	—	0.2 to 0.95 in steps of 0.05
		Power factor delay	Seconds	1 to 60 in steps of 1
ANSI 81O	Frequency protection ①	Over frequency pickup	% Hz	100.1 to 110 in steps of 0.1
		Over frequency delay	Seconds	0.05 to 60 in steps of 0.05
ANSI 81U		Under frequency pickup	% Hz	90 to 99.9 in steps of 0.1
		Under frequency delay	Seconds	0.05 to 60 in steps of 0.05
	Total harmonic distortion alarm ①	Current pickup		10 to 30 in steps of 1
		Current delay	Seconds	1 to 60 in steps of 1
		Voltage pickup		10 to 30 in steps of 1
		Voltage delay	Seconds	1 to 60 in steps of 1
ANSI 25	Synchronism check ②	Min live line volt	% Vnom	10 to 130 in steps of 1
		Max dead line volt	% Vnom	10 to 100 in steps of 1
		Min live load volt	% Vnom	10 to 130 in steps of 1
		Max dead load volt	% Vnom	1 to 100 in steps of 1
		Max volt difference	% Vnom	1 to 5 in steps of 1
		Max slip frequency	Hz	0.01 to 2 in steps of 0.01
		Max angle difference	Degrees	1 to 60 in steps of 1
		Volt dead time	Seconds	0 to 300 in steps of 1

Notes:
① See Table 5.
② Must be mapped to a programmable relay for use with additional customer circuitry for permissive close, using the optional spring release.

Table 5. Modes of operation.

Features	On/Off	Off/Alarm/Extended Capture	Off/Alarm/Trip
Overvoltage Trip	•		
Undervoltage Trip	•		
Overvoltage Alarm		•	
Undervoltage Alarm		•	
Overfrequency Trip	•		
Underfrequency Trip	•		
Overfrequency Alarm	•		
Underfrequency Alarm	•		
Voltage Unbalance Trip	•		
Current Unbalance Trip	•		
Voltage Unbalance Alarm	•		
Current Unbalance Alarm	•		
Forward Real Power			•
Forward Reactive Power			•
Reverse Real Power			•
Reverse Reactive Power			•
Apparent Power			•
Power Factor			•
Real Demand Power			•
Apparent Demand Power			•
Phase Rotation			•
Phase Loss			•

1.5 Current, voltage, frequency, power and energy data

All PXR trip units provide real time metering information regarding the loads being protected. The PXR 25 and PXR 35 include monitoring of voltage and phase angle to support power and energy data. All accuracy requires control power (AUX) to the trip unit and is specified for a 3-phase balanced load.

Table 6. Metering data.

			Accuracy		
Parameter		**Units**	**PXR 20**	**PXR 25**	**PXR 35**
Current metering	IA, IB, IC, IN ①, IG ②	Amperes	± 0.50% of reading	± 0.50% of reading	± 0.25% of reading ⑤
	Minimum IA, IB, IC, IN, IG	Amperes			
	Maximum IA, IB, IC, IN, IG	Amperes	Within the range of 20% to 120% x I_n at 25°C (77°F).	Within the range of 20% to 120% x I_n at 25°C (77°F).	Within the range of 125 A to 120% x I_n ③ at 25°C (77°F).
	Demand IA, IB, IC, IN, IG	Amperes			
	Demand Max. IA, IB, IC, IN, IG	Amperes			
Voltage metering ④	VAB, VBC, VCA (line to line)	Volts	—	± 0.50% of reading	± 0.25% of reading ④⑤
	Minimum VAB, VBC, VCA	Volts		Within the range of 102 to 690 Vac line-to-line at 25°C (77°F).	Within the range of 102 to 690 Vac line-to-line at 25°C (77°F).
	Maximum VAB, VBC, VCA	Volts			
	VAN, VBN, VCN (line to neutral)	Volts			
	Minimum VAN, VBN, VCN	Volts			
	Maximum VAN, VBN, VCN	Volts			
Power metering	Real	kW	—	± 1.00% of reading	± 0.50% of reading⑤
	Apparent	kVA		At PF = 1 within the range of 20% to 120% x I_n, 102 to 690 Vac line-to-line PF: at 25°C (77°F).	At PF =1 within the range of 125 A to 120% x I_n③, 102 to 690 Vac line-to-line at 25°C (77°F).
	Reactive	kvar			
	Real demand	kW			
	Apparent demand	kVA			
	Reactive demand	kvar			
	Real demand (peak)	kW			
	Apparent demand (peak)	kVA			
	Reactive demand (peak)	kvar			
Energy metering	Real total (forward + reverse)	kWh	—	± 1.00% of reading	± 0.50% of reading⑤
	Real net (forward – reverse)	kWh		At PF = 1 within the range of 20% to 120% x I_n, 102 to 690 Vac line-to-line at 25°C (77°F).	At PF = 1 within the range of 125 A to 120% x I_n ③, 102 to 690 Vac line-to-line at 25°C (77°F).
	Real forward (delivered by source to load)	kWh			
	Real reverse (delivered by load to source)	kWh			
	Apparent energy	kVAh			
	Reactive received (in quadrant 3 & 4)	kvarh			
	Reactive delivered (in quadrant 1 & 2)	kvarh			
	Reactive total (delivered + received)	kvarh			
	Reactive net (delivered – received)	kvarh			
Frequency metering		Hz	—	—	± 0.1 Hz
Power factor metering		—	—	—	✓

Notes:

① Neutral current accuracy for internal sensor (4 pole) breakers or 3 pole breakers with external Rogowski coil sensor. Neutral current accuracy for 3 pole breakers with external iron-core CT sensor is ± 2%.

② Ground fault accuracy applies to residual sensing method on all 3-wire systems, and 4-wire systems utilizing 4 pole breakers or 3 pole breakers with a Rogowski coil sensor. Ground fault accuracy for 4-wire systems with external iron-core CT sensor and source ground or zero sequence accuracy depends on the sensor selected.

③ From 50 to 125 A the current accuracy is ± 0.50% of reading and the power & energy metering is ± 1% of reading; below 50 A the accuracy is ± 1% of reading and the power & energy metering is ± 2% of reading.

④ The PXR 35 includes voltage metering for line and load. By default the line side is VDB1 (upper terminals) and has ± 0.25% accuracy. The accuracy for VDB2 (lower terminals) is ± 1%.

⑤ The trip unit accuracies shown are for factory installation and calibration in the circuit breaker. When replacing a trip unit in the field the accuracy may decrease up to ± 0.75% of reading for current and voltage, ± 1.50% of reading for power.

Please note that maximum and minimum values are held until reset using PXPM, the display, or communications. Demand values are on a configurable fixed or sliding window of 5 to 60 minutes.

The power and energy metering and the protection functions are calculated with the convention that power flow is from line to load through the circuit breaker. This assumes the top side conductor to be the line side. If the distribution system is configured such that the bottom side is the line side, the power values will indicate reverse power. This can be changed by using the display and navigation button. Navigate to the Edit Settings menu and under Power Feed, select Forward or Reverse.

2. Trip unit description

The PXR trip unit is located on the front and to the left side of the circuit breaker. It is enclosed in a housing which provides protection to the electronics as well as providing an interface for a user to configure and monitor operation. Details of the interface and operation are contained in the sections which follow.

Shown in Figure 1 are the PXR 20, PXR 25, and PXR 35 trip units. Both include a back-lit display (LCD) and menu navigation buttons below the display. The PXR 20 includes rotary switches for configuring the primary protection settings, other settings are configured through the menu system. The PXR 25 and PXR 35 configuration is entirely through the menu system. PXPM also provides an interface for configuring the trip unit's settings.

The pictures shown are examples of three styles, the PXR20VN30LGAM, PXR25VN30LGAM, and PXR35VN30LGAB. Certain features are available only in selected trip units.

Figure 1. Trip unit front panels.



2.1 Status indicator

All PXR trip units have an indicator in the top right labeled STATUS. During normal operation, this indicator blinks green (on and off approximately once each second), indicating that the trip unit is operating normally.

The status indicator blinks red if the trip unit detects an internal problem. This indicates a problem with the trip actuator coil, a firmware error, or a mechanism switch error. If the indicator remains on without blinking red or green, a hardware problem is indicated. Action must be taken to replace the trip unit or repair the breaker. See Section 7.1 - Troubleshooting for details.

When the status indicator remains off, either there is no auxiliary power applied or insufficient primary current to power the trip unit. PXR trip units will self-power at approximately 40 A of current in the breaker, per phase in three phase configuration.

2.2 Display and menu navigation buttons

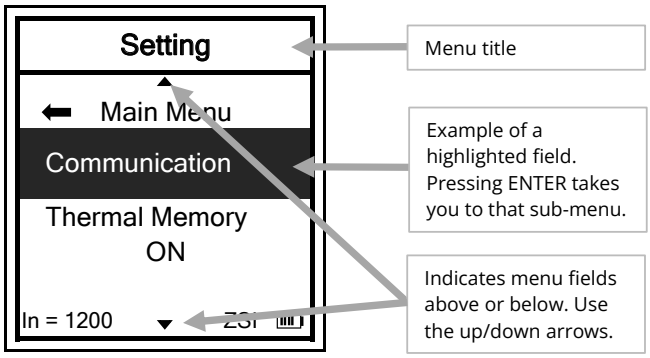
The PXR trip unit has a display on the front of the trip unit. This display provides information regarding the operation of the trip unit and the method to select certain configuration options. Information is presented on the display in either English, simplified Chinese, German, French, Spanish, or Portuguese. Back lighting is included with a power saver feature that after 30 seconds of inactivity will turn the backlight off.

When the PXR trip unit is initially powered-up, the display will briefly show a loading screen and then change to the Main menu. During this time, the trip unit is already functioning and performing protection operations. Depending on the trip unit style, submenu selections from the main menu provide information for monitoring or configuring the trip unit.

2.2.1 PXR 20/25 display

Each submenu can be accessed by pressing the Down or Up arrow buttons to highlight the appropriate submenu and then pressing the Enter button. Figure 2 shows a typical screen and navigation information.

Figure 2. PXR 20/25 display layout.

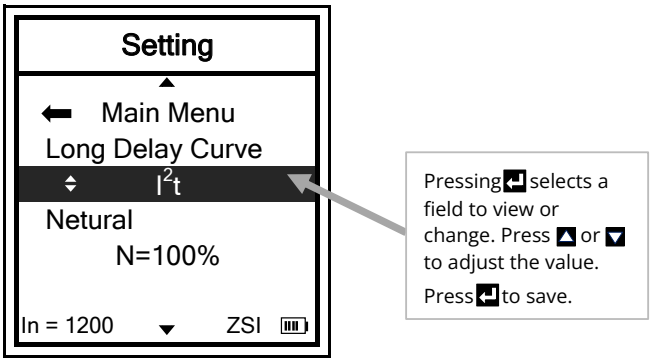


The top section of the display indicates the section of the menu system within which you are working. It will change as you navigate through the menus. The lower portion of the display indicates several pieces of information:

- The circuit breaker's continuous current rating is shown as I_n = XXXX on the left side.
- A check mark will appear in the lower part of the display as an indication of a received Zone Selective input signal. See Section 3.10 - Zone selective interlocking (ZSI) for more detail.
- The lower right has a battery symbol indicating remaining life of the battery. See Section 2.10 - Battery for more detail.

There are three buttons below the display (see Figure 3) that are used to control what information is shown on the display and to select certain configuration options.

Figure 3. PXR 20/25 Typical display navigation.



- The Up arrow button ▲ is used to move up in the menu display screen or increase an adjustment value.
- The Down arrow button ▼ is used to move down in the menu display screens or decrease an adjustment value.
- The Enter button ↵ is used to enter the menu or setting and can also go back the previous menu.

2.2.2 PXR 35 display

The PXR 35 includes a touchscreen display. Press the submenu on the screen to access it. Further navigation is done by pressing the arrows (right/left or up/down) as shown in each submenu. Figure 4 shows the home screen details and Figure 5 shows a typical screen and navigation information.

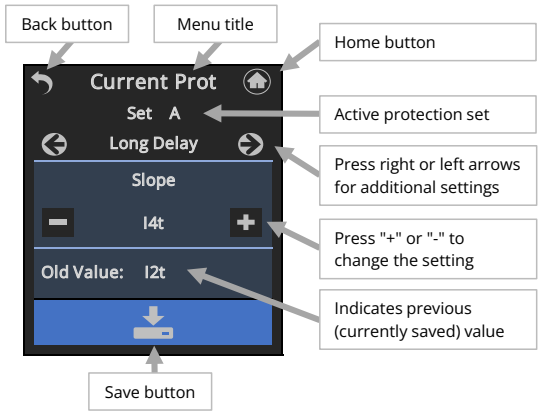
Figure 4. PXR 35 home screen.



The top row on the display indicates the active trip features and zone selective interlocking.

- The active trip features are shown in the upper left using L, S, I, and G. If a function is set to off, the corresponding letter will be removed from the display.
- Zone selective interlocking features are shown un the upper right. If enabled, the display will show "ZSI-SG", "ZSI-S", or "ZSI-G" to indicate which feature(s) it has been enabled on (short time and/or ground fault). See Section 3.10 - Zone selective interlocking (ZSI) for more detail.

Figure 5. PXR 35 display layout and navigation.



2.3 Cause-of-Trip indicators

All PXR family trip units record the Cause-of-Trip (CoT) in memory. The CoT is shown on LEDs on the front of the trip unit and is available through PXPM and communications interfaces. The four CoT indicators on the face of the trip unit are labeled: LONG, SHORT, INST, and GND.

The appropriate CoT indicator blinks when a current level pickup setting is exceeded. After a trip event, the indicator blinks (one second on, three seconds off) and the cause will be shown on the display if auxiliary power is applied. The indicators and the display can be cleared by pressing the RESET button.

- LONG – Long delay trip or over- temperature trip has occurred.
- SHORT – Short delay trip or an abnormal breaker condition has occurred.INST – Instantaneous, making current release, override, or maintenance mode trip.
- GND – Ground trip or ground alarm condition has occurred.

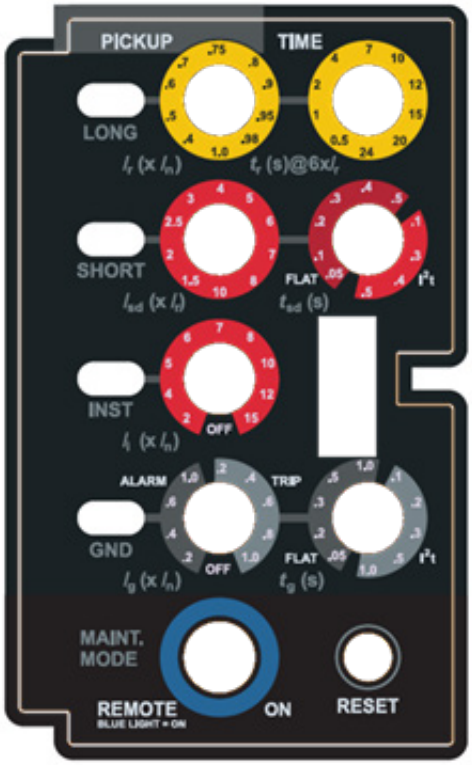
2.3.1 PXR 35 LEDs

PXR 35 includes 6 LEDs towards the bottom of the face of the trip unit. The LONG, SHORT, INST, and GND protection LEDs are lit with a white color when this protection is active (not set to off). The A/T LED indicates if an alarm or a trip has occurred: blinking for alarm and solid red for a trip. At the same time the protection LEDs will turn off, and in the case of a trip the associated LED will illuminate red until the trip unit is reset. The High Load alarm LED lights red anytime a high load alarm 1 or 2 condition is present.

2.4 PXR 20 – Configure using rotary switches

In the PXR 20, there are between five and eight rotary switches which configure the core protection settings. Each switch has ten positions and is set to achieve the appropriate trip curve response. Each switch can be set using a small screwdriver.

Figure 6. PXR 20 switch settings.



The rotary switches have legends above (PICKUP, TIME) and to the left (LONG, SHORT, INST, GND and MAINT. MODE) which indicate their use. For each rotary switch, there is a color-coded ring: yellow for long (L) settings, red for short (S) and instantaneous (I) settings, and grey for ground (G) settings. The legend within the ring and arrow on the switch indicate the value for each position. The formula that relates the setting to breaker ratings is below left of each switch.

At the bottom, the switch with the blue ring is used for enabling Maintenance Mode on trip units with the Arc Flash Reduction Maintenance System™ (ARMS). The blue ring illuminates when the trip unit is in Maintenance Mode.

Any time you change a switch position, the display will temporarily change and indicate the values of all switches to aid in configuring the correct settings. The menu system also provides access to additional configuration options beyond these core settings.

2.5 Configure using the display

2.5.1 PXR 25

PXR 25 settings are configured using the display and menu system. The same configuration options are present on both the PXR 20 and PXR 25, but in the PXR 25 the PICKUP and TIME settings can be fine-tuned to smaller increments between the minimum and maximum settings. Refer to Section 1.3 - Primary protection features to understand the flexibility in PXR 25.

The menu system provides access to each setting. From the Main menu, press the Down arrow button until you get to the Edit Settings field and then press the Enter button. You will be asked for the password and, if entered correctly, will be able to change the settings. You may also select View Settings in the Main menu if you only want to understand what settings have been configured.

2.5.2 PXR 35

PXR 35 settings are configured using the touchscreen display and menu system. The PXR 35 has the same configuration options as the PXR 25 as well as options for the features and settings specific to PXR 35. Refer to Sections 1.3 and 1.4 for primary and enhanced protection features to understand the flexibility in PXR 35.

The menu system provides access to each setting, except for GOOSE specific settings. From the home screen, press the Settings button to access the settings. Scroll left or right using the arrow buttons. Choose the setting you wish to change and adjust using the controls provided. You will be asked for the password to change a setting and, if entered correctly, the change will be saved.

Only the active protection set can be edited via the display. Other protection sets can be viewed, but not edited. Changing the protection set is only possible via communications or through the hardwired inputs. See section 3.7 for more information.

2.6 Maintenance mode switch

When included on the PXR 20, PXR 25, or PXR 35, the trip unit incorporates the Arc Flash Reduction Maintenance System™ (ARMS). This switch is labeled MAINT. MODE and has two positions labeled: REMOTE & ON. A blue LED on the trip unit is illuminated when ARMS is enabled. For the PXR 20/25 this is a ring surrounding the switch and for the PXR 35 the LED is to the left of the switch.

- ON – ARMS is enabled locally and cannot be disabled remotely.
- REMOTE – ARMS is controlled remotely by a dry contact or through communications. See Section 3.9 - Maintenance mode.

2.7 Reset button

The button labeled RESET can be depressed to reset the cause of trip indicators while the trip unit is powered. The button is in the lower right of the PXR 20/25 and lower left on PXR 35.

2.8 Tamper proof cover

Controlling physical access is an important element in your security policy. To support this, a clear plastic cover is provided which allows the core current protection settings of the PXR 20 to be viewed but not changed and to secure the Maintenance Mode switch when desired. In the PXR 25 and PXR 35 trip units, core current protection settings are protected using the password features.

Unauthorized access to make changes can be prevented by the insertion of a standard sealing wire through the security holes to meet applicable tamper-proof requirements. Also see Section 3.1 - Password security as part of providing tamper resistance.

2.9 Test and configuration port

Near the bottom of the PXR trip unit, there is a small door with the Universal Serial Bus (USB) icon printed on it. The door can be opened downward to expose the USB port utilizing a micro-B USB connection. This door can also be sealed using standard sealing wire in conjunction with the clear cover.

This USB connection is used in conjunction with the Power Xpert Protection Manger (PXPM) software to configure and monitor the trip unit. Note that a data cable is required, USB cables that provide power only will not work with PXPM. A USB connection will also provide power from the host side of the USB cable to power up the trip unit when another source of auxiliary power is not available. This connection is intended for temporary use while you are configuring or monitoring the trip unit.

2.10 Battery

At the bottom of the frame module is a small tray which holds the battery. When the trip unit is not powered, this battery supports the Cause-of-Trip indicators (CoT) and Real-time clock (RTC). When unpowered, the battery will support the CoT and RTC for 20 days. The battery plays no part in the protection functions of the trip system.

Status of the battery is displayed in the lower right corner of the LCD and within PXPM.

On the initial installation of the circuit breaker, remove and discard the insulating tab to enable the battery. Also press the reset button to ensure the cause of trip LEDs are clear. This battery is the standard lithium type CR 2032 coin-cell.

2.11 Breaker continuous current rating

On the lower right corner of the frame module is a label which shows the circuit breaker's continuous current rating (I_n).

2.12 Real-time clock

The trip unit contains a Real-time clock. The RTC is used as the timestamp for all logged events. It is helpful, when diagnosing the circumstances of an event, to have it set to the current date and time. The resolution is to the tenth (0.1) of a second.

The clock should be set during commissioning (remember to remove the battery insulating tab) to match a standard time reference for your location. PXPM can be used to quickly set the time, or the menu system via the front panel can be used. We recommend that you periodically check and correct the time should it experience drift during operation.

The Power Xpert Gateway and Power Distribution Monitoring and Control can be used to periodically update the RTC on all trip units connected to it. Alternately, the clock can be set by a client via Modbus RTU.

2.13 Operating temperature

All models of trip units are designed for industrial circuit breaker environments where the ambient temperatures can range from -20 to +65°C. (-4 to +149°F). However, if temperatures within the trip unit exceed this range, a factory set over-temperature protection which will trip the circuit breaker. PXR 20/25 trip units are set to trip at 95°C (203°F). PXR 35 trip units are set to trip at 105°C (221°F). In addition, the PXR 35 trip unit also reports humidity and includes over-temperature as a configurable alarm. The LONG cause of trip indicator will illuminate when a temperature trip occurs.

2.14 Storage temperature

Storage temperature is from -55 to +95°C (-67 to +203°F). Before putting the trip unit into operation, ensure it is within the operating temperature and no condensation is present.

2.15 Trip unit, frame, and voltage modules within the trip unit

There are three components of the system: the trip unit module, the frame module, and the voltage sensing module (PXR 25 and PXR 35 systems).

The electronic trip unit module (ETU) is the main component of the trip unit system. It contains the electronics which support the protection functions, metering capability, user interface, and communication features. It can be replaced in the field for the same style unit or to a different style. Note that upgrading may require the installation of other sub-system components and wiring to enable full feature support.

The trip unit module label is attached to the side and contains the catalog number, PXR 20, PXR 25, or PXR 35 and protection information (LSI, LSIG, LSID). Additionally, it contains the Eaton style number, unit serial number, and revisions of the hardware and firmware.

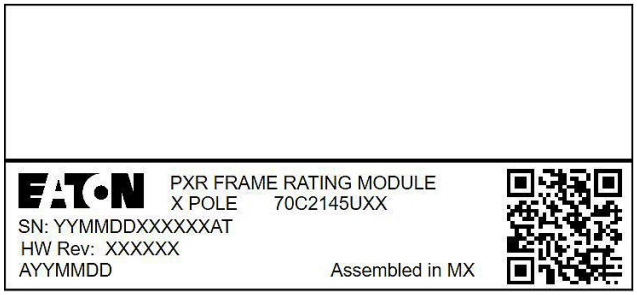
Figure 7. Trip unit label.



The frame module (FRM) is mounted permanently to the circuit breaker frame. It should not be removed or replaced under normal conditions. If it accidentally is damaged and requires repair, Eaton Field Service must be consulted to perform the work. The lower half of the plastic housing contains the electronics that hold factory configured frame information regarding the rating, calibration, settings, and operation. The interface circuit for the sensors as well as the configurable relays are mounted in the frame module.

A label is attached to the side of the frame module. It contains a brief description, the Eaton style number, serial number, and hardware revision.

Figure 8. Frame module label.



The voltage sensing module (VDB) is used with the PXR 25 and PXR 35 and is secured in the top half of the of the plastic housing, above the FRM. It converts the line voltage from the upper or lower breaker terminals for sensing by the trip unit electronics. With the PXR 25 trip unit, choice of upper or lower terminals as the sensed voltage can be changed in the field. With the PXR 35 trip unit, this is VDB1 and is connected to the upper terminals. Additionally, the PXR 35 utilizes a second VDB (VDB2) located behind the trip unit mounting plate that is connected to the lower terminals. For a 3 pole breaker feeding 4-wire loads, pin C49 can be wired to the neutral bus for higher line-to-neutral metering accuracy.

The three modules are interconnected via PCB-to-PCB connectors. In addition, the modules connect to the breaker and secondary terminals through a set of wiring harnesses.

The PXR modules (ETU, FRM, VDB) used in Magnum PXR and PD-SB circuit breakers are designed to be used as a set. As such, they shall not be interchanged with PXR 20/25 modules from the Series NRX or Power Defense NF/RF circuit breaker families. The frame module has a feature which will physically reject trip units not designed for it. The feature prevents the screw, which fastens the top of the control module to the frame module and mounting plate, from engaging the threads. **Do not tamper with or disable this rejection feature.**

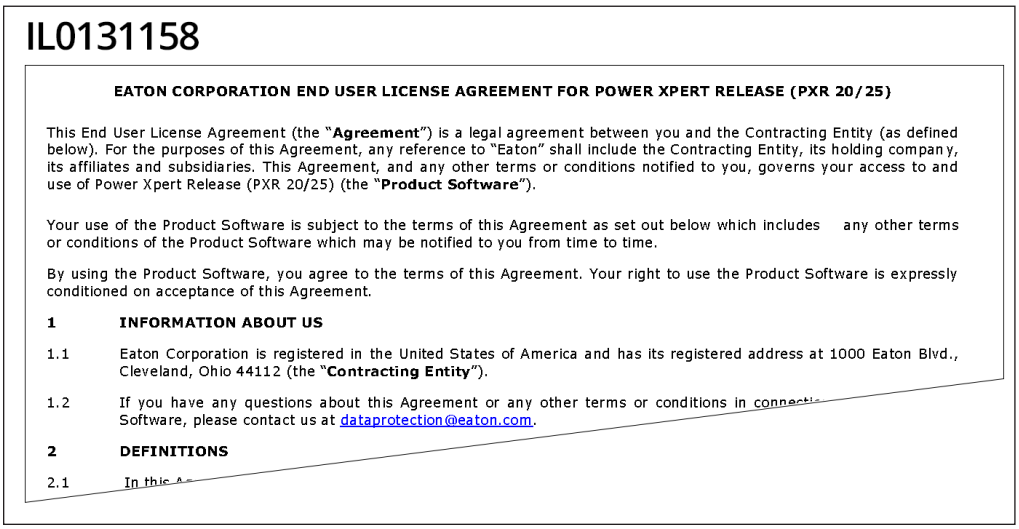
You can distinguish between Power Xpert Release trip units designed for Series NRX or Power Defense NF/RF circuit breakers and those designed for use in Magnum PXR and PD-SB circuit breakers based on the catalog numbers printed on the side label of the PXRs.

- PXR20V000… or PXR25V000… are designed to be used only with Series NRX or PD-NF/PD-RF circuit breakers (also known as v2 based on their firmware revision starting with 02.XX).
- PXR20NV30… or PXR25NV30… are designed to be used only with Magnum PXR or PD-SB circuit breakers (also known as v3 based on their firmware revision starting with 03.XX).

2.16 End user license agreement

The End User License Agreement (EULA), IL0131158EN is a legal agreement between you and Eaton which governs your use of the PXR trip units. Visit the Eaton Website – www.eaton.com – to review this important document.

Figure 9. EULA.



IL0131158

EATON CORPORATION END USER LICENSE AGREEMENT FOR POWER XPERT RELEASE (PXR 20/25)

This End User License Agreement (the "**Agreement**") is a legal agreement between you and the Contracting Entity (as defined below). For the purposes of this Agreement, any reference to "Eaton" shall include the Contracting Entity, its holding company, its affiliates and subsidiaries. This Agreement, and any other terms or conditions notified to you, governs your access to and use of Power Xpert Release (PXR 20/25) (the "**Product Software**").

Your use of the Product Software is subject to the terms of this Agreement as set out below which includes any other terms or conditions of the Product Software which may be notified to you from time to time.

By using the Product Software, you agree to the terms of this Agreement. Your right to use the Product Software is expressly conditioned on acceptance of this Agreement.

1 INFORMATION ABOUT US

1.1 Eaton Corporation is registered in the United States of America and has its registered address at 1000 Eaton Blvd., Cleveland, Ohio 44112 (the "**Contracting Entity**").

1.2 If you have any questions about this Agreement or any other terms or conditions in connecti... Software, please contact us at dataprotection@eaton.com.

2 DEFINITIONS

2.1 In this A...

3. Protection functions

The PXR trip unit protection settings are designed to be easily customized to any application. Settings for long delay pickup, long delay time, short delay pickup, short delay time, instantaneous pickup, ground fault pickup, and ground fault time are all independently configurable. These functions are set using the Power Xpert Protection Manager (PXPM) software, using the display and navigation buttons, or rotary switches on the front of the trip unit.

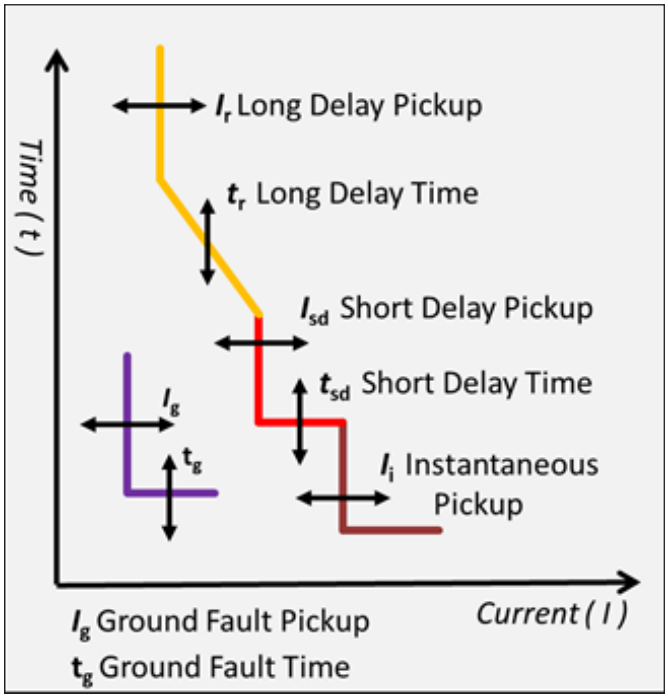


Figure 10. Settings and trip curve.

Before placing any circuit breaker in operation, set each trip unit protection setting to the values specified by the engineer responsible for the coordination study.

3.1 Password security

Protecting your system from cybersecurity threats is very important. PXR trip units have four-digit or six-digit passwords, depending on firmware version, used to secure settings and to enable secondary injection testing. To change a setting, which is not set by a physical switch, you will be required to enter the password. Authorization to make changes will timeout after ten minutes of inactivity. If three attempts are made with an incorrect password, all access will be denied for ten minutes.

Two levels of password are available: Admin and User. The factory default is the Admin level and the password is '0 0 0 0 0 0' or '0 0 0 0 0 0' depending on firmware version requiring four or six digit passwords. The Admin can, if desired, create a User and decide if the User can change protection setpoints or not. Password security is enforced when using the display, PXPM software, and when another device attempts a change via a communication network. Passwords should only be made available to those individuals who require it.

Changing the factory default password when the PXR is installed or commissioned is a key element of a comprehensive cybersecurity policy. Passwords are four-digits or six-digits long, numeric characters 0 to 9. When entering, the numbers will not be shown on the display. Instead, an asterisk (*) will be shown to protect the security of your password.

For additional information about best practices, please review Cybersecurity Considerations for Electrical Distribution Systems (WP152002EN): available at www.eaton.com.

3.2 Overload/long delay settings

The PXR trip unit offers a wide range of settings for overload protection. Long delay pickup (LDPU or I_r) setting ranges from 0.4 to 1.0 and is expressed as a multiple of the frame's current rating (I_n). I_r is also the basis for the short delay current setting. The tolerance for long delay pickup is 105% to 115% of the setting to ensure that the circuit breaker can carry the full rating of I_r, without tripping.

The long delay time (t_r) settings range from 0.5 to 24 seconds (as specified in Table 2. This represent the total clearing time when the current value equals six times (I_r) when an I^2t slope is selected. All times are referenced from the top of the tolerance band, ensuring that the time never exceeds that maximum setting. For the other slopes, please refer to the trip curves for total clearing time.

The I^2t setting is the factory default curve for long delay. PXR 25 and PXR 35 allow the curve to be changed to alternative slopes to better match the requirements for protection and coordination. Consult the time-current curves for available settings for each slope.

- I^2t - Inverse time current curve, standard distribution protection (factory default);
- I^4t - Extremely inverse time current curve, coordination with fuses or special types of loads;
- I^1t - Moderately inverse time curve; or
- $I^{1/2}t$ - Slightly inverse time curve.

The PXR 35 trip unit additionally includes the ANSI IEEE protective relay curve selection. The curves accurately follow the IEEE Inverse-Time characteristic equations. Curve shapes of Moderately Inverse, Very Inverse, or Extremely Inverse are available. Each of these curve shapes have a Pickup and Time selection to position the curve. The short delay function is included with the IEEE curves.

For PXR 35, the IEC protective relay curve selection is also possible. The IEC curves accurately follow the IEC protective relay curve equations. The curve shapes of IEC-A (normal inverse), IEC-B (very inverse), and IEC-C (extremely inverse) are available. Each of these curve shapes have a Pickup and Time selection to position the curve. The short delay function is included with the IEC curves.

If a long delay causes the circuit breaker to trip, the LONG

indicator will be illuminated and the long delay message will be displayed.

3.2.1 Long delay thermal memory

In addition to the standard long delay protection, a long thermal memory (LTM) function is supported. This protects load circuits from the effects of repeated overload conditions. LTM is enabled or disabled using the display and navigation buttons or using the Power Xpert Protection Manager software.

As an example, if a circuit breaker is closed soon after a long delay trip, and the current again exceeds the long delay setting (I_r), the LTM automatically reduces the time to trip to allow for the fact that the load conductor temperature is already higher than normal because of the prior overload condition. Each time the overload condition is repeated, the LTM causes the circuit breaker to trip in a progressively shorter time. When the load current returns to normal, the LTM begins to reset (after ten minutes it will have reset fully) so the next long delay trip time will again correspond to the setting value. During sequential field testing, it may be desirable to disable the LTM function.

3.3 Short circuit settings

Settings for short delay pickup (SDPU or I_{sd}) are expressed as multiples ranging from 1.5 to 10 times (PXR 20/25) or 14 times (PXR 35) the long delay pickup current setting (I_r).

The short delay time (t_{sd}) settings range from 0.05 seconds to 0.5 seconds. The I^2t response curve will provide a longer time delay for currents below eight times I_r as compared with a flat response curve. For currents greater than eight times I_r, the I^2t response flattens out to the flat response.

The optional zone selective interlocking (ZSI) feature may affect the tripping times for the short delay protective function. More information can be found in Section 3.10 - Zone Selective Interlocking (ZSI).

If a short delay causes the circuit breaker to trip, the SHORT indicator will be illuminated, and the short delay trip message will be displayed.

3.4 Instantaneous protection setting

The instantaneous (I_i) setting is expressed as multiples ranging from 2 to 15 of the (I_n) value or can be set to OFF. The instantaneous protection trips the breaker with no intentional time delay.

The instantaneous (INST) indicator shows this cause of trip and the instantaneous delay trip message will be displayed.

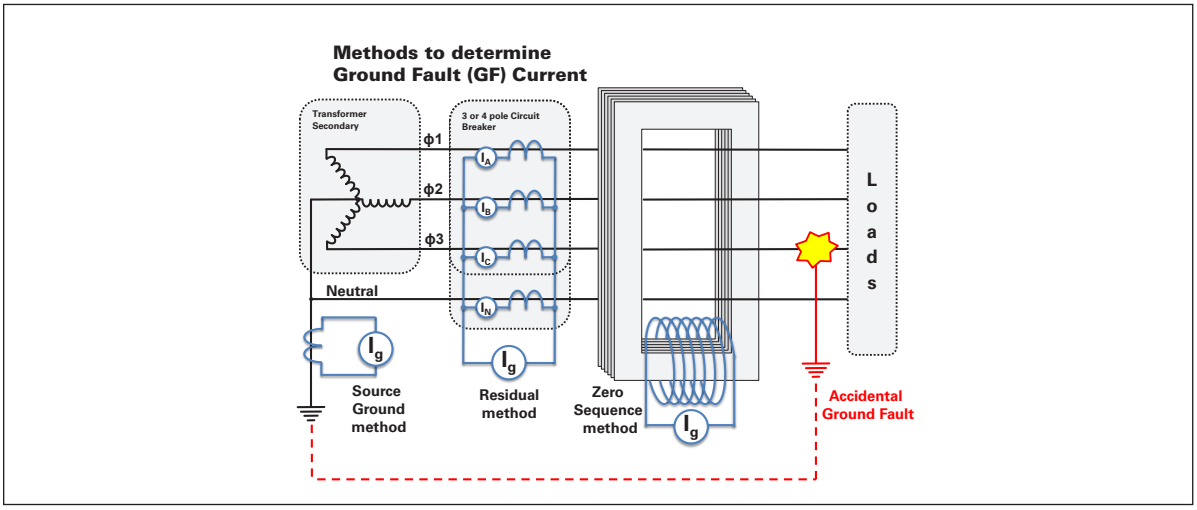
3.5 High instantaneous (withstand) trip

The PXR trip unit provides a factory set function that will trip the circuit breaker at currents above the short time withstand rating of the circuit breaker frame. The function is always active on breakers with interrupting capacity higher than the short time withstand rating and reacts to the peak current level.

3.6 Ground fault settings

When the PXR trip unit includes ground fault protection features, the distribution system characteristics (such as system grounding, number of sources, and number and location of ground points) must be considered along with the manner and location in which the circuit breaker is applied to the system (see Figure 11). To ensure correct ground fault equipment performance and compliance, you must conduct the field testing required to comply with country or regional requirements.

Figure 11. Ground fault examples.



The PXR trip unit provides flexibility in detecting and acting on ground currents. A ground fault alarm can provide a warning of a ground fault condition and a ground fault trip can provide protection under these conditions. For the PXR 25/35, settings options are the same for alarm or trip and can be set using display (Main -> Edit Settings -> Protection settings) or using PXPM. On the PXR 20, the settings are done via rotary switches.

On a PXR trip unit with ground fault protection, there are three modes of operation which are selectable:

- Ground fault protection may be turned OFF;
- Ground fault protection with an Alarm only action; or
- Ground fault protection with a Trip action.

The PXR trip unit provides for two options for sensing to detect ground fault currents: residual and source ground/zero sequence. The option is selected using the display and navigation buttons or by using PXPM configuration software.

3.6.1 Residual current method

Residual sensing is the factory default method of ground fault protection in Magnum PXR and PD-SB circuit breakers. This method is used for 3-wire systems with one current sensor on each phase conductor or 4-wire systems with one current sensor on each phase conductor and one on the neutral. In this method of sensing, the trip unit sums the outputs of the three individual current sensors (IA, IB, IC) and the neutral (IN) if present. If the sum is zero, then no ground current exists. When the sum is greater than the ground fault pickup for the time delay, the breaker will trip or alarm.

The pickup (I_g) is expressed as a multiple of the breaker's current rating (I_n) for circuit breakers less than 1200 A. For breakers 1200 A and greater, the pickup setting is expressed as a multiple of 1200 A (the NEC maximum) (see Table 6).

Table 7. GF residual settings.

Setting / Rating	PXR 25/35	From 0.2 to 1.0 in steps of 0.01				
	PXR 20	**0.2**	**0.4**	**0.6**	**0.8**	**1.0**
Less than 1200 A	I_g =	0.2 x (I_n)	0.4 x (I_n)	0.6 x (I_n)	0.8 x (I_n)	1.0 x (I_n)
1200 A or greater	I_g =	240	480	720	960	1200

If the 1200A bypass license has been purchased, then the residual current (I_g) uses the same formula as shown for under 1200A. For example, if the bypass license is purchased for a breaker with I_n = 3000A and residual ground is set to 0.6, then I_g = 0.6 x I_n = 1800A.

The time delay has a range from 0.05 to 1.0 seconds. Two ground fault protection curve slopes are available: a fixed time (flat) or I^2t response. The slope should be chosen to match the selective coordination requirements. The I^2t response provides a longer time delay for current below 0.625 x I_n than a fixed time (flat) response.

For 3-pole circuit breakers used in a 4-wire system, an external neutral sensor is used. It is connected to the secondaries at N & N•. See Section 4.1 - Neutral sensors for the appropriate sensors.

3.6.2 Source ground and zero sequence method

By setting the ground fault type to source ground/zero sequence, either sensing method is enabled in the trip unit.

Source ground sensing uses a current sensor on the equipment's ground-bonding jumper. The output of the sensor directly measures the total ground current flowing in the grounding conductor. When the current measured by the trip unit exceeds the ground fault settings, the breaker will trip or alarm.

Zero sequence sensing uses a single sensor that surrounds all the phase conductors and the neutral conductor in a 4-wire system. If a ground fault occurs, the vector balance of all the magnetic fields will not sum to zero and the sensor will generate an output signal equivalent to the current flowing to ground. When the current measured by the trip unit exceeds the ground fault settings for the time delay, the breaker will trip or alarm.

The sensor used for both sensing methods is an iron-core CT, connected to the secondary wiring terminals at G1 and G2. See Section 4.6 - Secondary wiring terminals for for terminal numbering.

The input to the trip unit at G1 and G2 is 100 mA per unit. To use a CT that has a 1 A per unit secondary, you must convert the 1 A level to 100 mA by using an auxiliary CT with a 10:1 equivalent ratio. Select the SG/ZS ratio on the display or in PXPM assuming the 10:1 auxiliary CT is being used. Select the SG/ZS ratio on the display or in PXPM assuming the 10:1 converter module is being used.

Alternately, you can use a CT without an additional 10:1 auxiliary CT by using a transformer with the appropriate ratio. For example, a CT with a 1000:1 ratio could be used without an auxiliary CT, providing a 100 mA output with 100 A on the primary.

Table 8. GF source and zero sequence settings.

Typical 1A per unit current transformer	10:1 secondary transformer	Setting / SG/ZS ratio	PXR 25/35 / PXR 20	From 0.2 to 1.0 in steps of 0.01				
				0.2	0.4	0.6	0.8	1.0
1200:1	Yes	1200/1	I_g =	240	480	720	960	1200
1000:1	Yes	1000/1	I_g =	200	400	600	800	1000
800:1	Yes	800/1	I_g =	160	320	480	640	800
400:1	Yes	400/1	I_g =	80	160	240	320	400
4000:1	No	400/1	I_g =	80	160	240	320	400
200:1	Yes	200/1	I_g =	40	80	120	160	200
2000:1	No	200/1	I_g =	40	80	120	160	200
100:1	Yes	100/1	I_g =	20	40	60	80	100
1000:1	No	100/1	I_g =	20	40	60	80	100
50:1	Yes	50/1	I_g =	10	20	30	40	50
500:1	No	50/1	I_g =	10	20	30	40	50
25:1	Yes	25/1	I_g =	5	10	15	20	25
250:1	No	25/1	I_g =	5	10	15	20	25
10:1	Yes	10/1	I_g =	2	4	6	8	10
100:1	No	10/1	I_g =	2	4	6	8	10

The time delay settings are the same for source ground/zero sequence as they are for residual sensing.

3.6.3 Ground fault thermal memory

The PXR trip unit additionally has a ground fault thermal memory function that serves to protect loads in the event of an intermittent or sputtering arc to ground. Without this function, the ground fault protection timer resets each time the arc goes out and the ground fault current is below pickup. Therefore, a sputtering arc fault may not trip the circuit breaker. With the ground fault memory function, the trip unit "remembers" the sputtering ground current. The memory decays with time when the ground current is below pickup. The reset time interval equals six times the ground fault time. For example, with a 0.4 second setting, the function will reset in 2.5 seconds.

3.6.4 Ground fault zone selective interlocking

In addition to Short Delay phase ZSI, the PXR trip units have ground fault protection ZSI. More information can be found in Section 3.10 - Zone selective interlocking (ZSI) of this manual.

3.6.5 Ground fault pre-alarm

A unique feature of the PXR trip units is the ability to set a pre-alarm signal when the level of ground fault current gets to a desired level. The pre-alarm activate level can be set from 50% to 100% of the I_g pickup value. The pre-alarm notification can also be mapped to the one of the three trip unit relays. See Section 4.2 - Configurable relays for additional details.

3.7 Enhanced protection settings

The PXR 25 and PXR 35 offer an additional set of functions which can be used to generate an alarm or trip the breaker. Each has specific setpoints for pick-up and time delay (minimum 1.0 second). In addition, the actions are configurable to OFF (default), ALARM, or TRIP depending on the feature. The ranges for each setpoint are shown in Section 1.4 - Enhanced protection features. Control power (AUX) to the trip unit is recommended when using enhanced protection features.

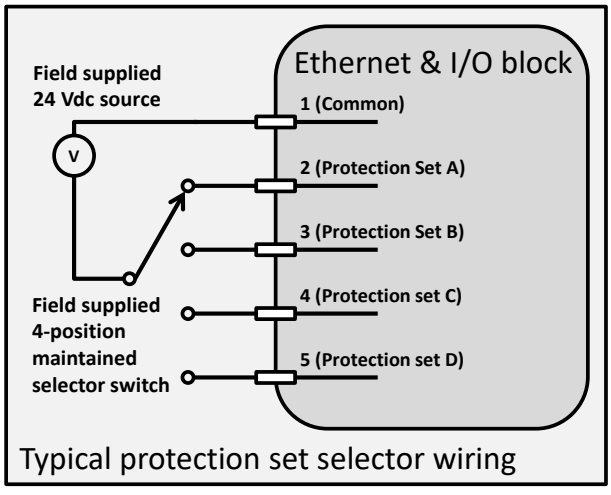
For PXR 35 trip units, specific groups of protection types can be enabled, allowing multiple protection features. The protection group must be enabled for any of the individual protection features to be active. The following are available:

- Motor protection - Over voltage, under voltage, voltage unbalance, phase rotation (sequence), current unbalance, phase loss, reverse power, reverse reactive power;
- Power protection - Real power, apparent power, reactive power, real power demand, apparent power demand, power factor;
- Frequency protection - Over frequency, under frequency;
- Total harmonic distortion alarm - Current, voltage;
- Synchronism check.

3.7.1 Protection sets (PXR 35 only)

There are four protection sets that can be fully configured for protection and alarm settings. This provides separate groups of settings that can be switched between using hardwired inputs to the trip unit or over communications. Figure 12 shows a typical hard-wired switch connection to select one of four protection sets. The inputs are mutually exclusive; only one can be connected at a time.

Figure 12. Protection Set Selector.



3.7.2 Over voltage (PXR 25/35)

Line-to-line RMS voltages (Vab, Vbc, Vca) are continuously monitored. If any line-to-line voltage is greater than the pickup setting for the time delay, then the configured action will be taken.

3.7.3 Under voltage (PXR 25/35)

If any line-to-line voltage is less than the pickup setting for the time delay and the current in at least one phase is greater than 0 A, then the configured action will be taken. Minimum voltage must be applied for this to be active, see section 3.7.4.

Note: PXR 25 setpoint ranges for under voltage and overvoltage overlap: If both are used in an application, under voltage pickup should always be set less than over voltage pickup. If under voltage pickup is set higher than overvoltage pickup and the breaker is closed, the trip unit will see one or both conditions pickup satisfied, begin timing and trip.

3.7.4 Voltage unbalance (PXR 25/35)

If the difference between the maximum and minimum of any of the three line-to-line voltages is greater than the pickup setting for the time delay, then the action will be taken. Action will be taken only when at least one line-to-line voltage is greater than 84 V.

The calculation for the voltage unbalance pickup is:

$$\frac{Max(Vab,Vbc,Vca)-Min(Vab,Vbc,Vca)}{Max(Vab,Vbc,Vca)} \times 100\%$$

3.7.5 Current unbalance (PXR 25/35)

The RMS current in each of the phases (Ia, Ib, Ic) is continuously monitored. Unbalance protection will protect against partial or full loss of one or two phases. If the difference between the maximum and minimum of any of the three phase currents is greater than the pickup setting for the time delay, then the protection action will be taken. Action will be taken only when at least one phase current is greater than 50% of the long delay pickup (LDPU or I_r) setting.

The calculation for the current unbalance and phase loss pickup is:

$$\frac{Max(Ia,Ib,Ic)-(Min(Ia,Ib,Ic)}{Max(Ia,Ib,Ic)} \times 100\%$$

3.7.6 Phase loss (PXR 25/35)

Phase loss protection is used for complete loss of one or two phases. If the difference between the maximum and minimum of any of the three phase currents is greater than 75%, for the time delay, then the protection action will be taken. Action will be taken only when at least one phase current is greater than 50% of the Long Delay Pickup (LDPU or I_r) setting.

3.7.7 Reverse power (PXR 25/35)

Power flowing through the breaker is continuously monitored. If the reverse real power is greater than the pickup setting for the time delay, then the configured action will be taken. Additionally, with the PXR 35, if the reverse reactive power is greater than the pickup setting for the time delay, then the configured action will be taken.

3.7.8 Over power (PXR 35 only)

Real power (kW), apparent power (kVA), reactive power (kvar), and power demand (kW) are continuously monitored. If any of these parameters is greater than the associated pickup setting for the time delay, then the configured action will be taken.

3.7.9 Over frequency (PXR 35 only)

Frequency is continuously monitored. If the frequency is greater than the pickup setting for the time delay, then the configured action will be taken.

3.7.10 Under frequency (PXR 35 only)

Frequency is continuously monitored. If the frequency is less than the pickup setting for the time delay, then the configured action will be taken.

3.7.11 Power factor alarm/trip (PXR 35 only)

The 3-phase power factor is continuously monitored. If the power factor is less than the pickup setting for the time delay, then the configured action will be taken.

3.7.12 Total harmonic distortion alarm (PXR 35 only)

The total harmonic distortion for phase and neutral currents, and for line-to-line voltages is continuously monitored. If the configured parameter is greater than the pickup setting for the time delay, then the trip unit will issue an alarm.

3.7.13 Synchronism-check (PXR 35 only)

Both the line and load voltages are continuously monitored. This allows the PXR 35 to provide both live bus and dead bus synchronism-check options. This can be mapped to a programmable relay and used in conjunction with the spring release. The live bus setting is the minimum line voltage required to be active. The dead bus setting is the maximum voltage that can be measured and considered to be "dead".

The following configurations are possible:

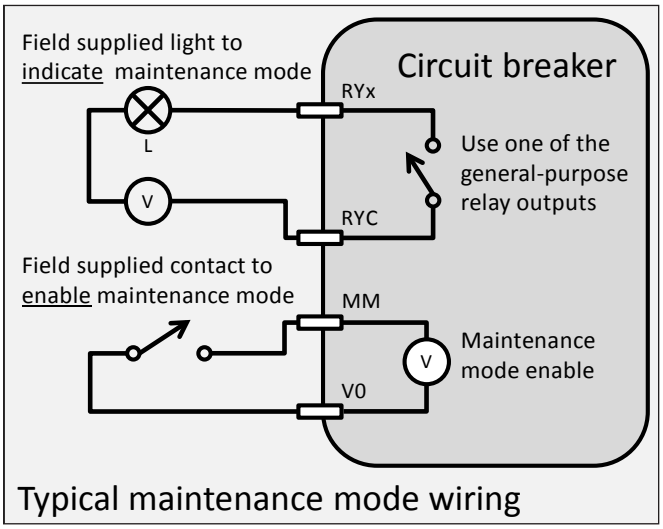
- Live source – Live load
- Dead source – Live load
- Live source – Dead load
- Dead source – Dead load

3.8 Maintenance mode

The PXR trip units support Eaton's Arc Flash Reduction Maintenance System™ (ARMS), also known as maintenance mode. When active, the trip unit will trip the breaker with no intentional delay whenever the configured pickup level is exceeded. The maintenance mode function operates regardless of the instantaneous settings. If maintenance mode

causes the circuit breaker to trip, the INST indicator will be illuminated and the maintenance mode trip message will be displayed.

Figure 13. Typical maintenance mode wiring.



The maintenance mode pickup level setting is configured using the display and navigation buttons or PXPM software. They range from 1.5 (most protective) to 10, expressed as a multiplier of I_n. The adjustable current settings allow for different levels of protection. A higher level may be needed when, for example, another load fed from this breaker may contain motors that are being started and create large inrush currents over the lowest trip current level. The selection of one of the reduction settings should be determined and selected by a person who is experienced in power system analysis.

3.8.1 Activation and indication of maintenance mode

A blue colored ring surrounding the MAINT. MODE switch on the front of the trip unit always illuminates to confirm when the function is active. Additionally, a relay can be mapped which will close to indicate when maintenance mode is active. See Figure 12 Typical maintenance mode wiring.

When the switch on the front of the trip unit is in the ON position, maintenance mode is active and cannot be turned off remotely.

When the switch is in the REMOTE position, maintenance mode can be remotely activated:

- By a contact wired to the secondary terminal block of the breaker's MM terminal. See Section 4.6 - Secondary wiring terminals for wiring information and contact requirements.
- Via communications. This can be done through the Modbus register map. See Section 9 - Modbus register map for details.
- Using PXPM software connected to the trip unit via USB.

Maintenance mode will be active when any of the methods is calling for activation. If you have enabled maintenance mode using multiple methods, to disable it requires removing each of those methods.

3.9 Zone selective interlocking (ZSI)

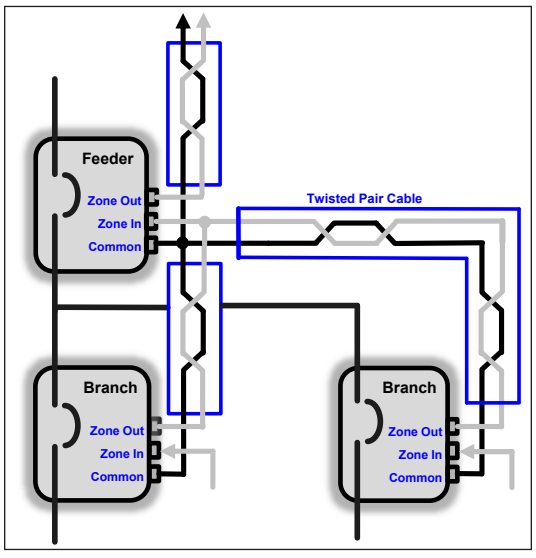
The zone selective interlocking (ZSI) functions in conjunction with the short delay and ground fault protection functions. ZSI provides the fastest possible tripping for faults within the zone of protection of the circuit breaker and provides positive coordination among all circuit breakers in the system (mains, ties, feeders, and downstream circuit breakers).

When ZSI is enabled and a fault occurs, all circuit breakers that see the fault current will send a signal upstream that they "see" the fault. When the fault current is above the pickup level of the ground fault or the short delay settings, the circuit breaker will check its input to check if any breaker below it has sent a restraining signal. If there is an input signal, the circuit breaker will follow its normal time delay waiting for the downstream breaker to clear the fault. If there is no input signal, and after a double check of the signal, the PXR breaker will immediately trip since it is the closest circuit breaker to the fault.

The ZSI is wired using a set of three wires labeled Zone In (ZI), Zone Out (ZO), and Zone Common (ZC) on the secondary terminals above the circuit breaker. These signals are compatible with all Eaton circuit breakers which have the ZSI function. The zone out signal is sent whenever the ground fault pickup is exceeded or when the short delay pickup value is exceeded. This provides maximum selectivity for coordination with larger upstream circuit breakers.

When the PXR trip unit includes IEC 61850 GOOSE functionality the ZSI signal can be sent via GOOSE messaging over Ethernet in conjunction with, or in place of, the hardware ZSI signaling.

Figure 14. Typical ZSI wiring.



The PXR trip unit display provides unique new features to provide confidence that ZSI is enabled, working, and for testing ZSI. For PXR 25 trip units, ZSI applies to both short time and ground faults (when ground is included) and the display of the trip unit will show the letters ZSI on the bottom when ZSI is enabled. For PXR 35 trip units, ZSI can be separately enabled for short time and ground fault (when ground is included), and the display of the trip unit will show the letters ZSI at the top with an S (short), G (ground), or SG (short and ground) to indicate the functions enabled. When there is a ZSI input signal detected on the input terminal, the display of the PXR trip unit will show a checkmark at the bottom next to the letters ZSI. The check mark will remain on the display until the RESET pushbutton is pushed to clear it.

Figure 13 shows an example of the feeder 11 breaker being tested. At the start of a test, press the RESET pushbutton on all trip units and all trip units will start off with the checkmark cleared.

In Figure 14, feeder 11 is tested with 4000 A of secondary injection current to simulate a fault above its short delay pickup setting. Feeder 11 shows a checkmark because it is self-interlocked. Main 1 breaker shows a checkmark, so it has received the ZSI input signal.

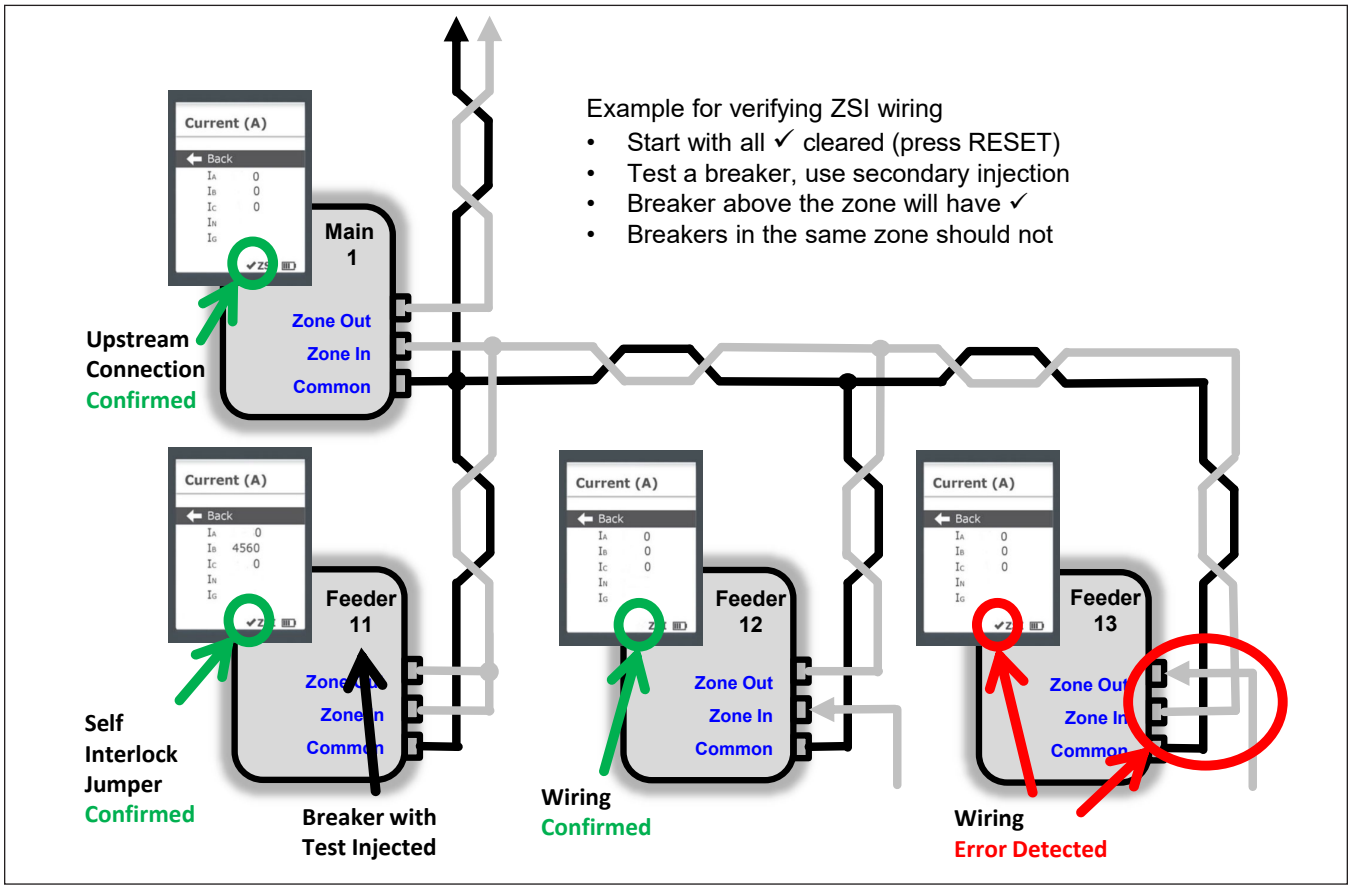
Feeders 12 and 13 are in the same zone and did not see the fault (the simulated fault is on feeder 11) and should not see the ZSI input signal. Feeder 12 will not show a checkmark, which is correct. But feeder 13 shows a checkmark. Further inspection will find that feeder 13 has its input and output signals wired incorrectly.

Additional information on ZSI can be found in the Eaton application note: AP02602002EN.

3.10 Making current release (MCR)

All PXR trip unit styles have a making current release (MCR) function. This safety feature prevents the circuit breaker from being closed and latched-in on a faulted circuit. The MCR is enabled only for the first two cycles of current following the initial circuit breaker closing operation. The circuit breaker will trip with no delay and the instantaneous (INST) indicator will show the cause of trip. This non-adjustable release is set at the factory based on the sensors and frame of the circuit breaker.

Figure 15. Verifying ZSI operation (example).



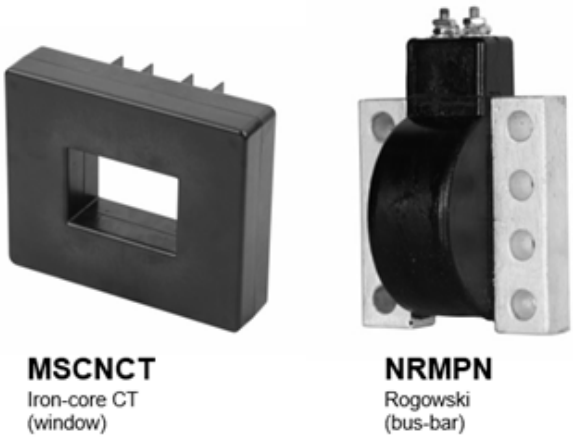
4. Complementary features

The PXR has features which allow you to integrate within a larger system. An external neutral sensor can be added to 3-pole breakers. A control power supply will keep the trip unit active when there is insufficient current in the phases to self-power. Relays can be used for signaling conditions within the breaker. Communications can be used to monitor real time operation. Testing functionality eliminates the need for separate secondary injection equipment. The event capture information is extremely valuable in investigating the cause of an alarm or trip. Finally, the health monitoring functions allow you to understand the impact of each event and guide you in servicing the circuit breaker.

4.1 Neutral sensors

For a 3-pole breaker feeding 4-wire loads, an external neutral sensor is connected to the secondaries at N & N•. See Section 4.6 - Secondary wiring terminals for terminal numbering. The proper polarity of the sensor and conductor, both marked on the sensor, is important. The external sensor may be selected as Rogowski or iron-core CT (factory default) based on your application requirements. The system does not harvest energy to power-up the trip unit from the external neutral sensor.

Figure 16. External neutral sensors.



If neutral sensing is not needed in your application, select the iron-core CT setting or jumper N to N•.

Table 9. External neutral sensor part numbers.

Frame	Neutral sensor type	Sensor	Instruction leaflet
All	Rogowski (bus bar)	NRMPN	IL0131154EN
	Iron-core CT (window)	MSCNCT	

If the sensor installed does not match the selection in the trip unit, the reading will not represent the actual current flowing in the neutral. Power up the trip unit by applying 24 Vdc to the breaker secondary (see Section 4.6 - Secondary wiring terminals) or connecting to the micro-USB port (see section 2.9 - Test and configuration port), then apply a primary current through the sensor to verify. A settings mismatch will result in either a much higher or lower (10x or more) reading than actual. Change the setting to match the sensor and re-check.

4.2 Configurable relays

The PXR contains three relays which can be configured to indicate various conditions within the circuit breaker. They have factory default configuration (depending on style) relay #1 = Maintenance Mode Active or Health Warning, relay #2 = High Load 1 Alarm or Ground Fault Alarm, and relay #3 = Breaker Trip (all current triggered trips).

The contacts of the relays are wired at the circuit breaker's secondary terminal block. Refer to Section 4.6 - Secondary wiring terminals for terminal locations and contact ratings.

The relays may be reconfigured using PXPM to indicate any of the conditions in Table 9. For PXR 20/25, up to two conditions can be mapped to the same relay. For PXR 35, multiple triggers (beyond just two) can be mapped to the same relay.

Table 10. Configurable relay function.

Condition/function name	Description of relay operation:	
	The relay will close when ...	The relay will open when ...
Aux. contact	Breaker is closed	Breaker is open
Bell contact	Breaker is tripped	Breaker is not tripped (open or closed)
High load 1 alarm	Current flow is greater than setpoint (adjustable from 50% to < high load 2 of I_r)	Current flow falls 5% below the setpoint
High load 2 alarm (relay #2 default, non-GF styles)	Current flow is greater than setpoint (adjustable from > high load 1 to 120% of I_r)	Current flow falls 5% below the setpoint
Maintenance mode active (relay #1 default, ARMS styles)	Trip unit is in the maintenance mode	When the trip unit exits Maintenance Mode
ZSI active	ZSI function active	ZSI is not active
ZSI input received	A ZSI INPUT signal is received	RESET button is pressed or communications reset command received
ZSI output sent	A ZSI OUTPUT signal is sent	RESET button is pressed or communications reset command received
High temperature alarm	Temperature exceeds 5°C (41°F) below the level of the temperature trip setting	Temperature falls 5°C (41°F) below the trip setting
Ground fault pre-alarm	Ground current is greater than the setpoint (adjustable from 50% to 100%)	Ground current falls 5% below the setpoint
Thermal memory alarm	Thermal memory value is >75%	Thermal memory value is <70%
Watchdog alarm	Auxiliary power is active, and the trip unit is healthy and operating	There is an error in the trip unit from any of the self-diagnostics
Low battery alarm	Battery is below 1 bar (25%)	Battery value is 1 bar (25%) or higher

Table 10. Configurable relay functions (continued).

Condition/function name	Description of relay operation:	
	The relay will close when ...	The relay will open when ...
Internal alarm	There is an internal fault detected	RESET button is pressed or communications reset command received
Health warning (relay #1 default, non-ARMS styles)	Health value is below setpoint (adjustable from 10 to 50%)	Health value is at or above setpoint
Communication alarm	Any external communications error occurs	RESET button is pressed or communications reset command received
Alarm present	Any of the internal, health, communication, or battery alarms are active	All of the referenced faults are inactive
Over voltage	Voltage is greater than or equal to setpoints for pickup and time (ANSI 59)	If set as alarm: voltage is less than pickup; If set as TRIP: RESET pressed or reset command received.
Under voltage	Voltage is less than or equal to setpoints for pickup and time (ANSI 27)	If set as ALARM: voltage is greater than pickup; If set as TRIP: RESET pressed or reset command received.
Voltage unbalance	Voltage unbalance is greater than or equal to setpoints for pickup and time (ANSI 47)	If set as ALARM: voltage unbalance is less than pickup; If set as TRIP: RESET pressed or reset command received.
Current unbalance	Current unbalance is greater than or equal to setpoints for pickup and time (ANSI 46)	If set as ALARM: current unbalance is less than pickup; If set as TRIP: RESET pressed or reset command received.
Reverse active power	Reverse active power is greater than or equal to setpoints for pickup and time (ANSI 32R)	If set as ALARM: reverse active power is less than pickup; If set as TRIP: RESET pressed or reset command received.
Over power demand	Real power demand is greater than or equal to setpoints for pickup and time	If set for ALARM: real power demand is less than pickup; If set as TRIP: RESET pressed or reset command received.
Phase loss (current)	At least one phase is less than or equal to 75% of another phase for configured time setpoint	If set as ALARM: no phase is less than or equal to 75% of another; If set as TRIP: RESET pressed or reset command received.
Overload trip	There is a long or over-temperature trip	RESET button is pressed or communications reset command received
Neutral current trip	There is a neutral current trip	RESET button is pressed or communications reset command received
Short delay trip	There is a short delay trip	RESET button is pressed or communications reset command received
Instantaneous trip	There is an instantaneous trip or withstand (override) trip	RESET button is pressed or communications reset command received

Table 10. Configurable relay functions (continued).

Condition/function name	Description of relay operation:	
	The relay will close when ...	The relay will open when ...
Short circuit trip	There is a short, inst. or withstand (override) trip	RESET button is pressed or communications reset command received
Ground fault trip (relay #2 default, GF styles)	There is a ground fault trip	RESET button is pressed or communications reset command received
Maintenance mode (ARMS) trip	There is a maintenance mode trip	RESET button is pressed or communications reset command received
Any current protection trip (relay #3 default)	There is any type of protective current trip (overload, neutral, short delay, instantaneous, short circuit, ground fault, maintenance mode)	RESET button is pressed or communications reset command received
Mechanism trip	There was a "sneaker" trip because the mechanism position switch indicates open but there is measurable current	RESET button is pressed or communications reset command received
Open breaker pulsed	An OPEN breaker command from any of the communications channels is received	Two seconds after the OPEN breaker command is received
Close breaker pulsed	A CLOSE breaker command from any of the communications channels is received	Two seconds after the CLOSE breaker command is received
Remote control	An output ON command for the relay specified in the command is received on any of the communications channels	An output OFF command for the relay specified in the command is received on any of the communications channels
AQD Pre-warning	Immediately before a trip for any reason	Two seconds after closing
Over frequency (PXR 35 only)	Frequency is greater than or equal to set points for pickup and time	Frequency is less than pickup if function set or RESET pressed or reset command received if function set for TRIP
Under frequency (PXR 35 only)	Frequency is less than or equal to set points for pickup and time	Frequency is greater than pickup if function set or ALARMRESET pressed or reset command received if function set for TRIP
Phase rotation (PXR 35 only)	Phase rotation (sequence) does not match configured set point	Phase rotation matches configured set point if function set or RESET pressed or reset command received if function set for TRIP

Table 10. Configurable relay functions (continued).

Condition/function name	Description of relay operation:	
	The relay will close when ...	The relay will open when ...
Setpoint mismatch (PXR 35 only)	Setpoint(s) in the trip unit do not match those in the Cassette ID board. This typically occurs when replacing circuit breakers or trip units in draw-out applications.	RESET button is pressed or if a reset command sent by any communication. But if the faulty condition remains then the relay will pick up again. Customer should enter the setpoint programming menu and transfer setpoints from cassette to breaker or breaker to cassette to clear this error
Reverse reactive power (PXR 35 only)	Reverse reactive power is greater than or equal to set points for pickup and time	Reverse reactive power is less than pickup if function set or RESET pressed or reset command received if function set for TRIP
Total harmonic distortion (PXR 35 only)	THD is greater than or equal to set points for pickup and time	THD is less than pickup if function set or RESET pressed or reset command received if function set for TRIP
Under power factor (PXR 35 only)	Power factor is less than or equal to set points for pickup and time	power factor is greater than pickup if function set or RESET pressed or reset command received if function set for TRIP
Real power (PXR 35 only)	Real power is greater than or equal to set points for pickup and time	Real power is less than pickup if function set or RESET pressed or reset command received if function set for TRIP
Reactive power (PXR 35 only)	Reactive power demand is greater than or equal to set points for pickup and time	Reactive power demand is less than pickup if function set or RESET pressed or reset command received if function set for TRIP
Apparent power (PXR 35 only)	Apparent power demand is greater than or equal to set points for pickup and time	Apparent power demand is less than pickup if function set or RESET pressed or reset command received if function set for TRIP
Sync check (PXR 35 only)	Line and load voltages and frequencies are synchronized based on the configured set points for the specified time	Line and load voltages and frequencies are not synchronized for the specified time

4.2.1 Wiring and configuration for remote open/close

The on-board relays may be used to energize a shunt trip (ST) to open the circuit breaker or a spring release (SR) to close the circuit breaker when commanded via Modbus. Consult IL2C14762 & IL2C14761 for details on the shunt trip (ST) and spring release (SR) accessories.

To accomplish this requires three elements:

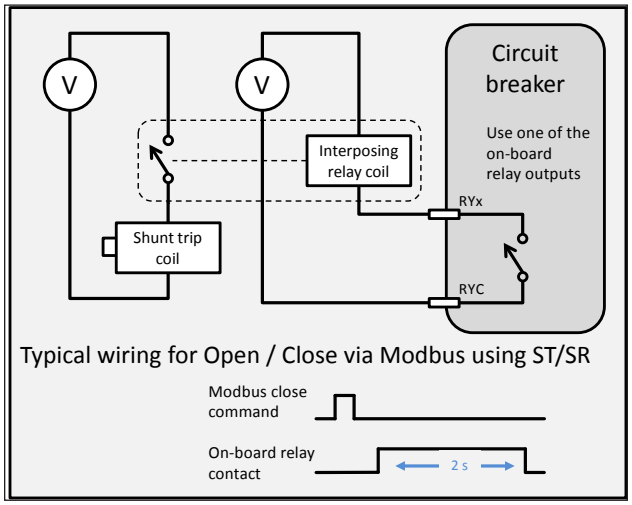
- Mapping the on-board relays to act using the open breaker pulsed or close breaker pulsed function;
- Wiring the on-board relays to an interposing relay to subsequently connect a power source to the ST/SR; and
- Programming a Modbus client to send the open breaker pulsed or close breaker pulsed commands.

To open the breaker, an on-board relay should be chosen and mapped to the open breaker pulsed function. Do not map the same relay to any other functions or the ST will energize unexpectedly when that function causes the relay to close. The mapping can be accomplished using the display: Main menu -> Edit Settings -> Relay Config -> Open Command.

PXPM can also be used by navigating: Setpoint Configuration -> Connect to Unit -> Expand the System Settings bar -> Relay Function Configuration select "Other" and "Open Breaker Pulse".

Wire the chosen on-board relay to complete a circuit to the ST. The on-board relay should energize the coil of an interposing relay using control power. The on-board relay contacts are not designed to switch high current inductive loads. The wiring terminals and relay specifications are found in Section 4.6 - Secondary wiring terminals. The interposing relay contacts should be chosen to match your ST specifications and the power source used to energize the ST.

Figure 17. Typical wiring to open/close.



The Modbus client is then programmed to send a remote control command to the trip unit. The command and its complement are written to three consecutive registers. This triggers the trip unit to close the on-board relay for two seconds which has been mapped to the open breaker pulsed function. In turn, this energizes the ST and opens the breaker. See Section 9.5 - Remote control commands for details of the command encoding.

To close the breaker, substitute Close for Open and spring release (SR) for shunt trip (ST). You may want to include a motor-operator accessory in the circuit breaker to charge the closing springs for fully remote operation.

4.3 Integrated Modbus

A Modbus RTU communication port is integrated into the PXR trip unit for certain styles of PXR 20 and all PXR 25/35 styles. Modbus is a communication protocol used for transmitting information between devices. The PXR is a responding device (server), using the remote terminal unit (RTU) protocol, it will process commands from another device and respond with the function or data requested. Breaker status (closed/tripped/open), setpoints, and operating information are all available via Modbus.

The PXR 35 also includes Modbus TCP via the Ethernet connection if this option is installed.

The trip unit supports Modbus RTU function codes 2, 3, 4, 6, and 16. Single and multiple read or write operations can transact up to 122 registers (244 bytes) in a single Modbus message. The factory default configuration for PXR 20/25 trip units is device address = 1, baud rate = 9600, parity = even, stop-bits = 1. The factory default configuration for PXR 35 trip units is device address = 7, baud rate = 38400, parity = none, stop-bits = 1. The physical layer is based on 3-wire RS-485 and connects at the circuit breaker's secondary terminals labeled B(+), A(-), and CMN. See Section 4.6 - Secondary wiring terminals for details on the connector locations and recommended cable. For PXR 35, Modbus TCP settings follow the Ethernet settings. The detailed Modbus register map including network configuration options is shown in Section 9 - Modbus register map. The same map is used for both Modbus RTU and TCP. Also refer to TD013198EN "Modbus Real Time Data Register Map.

4.4 Ethernet Connectivity

4.4.1 Communication adapter modules (CAMs)

All styles of the PXR 20 and PXR 25 trip units support communication adapter modules (CAMs). These modules provide communication to either Ethernet or Profibus field networks.

Figure 18. ECAM module.



Table 11. Communication adapter modules.

Network	Module	Instruction leaflet
Ethernet (Modbus TCP)	PXR-ECAM-MTCP	IL0131132EN
Profibus DP	PXR-PCAM	IL120009EN

These modules are remotely mounted on a DIN rail and field-wired into the trip unit using the circuit breaker's secondary terminal block. The field bus is then wired to a connection on the CAM module. See Section 4.6 - Secondary wiring terminals for details.

Please consult the respective Instruction leaflets for details of each CAM's functionality and configuration.

Note: PXR 35 uses integrated Ethernet and does not support CAMs.

4.4.2 Integrated Ethernet Connectivity (PXR 35 only)

Ethernet connectivity is integrated into all PXR 35 trip unit styles. An Ethernet and I/O terminal block is available that includes two Ethernet ports and Inputs/Outputs connected to the high-density block in the breaker secondary connector. This provides communication from the trip unit to a field Modbus TCP or IEC 61850 network using the Ethernet physical layer. It also allows connection to a webpage via the IP address of the Ethernet port.

The PXR 35 trip unit contains two Ethernet ports that both respond to the trip unit's single MAC and IP address. Internally, the trip unit utilizes a simple switch that passes traffic through to both physical ports. This requires the trip unit to be powered, and in the case of a draw-out installation, the breaker to be racked in for the two ports to operate. When powered, the trip unit can communicate via the Ethernet ports as a responding (server) device.

4.4.2.1 IEC 61850 GOOSE Messaging

Starting with firmware version 2.0.0 the PXR 35 provides IEC 61850 GOOSE functionality with the following features:

- Zone selective interlocking (ZSI) in conjunction with or in place of hardwired ZSI signaling.
- Global Event Captures to collect data simultaneously across multiple trip units in a network.
- Microgrid operation with a user-provided external microgrid controller for on/off breaker control and protection set selection.
- Transmitting current, voltage, frequency, power, and power factor measured values.

Refer to the TD150066EN "IEC 61850 GOOSE Functionality for Magnum PXR 35" for full details.

4.5 I/O Block (PXR 35 only)

The I/O block contains 6 binary inputs and 2 binary outputs with integral currently limiting designed for nominal 24 Vdc operation. The inputs draw approximately 2.4 mA and the outputs can switch approximately 2 mA at nominal voltage. Additional connections are provided for common input returns, chassis ground wiring, and future functionality. See MN013016EN "Magnum PXR and Power Defense SB low voltage power circuit breakers user manual" for complete details.

4.6 Cassette ID (PXR 35 only)

The Cassette ID board is located in the center of the secondary terminal connector and stores a separate copy of setpoints. When the trip unit is powered the setpoints within the trip unit and frame module are compared to the setpoints in the Cassette ID board. If a difference is detected the user is prompted by the touchscreen to select which setpoints to use. This allows the user to quickly reconfigure a system when replacing circuit breakers in draw-out applications.

4.7 Bluetooth (PXR 35 only)

Bluetooth is integrated for certain PXR 35 trip unit styles. The presence of Bluetooth is indicated by the Bluetooth symbol on the front of the trip unit. Bluetooth can only be used in conjunction with PXPM via a computer. No other connection type is possible.

4.8 Auxiliary control power

Providing auxiliary control power to the PXR trip unit will provide full functionality even when the circuit breaker is open or when the circuit breaker is under very light load such that the self-powering current transformer cannot provide energy to power the trip unit. The enhanced protection features, relays, and communication channels require auxiliary control power for operation. Auxiliary control power is connected to the circuit breaker's secondary terminal block. The PXR draws up to 100mA steady state and 700mA inrush during power up and test trips. We recommend an isolated supply such as the Eaton PSG family rated for 24 Vdc output and a minimum of 0.5A.

Figure 19. Control power supply.



Remember: auxiliary control power is not required to provide current protection features. Protection is active well before any overload. The trip unit begins to power-up at very low levels of current (approximately 40 A per phase in three phase configuration).

In situations where auxiliary control power may be intermittent or sag during a transfer, an uninterruptable power supply (UPS) is recommended to ensure continuous operation of the relays and communication.

4.9 Secondary wiring terminals

Table 11 shows the circuit breaker secondary wiring terminals by feature and terminal number. The name column refers to the abbreviation used in this manual and on the terminal label. The alternate name is sometimes seen on devices which may be used within the larger system. Please refer to MN013016EN which includes the secondary terminals wiring diagram for additional circuit breaker wiring information, including terminal numbers for installed accessories.

Table 12. Wiring terminals.

Feature	Terminal	Name	Notes	Alt. name
Auxiliary control power	C14	+24 V	Connect a high quality, isolated source capable of supplying 100mA steady state and 700mA inrush during power-up and test trips. Eaton PSG family rated for 24 Vdc output and a minimum of 0.5A recommended. See Section 4.8 - Auxiliary control power.	
	C15	0 V (common)		
Zone selective interlocking (ZSI)	C19	ZI	Connect to other ZSI enabled breakers in the system. Maximum length of 75 meters (255 feet) using AWG #22 wire. Provide a self-interlocking jumper (C19-C21) on ZSI enabled breakers if coordination is desired with other downstream circuit breakers without ZSI capability. See Section 3.10 - Zone selective interlocking (ZSI).	ZIN
	C20	ZC		ZCOM
	C21	ZO		ZOUT
Maintenance mode enable input	C18	MM	Connect 0 V (C15) to this terminal through an external dry contact to enable ARMS. This is a low-voltage signal. Use a high-quality gold contact, maximum length 3 meters (15 feet). See Section 3.9 - Maintenance mode.	
Neutral voltage reference	C49	VN	Connect to the neutral bus for line to neutral voltage readings. Connection is internal for 4 pole breakers.	
External neutral current sensor for overcurrent or residual ground	C16	N	Connect to the external neutral current sensor (IL0131154EN). See Section - 4.1 Neutral sensors.	
	C17	N•		
Ground fault sensor for source ground or zero sequence	C7	G1	Connect to the source ground or zero sequence sensor when used. See Section 3.6.2 - Source ground and zero sequence.	
	C8	G2		
Configurable relays	C10	RY1	Normally open (NO) contacts rated 6 A at 250 Vac/30 Vdc. Contact closes per the associated function. RYC (C13) is the common connection. Requires control power to actuate. See Section 4.5 - Auxiliary control power.	
	C11	RY2		
	C12	RY3		
	C13	RYC		
Modbus RTU	C22	B (+)	Connect other Modbus RTU devices in a daisy-chain to these terminals. Recommended cable is twisted-pair, foil shielded w/drain, 120 ohm impedance.	
	C23	A (-)		
	C24	CMN		
Communication adapter module (CAM)	C29	CM1	Connect to the selected CAM module referring to CAM instruction leaflets for detail. CAM signal names are listed to the right. See Section 4.4 - Communications adapter modules (CAMs).	RX+/CMM1
	C30	CM2		RX-/CMM2
	C27	CM3		TX+/CMM3
	C28	CM4		TX-/CMM4
	C15	0 V (common)		0 V/C/AGND

5. Power Xpert Protection Manager software

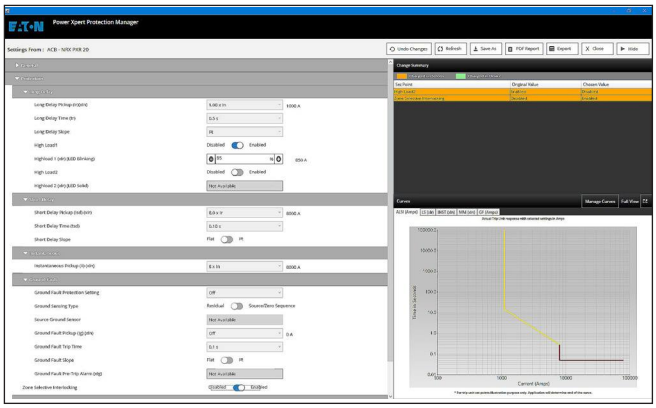
Eaton's PXPM is a Microsoft® Windows-based software that configures, controls, monitors, and tests Eaton PXR trip units. You can create, modify, and save configurations for a PXR trip unit. The software further allows the user to reset trip units, adjust trip unit's date and time, capture current or voltage waveforms, and perform trip or no-trip tests.

The software is available as a download from the following link: www.eaton.com/PXPM. The Power Xpert Protection Manager's main screen allows you to complete activities as described below.

5.1 Setpoint configuration

Key to the protection settings in your trip unit is the configuration screen, which allows you to view and edit setpoints. In the PXR 20, key protection settings are determined by physical switches which cannot be changed using PXPM. The PXR 25/35's settings are all configured electronically, offering a finer resolution and configuration through PXPM. Settings are arranged in the left half of the screen under categories for system, protection, communication, and relay configuration. The upper right part of the screen includes click-able buttons which take the described action. The mid-right part of the screen lists the changes you are making, and the lower right holds a trip-curve which dynamically changes based on your setting changes.

Figure 20. PXPM setpoint screen.



Typical actions available from the configuration screens include:

- View and edit setpoints - For each setpoint, its range, step size, and description are shown in the tool tip when a user hovers the mouse cursor over that setpoint.
- Save as - Saves setpoints to a configuration file. Users will be prompted to select a location and a name for the configuration file.
- Export - Sends the setpoints to a trip unit. The trip unit must be connected to the computer through USB for successful operation.
- PDF report - Exports all setpoints to a portable document format (PDF) file. Modified setpoint parameters are highlighted in the exported PDF file.
- Undo changes - Resets all setpoints to their original values.
- Curves - Shows a dynamic representation of the trip curve as you are configuring the setpoints. It displays long and short delay protection curves, as well as ground (earth) and instantaneous protection curves.
- Change summary - Displays a summary of setpoints that have been changed in the present session. Both original and changed values are displayed.

5.2 Device settings

The Device Settings screen allows you to change the password, set the date/time, reset the trip unit max./min. values, counters, and cause of trip indicators.

5.3 Test mode

This set of screens supports the PXR's internal hardware secondary injection system. The screens guide you through testing of the sensors, trip actuator, and tripping performance based on your settings and the secondary injection current selected. Using test mode with the PXR trip unit is described in Section 6.4.1 - Secondary injection test.

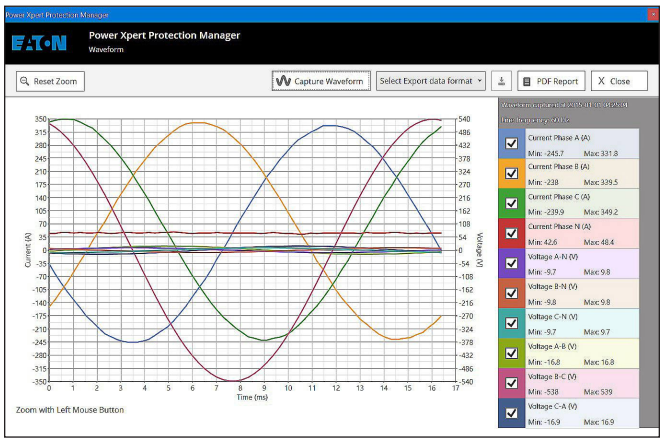
5.4 Breaker information

Here you will find the breaker catalog number and serial number, the trip unit style and serial number, as well as date of manufacture.

5.5 Real time data

The Real Time Data screens allow you to see the sensed information being evaluated by the trip unit. Expand any of the groupings of information on this screen to see the data. All the currents, voltages, power, and energy can be set for a one-time view, click to refresh or automatic refresh.

Figure 21. PXPM waveform display.



From this screen you can also capture a 1-cycle waveform of the currents and voltages being sensed by the trip unit. The capture can be viewed on-screen and then either converted to a PDF document or to one of several export formats. You can trigger a waveform capture at any time using PXPM.

5.6 Event summaries

Any of the events that are logged by the trip unit can be viewed using this screen. Each type of event is categorized and viewable in the order they occurred. Events which capture a waveform also show an icon which can be used to view that waveform.

5.7 Reports

The software allows you to create a PDF file which contains information about the circuit breaker, its real time operating conditions, the events which have been recorded and the setpoints it is using. You can also add your application information details including customer name, plant/gear location identification, equipment condition, and general notes.

5.8 Application settings

Here you can change the software's language and see the installed version. This screen provides the important contact information for EatonCare technical support. PXPM also has a feature that allows it to check for updates and download them from the Eaton website every time it runs. We recommend that you use this option to ensure that you always have the latest software features available to you.

5.9 PXPM feature licenses

The Application Settings screens also provide the mechanism to request value-added feature licenses. These licenses can support a wide range of activities including testing sessions, waveform viewing, troubleshooting assistance, health analytics, and changing the rated current (I_n).

6. Monitoring and testing

6.1 Monitoring from a supervisory system

The PXR trip unit is easy to integrate within a system using the integrated Modbus RTU or Ethernet w/ Modbus TCP module. The trip unit provides real time status and data for use by your supervisory system. The detailed Modbus register map including network configuration options is shown in Section 9 - Modbus register map and TD013198EN "Modbus Real Time Data Register Map".

The three on-board relays may also be used to provide discrete signals to an event recording or monitoring system. See Section 4.2 - Configurable relays for details on the configuration and behavior of the relays.

6.2 Event recording and waveform capture

The PXR trip unit will automatically record information surrounding events, alarms, and trips into a set of logs. For simple events, only the cause of the event and a timestamp (based on the trip unit's real time clock) are stored. More important events additionally store a snapshot of real time values of currents, voltages, and power. The most significant also store waveforms of current and voltage experienced during the event. The event records and waveforms are viewed and printed using the Power Xpert Protection Manager software.

Each log can store a certain quantity of events and is managed as a first-in first-out buffer (FIFO). As the information is stored for the most recent event, the information from the oldest event is eliminated.

Waveform captures can be initiated by the user as well as triggered by an alarm or trip event. PXR trip units will store up to the most recent 20 alarm and 24 trip event waveforms. These can be accessed via PXPM or PDMC via Modbus RTU/TCP (depending on the included trip unit communications protocols).

In addition to the user initiated, alarm, and trip event waveforms, the PXR 35 trip unit has the capability of capturing waveforms for "close call" captures. A "close call" capture occurs when the trip unit enters a high load alarm condition. It will capture 0.6 seconds of real current data, then 6 seconds of one cycle RMS current data, then followed by 60 seconds of 200ms RMS current data. The associated logged event includes duration of the event, highest current measured, and average current measured.

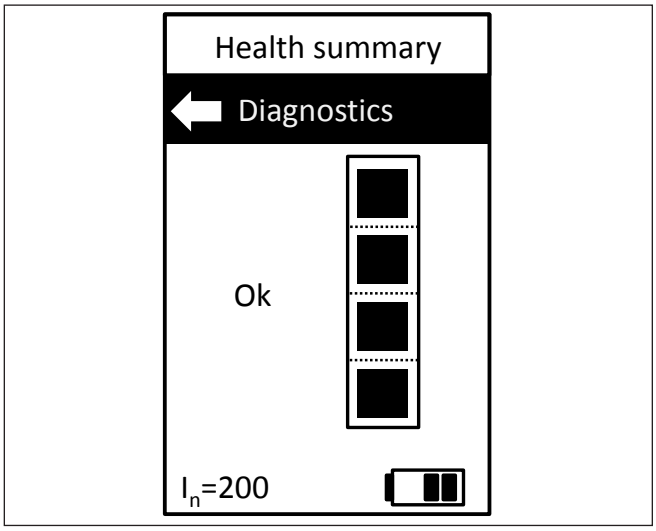
Table 13. Event log data.

What data is captured:	The most recent 200 events are stored. The data includes: Cause and timestamp	The most recent 10 trips and alarms, the data includes: Cause and timestamp; Current: IA, IB, IC, IN, IG; Voltages: VAB, VBC, VCA, VAN, VBN, VCN (PXR 25 and PXR 35 only); Power and demand: Watts, Vars, VA (PXR 25 and PXR 35 only); Power factor (PXR 25 and PXR 35 only); Line frequency; Breaker operations count; Trip unit internal temperature	The last alarm and trip also store waveforms: IA, IB, IC, IN, IG; VAB, VBC, VCA, VAN, VBN, VCN (PXR 25 and PXR 35 only)	
What triggers a capture:				
Event - power up - clock OK	•			
Event - power up - clock bad	•			
Event - setpoints download	•			
Event - enter test mode	•			
Event - exit test mode	•			
Event - test complete	•			
Event - enter maintenance mode	•			Indicator also illuminates.
Event - exit maintenance mode	•			Indicator un-illuminates.
Event - time change (if > 60 seconds)	•			Previous time is recorded.
Event - admin. password entered	•			
Event - admin password changed	•			
Event - user password entered	•			
Event - user password changed	•			
Event - wrong password entered	•			
Alarm - calibration	•			
Alarm - setpoints fault	•			
Alarm - battery low voltage	•			See Section 7.3 - Replacing the battery.
Alarm - low control voltage	•			See Section 4.5 - Auxiliary control power.
Alarm - RTC error	•			Real time clock error.
Alarm - NV memory error	•			Non-volatile memory error.
Alarm - watchdog restart	•			
Alarm - long delay pickup (test mode)		•		
Alarm - ground fault (test mode)		•		
Alarm - TA open	•			Trip actuator fault.
Alarm - operations count	•			
Alarm - long delay pickup	•			
Alarm - ground fault		•	•	Alarm waveforms include one cycle before the alarm.
Alarm - high load		•	•	
Alarm - neutral current		•	•	
Trip - over temperature		•		
Trip - test		•		
Trip - long delay		•	•	Trip waveforms include six cycles before the trip and four cycles after the trip.
Trip - short delay		•	•	
Trip - instantaneous		•	•	
Trip - ground		•	•	
Trip - maintenance mode		•	•	
Trip - neutral		•	•	
Trip - hardware instantaneous		•		
Trip - making current release trip		•		

6.3 Health monitor

The PXR trip unit includes an innovative algorithm to monitor the health of the circuit breaker. The parameters that are monitored by this algorithm include short circuits, overloads, operations, temperature, and run-time. These parameters can provide a host of individual insights while the combination of all of them provide an overall picture of the condition of the breaker that can be used for preventative maintenance and improved system reliability. This monitor is a tool and is not a replacement for established breaker maintenance or replacement practices.

Figure 22. Health summary.



The health monitor feature is comprised of indices which track the accumulated service performed by the circuit breaker. The indices start at zero and are increased up to 10,000 points based on the conditions which the breaker is operating within. The indices are retained in the frame module of the circuit breaker, changing the trip unit will not change the accumulated points.

The three indices tracked are:

- Interruption wear – Trip events where there is an overload or short-circuit of any level which opens the breaker. This index increases as a function of current and system voltage. The current is the RMS value used by the protection routines and is measured immediately before the breaker is opened. The voltage is the measured system voltage in PXR 25/35 or the system voltage setpoint in PXR 20.

  The PXR 35 can calculate the energy of a trip event, estimate the amount of interruption wear from this value, and factor this into the health points. Depending on the severity of the event, the number of health points applied to this index may vary across trips. The energy calculated is a function of voltage, current, and the duration of the trip event. This is used to augment the interruption wear index to improve its accuracy.

- Operations – This index increases when the breaker is mechanically operated (open-close or close-open) independent of the current level. The breaker might be operated due to an on-board secondary injection test, or a communication command to open/close, or an accessory or mechanical button release. Note that only operations when the trip unit is powered (self-powered, via USB or auxiliary power) will be counted.

- Environmental – Time and temperature contribute to this index. It increases more quickly when the temperature is higher. Note that temperature is measured within the trip unit. It increments by a small amount for every week that the breaker is in operation. If the breaker is not operated within a full year, additional points are added.

The health of the circuit breaker is summarized with a health percentage calculated from the indices, with new breakers reporting 100% health corresponding to 100% remaining service life.

Note: PXR trip units in NRX circuit breakers and the initial release in Magnum circuit breakers report 0% for a new breaker, corresponding to 100% remaining service life.

The three indices are combined, using the highest of the three indices when viewing on the display in the Diagnostics Summary screen (bar graph) or via a Modbus register (percentage), while in PXPM you may observe each independently.

The health percentage may be mapped to one of the configurable relays, closing the relay when the remaining percentage is between 10 to 50% (configurable).

Circuit breakers operating under normal conditions and reporting a health percentage below 100% have accumulated service that is within its normal useful life. Periodic maintenance should be performed throughout this period, and in the absence of any signs of abnormal or excess wear, damage or adverse environments, the circuit breaker is considered suitable for continued use. As the health percentage approaches zero, the circuit breaker should be carefully evaluated to determine its suitability for continued service. Based on accumulated service history, the circuit breaker may be nearing the end of its normal service life and a careful inspection should be performed to determine actions required. The best practice is to plan for the replacement of the circuit breaker, depending on its condition, maintenance performed, and your site-specific maintenance guidelines and standards.

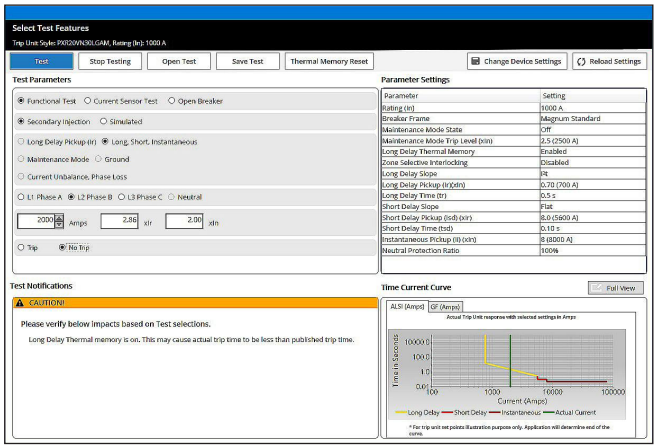
If maintenance and inspections are preformed which prolongs the serviceability of the circuit breaker, then you may reset the indicies. Remember that the event counters are not reset by the action of resetting any of the three indices. Accumulated service will then be tracked by incrementing the indices from the new values, and the health percentage will be recalculated.

6.4 Testing the circuit breaker and trip unit

The test functions do not require a separate test kit. A battery of testing possibilities is available via the front panel and using PXPM. Four built-in test modes are available for use:

- Functional test using secondary injection;
- Functional test using simulated current;
- Current sensor continuity test; and
- Open breaker functional test.

Figure 23. PXPM secondary injection test.



Secondary injection and simulated current can be configured for trip or no-trip. Complete testing of the trip unit system can be accomplished when a current sensor test is used in conjunction with secondary injection test.

Testing prior to startup should always be done with the circuit breaker either in a de-energized system, or in TEST or DISCONNECTED cassette position, or WITHDRAWN from cell. The system will prevent a test if more current than 5% of I_n is measured in the breaker phases.

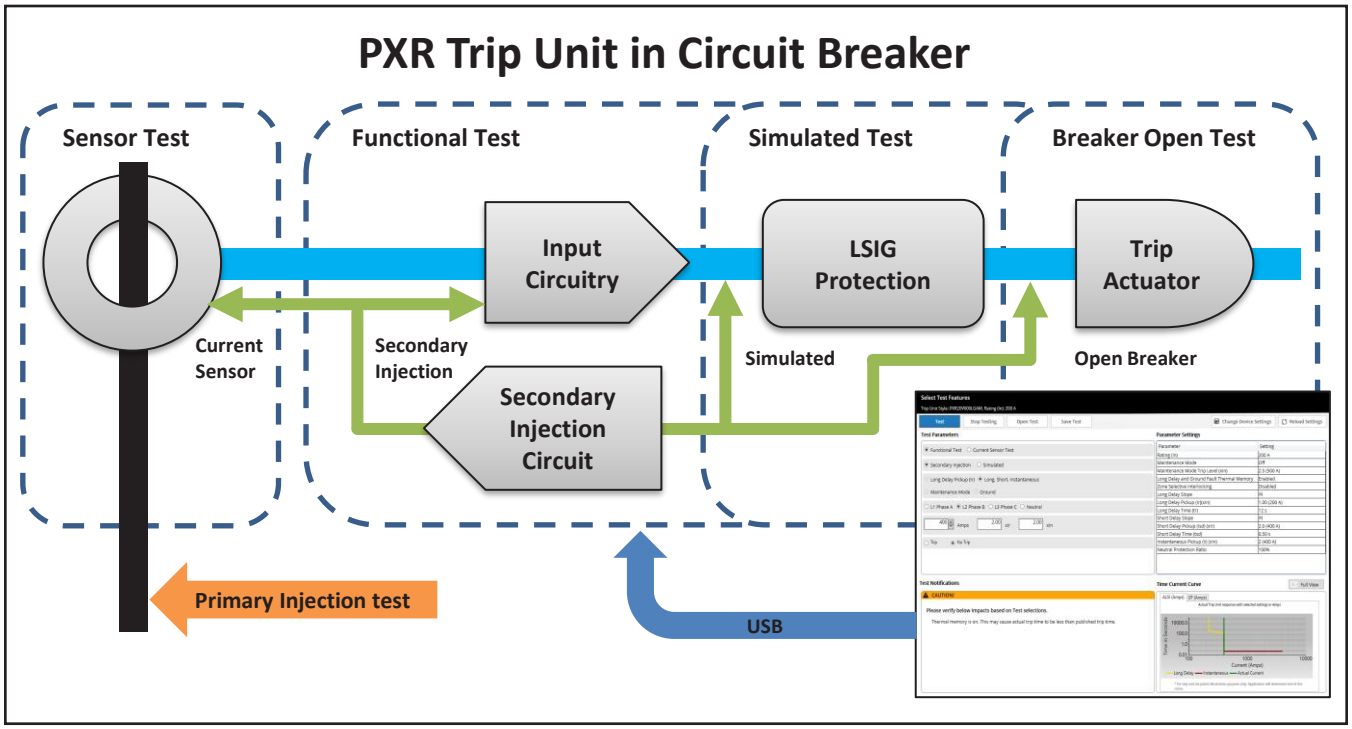
Note: Because time-current settings are based on desired system coordination and protection schemes, the protection settings, if altered during any test sequence, should be reset to their as-found conditions.

6.4.1 Secondary injection test

Secondary injection testing utilizes a separate circuit that injects a signal in parallel with the output of the current sensor. This signal is measured the same as a signal from the sensor would be measured. The protection circuitry and trip algorithm operate with no knowledge of the secondary injection circuit. The trip unit and circuit breaker respond, and configured operation is verified.

PXPM software initiates the functional testing using secondary injection. Note that the circuit breaker must be carrying less than 5% of I_n prior to the test, and the password must be entered to initiate the test. Testing for overload, short circuit, and instantaneous performance is accomplished by selecting the equivalent level of primary current to inject in either absolute amperes or as a multiplier of I_r. When a level is selected, remember to consider pickup and tolerances per the trip curves. You can test any phase, including neutral. Note that neutral testing is for internal neutral (4- or 8-pole breakers), external Rogowski sensors testing is limited, and external iron-core CT sensors are not supported. The trip unit's display can be used to observe the current being injected, and the time until trip.

Figure 24. On-board test features.



For detailed information, please see the Quick Start Guide supplied with the Power Xpert Protection Manager (PXPM) software.

6.5 Testing for ground fault trip units – primary injection

Many local or country building codes require that any ground fault protection system be performance tested when first installed. Conduct tests in accordance with the approved instructions provided with the equipment. Make a written record of this test and make the results available to the authority having inspection jurisdiction.

For basic performance requirements stipulated by the NEC and UL Standard No. 1053, the test requires that certain minimum instructions must accompany each ground fault protection system. These statements, plus a copy of the record forms, are included in Section 10 - Records and forms.

The interconnected system must be evaluated only by qualified personnel and in accordance with the equipment assembler's detailed instructions.

⚠ WARNING

Electrical shock or burn injury can occur when working on power systems. Always turn off the main power that is supplying the circuit breaker before conducting tests. Test out of the cell, if possible.

To avoid improper operations following apparently correct simulated test operations, the polarity of the neutral sensor connections (if used) must agree with the equipment assembler's detailed instructions. Where a question exists, consult the specifying engineer and/or equipment assembler.

Verify the grounding points of the system using high voltage testers and resistance bridges to ensure that ground paths do not exist that could bypass the sensors.

Use a low voltage (0 to 24 volt), high-current, AC source to apply a test current of 125% of the ground pickup setting through one phase of the circuit breaker. This should cause the circuit breaker to trip in less than one second and operate the alarm indicator, if one is supplied. Reset the circuit breaker and the alarm indicator. Repeat the test on the other two phases.

Apply the same current as described above through one phase of the circuit breaker, returning through the neutral sensor if a neutral sensor is used. The breaker should not trip and the alarm indicator, if one is supplied, should not operate. Repeat the test on the other two phases.

See IL012125EN for additional details on performance testing for ground fault circuit breakers and AP013002EN for field testing instructions for ground fault and maintenance mode protection systems.

6.6 Testing within the switchgear

When the circuit breaker is integrated within the switchgear, the unique features of the PXR trip unit should be considered.

- When testing insulation, the voltage sensing circuit (VDB) will be seen by a megohmmeter as a very high resistance (> 6 megaohms) between line and neutral. If your testing requires eliminating this effect, then test with the breaker in the racked-out position, there is no voltage disconnect plug.
- If you are testing with primary current, the trip unit will self-power only from the sensors within the breaker frame. It will not self-power from external neutral or ground fault sensors.
- With primary injection, your protection and enhanced protection settings may trip the breaker. For example, ground faults or phase loss trips may be generated when current is applied on individual phases.
- When testing, a breaker may trip outside of its own settings because of ZSI signals. See Section 3.10 - Zone selective interlocking (ZSI) for details.
- Thermal memory, when enabled, will affect trip times on repeated testing of the same circuit breaker. See Sections 3.2.1 - Long delay thermal memory and 3.6.3 - Ground fault thermal memory which explain this behavior.
- The VDB (voltage sensing module) located within the frame rating module does not need to be disconnected for high potential (hi-pot) testing.

7. Maintaining the PXR trip unit

7.1 Troubleshooting

Table 14. Troubleshooting matrix.

Symptom	Probable cause	Possible solutions	Reference
The trip unit status LED is not blinking.	No auxiliary (control) power or sufficient current is not flowing through the circuit breaker sensors to power the trip unit.	Connect +24 Vdc and ground auxiliary power to circuit breaker terminals C14 and C15 and observe the status LED.	See Section 2.1 - Status indicator and 4.5 - Auxiliary control power.
Breaker trips on ground fault.	There actually is a ground fault.	Find the location of the fault.	
	On 3-pole, 4-wire residual systems, the neutral current sensor may not match the trip unit setting or be properly connected, or an incorrect sensor is being used.	Check the ground fault setting is residual or source/zero ground. Check that the connections from the neutral current sensor to the breaker are not reversed. Check that the sensor type matches the trip unit selection (iron core vs Rogowski).	See Section 3.6 - Ground fault settings and Section 4.1 - Neutral sensors.
	High inrush phase currents may cause fictitious ground pickup momentarily.	If the zone interlocking function is employed, connect a jumper from Z_{out} to Z_{in} to provide some time delay.	See Section 3.6.4 - Ground fault zone selective interlocking.
	The trip unit is malfunctioning.	Replace the trip unit.	See Section 7.2 - Replacing the trip unit.
Breaker trips too rapidly on ground fault or short delay (zone selective interlocking not used).	ZSI function is ON.	Check the ZSI setting in the Setting menu is OFF.	See Section 3.10 - Zone selective interlocking (ZSI).
	The trip unit settings are not correct. Is I^2t slope or flat selected?	Change the ground fault or short delay settings.	See Section 3.3 - Short circuit settings and 3.6 - Ground fault settings.
	The trip unit is malfunctioning.	Replace the trip unit.	
Neutral readings incorrect or tripping unexpectedly.	Neutral sensor connected is not matched to selected option or an incorrect sensor is being used.	Make sure that neutral sensor selection is set for Rogowski or iron-core CT to match what is connected.	See Section 4.1 Neutral sensors.
The breaker trips too rapidly on long delay.	Long time memory is selected.	Disable the long time memory for long delay times setting.	See Section 3.2.1 - Long delay thermal memory.
	The trip unit settings are not correct.	Change the long delay settings.	See Section 3.2 - Overload/long delay settings.
Breaker trips when control power is applied to the trip unit.	The secondary of the neutral sensor is grounded.	Remove ground connection from the neutral sensor secondary.	
The primary injection current source is not putting out the correct current.	Primary injection testing pickup and trip times are not correct.	Use an oscilloscope with a current probe to accurately verify current values, times, and that there are no inrush spikes of current observed.	
	Single phase testing.	When testing single phase, the current can "bleed" into other un-energized phases and reduce the value of current in the testing phase. You can run current through 2 or 3 poles in series.	
	It is possible, when using the high current "pulse" testing technique, to create a long time memory trip as a result of the cumulative action of the short duration currents.	Disable the long time memory feature for long delay time setting.	See Section 3.2 - Overload/long delay settings and 3.2.1 - Long delay thermal memory.
	Precise pickup levels are difficult to control and repeat for primary injection testers as high current levels.	Use functional current testing (remote) via USB/PXPM.	See Section 6.4.1 - Secondary injection test.
Cause of trip LEDs are flashing and breaker is closed.	The trip unit was not reset from a previous event.	Depress the Reset push button to clear the LED or test flashing.	See Section 2.7 - Reset button.
	The battery voltage is too low to reset the latch circuit on the cause of trip LEDs.	Replace the battery then press the Reset push button.	See Section 7.3 - Replacing the battery.
The LCD display is not energized.	No auxiliary power when the current through the breaker is less than the minimum current requirement for the LCD display.	Apply auxiliary power.	See Section 4.5 - Auxiliary control power and 4.6 - Secondary wiring terminals.
Setpoints fault alarm	Memory failure	Replace the trip unit and/or frame module.	
TA Open alarm and status LED is blinking green.	The trip actuator open alarm may appear when control power is applied. Note that as long as the status LED is blinking green the trip actuator is functioning correctly and protection features are active.	Reset the trip unit.	2.7 - Reset button.
TA Open alarm and status LED is blinking red.	The trip unit senses that the trip actuator is not connected or is malfunctioning.	Contact Eaton Electrical Services & Systems.	

Table 13. Troubleshooting matrix (continued).

Symptom	Probable cause	Possible solutions	Reference
Battery low voltage alarm.	The battery tab is still installed.	Pull the battery tab to put it into service.	See Section 7.3 - Replacing the battery.
	The battery voltage is low.	Replace the battery.	
Calibration alarm.	Calibration error.	Replace trip unit. If the issue persists, contact Eaton Electrical Services & Systems.	

7.2 Replacing the trip unit

The PXR trip unit is designed as a field replaceable unit. The breaker must be in the open position to replace the trip unit. There is an interlock that will cause the breaker to open if you attempt to remove the trip unit while the breaker is closed.

Before replacing the trip unit, record the position of the setting dials for the PXR 20. It may be convenient to use PXPM to save or print all the settings for validation with the new trip unit after replacement.

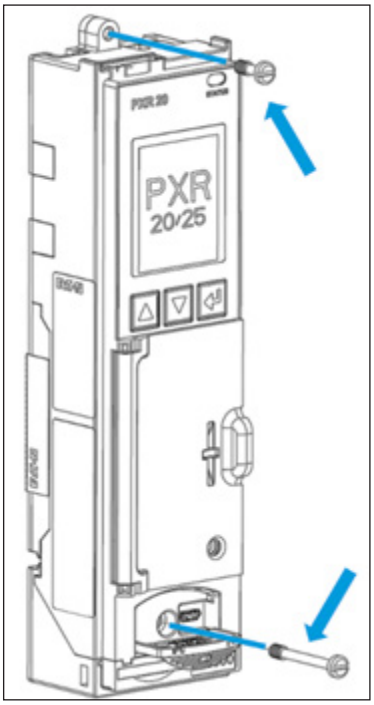
The frame module retains the circuit breaker's configuration, the primary and enhanced protection settings, relay configuration, and Modbus configuration. They will be restored into the new trip unit except for the rotary switch settings of the PXR 20 which must be set manually.

Remove the circuit breaker from service.

1. Remove all control power from the breaker and remove any connected USB cable. The trip unit must not be powered during removal or installation.
2. Remove the circuit breaker front cover. Then remove the trip unit from the circuit breaker's frame rating module using the following steps:
   - Break the factory seal.
   - Then use a screwdriver to loosen the top and bottom screws.

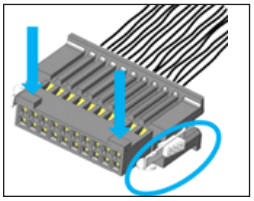
   Note that the bottom screw is located left of the USB port and covered by the door.

Figure 25. Removing the ETU.

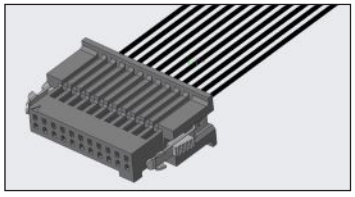


3. Pull the unit outward to disengage the molded, 40-pin connector from the frame module printed circuit board and the small 10-pin connector from the voltage divider board (PXR 25/35). The 18-pin wiring connector at the top of the trip unit is keyed (arrows in Figure 25) and has locking tabs on the sides (circled in the Figure 25). Squeeze the locking tabs inward and gently disengage the connector.

Figure 26. Installing the ETU wire harness.



PXR 20/25/35



PXR 35 high density connector

4. To install the new trip unit, first connect the single wiring harness (PXR 20/25) or two harnesses (PXR 35) at the top of the trip unit. Both are keyed connectors and you should hear a slight click as the locking tabs engage. Then, to mount the trip unit, align it with the guides of the frame module. Keep it parallel with the receiving, molded housing.
5. Use caution to ensure that the interlock that prevents the breaker from closing if the trip unit is removed is engaged with the boss on the side of the housing and is being depressed. Press the trip unit into the circuit breaker until the trip unit seats firmly into the receiving housing. Secure it in place with the two screws supplied.
6. Replace the circuit breaker front cover.
7. Set the rotary settings switches if using a PXR 20 and validate your settings using your records or PXPM.
8. Restore control power if available. A secondary injection test is recommended before placing the circuit breaker back into service. See Section 6.4.1 - Secondary injection test.
9. Restore the circuit breaker to service.

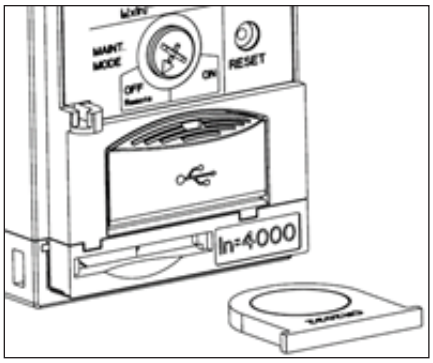
7.3 Replacing the battery

The battery is provided to maintain the LED indication of the cause of the trip and the real time clock. A battery icon at the bottom of the display indicates remaining battery life. The battery plays no part in the protection function of the trip system. The battery can be replaced at any time, including while the circuit breaker is in-service, without affecting the operation of the circuit breaker or its protection functions.

On the initial installation of the circuit breaker, remove and discard the insulating tab and then re-insert the battery tray. Press the reset push button to extinguish any cause of trip LEDs which may illuminate when the battery is inserted. Check to ensure the battery icon is showing full remaining life in the lower right of the display.

The 3 V lithium battery (CR 2032) is easily removed and replaced; pull to remove the battery tray, remove the old battery from the holder, replace with new one (observe proper polarity as marked on the tray), and then re-insert the battery tray into the slot on the frame module. Accidentally installing the battery in the reverse direction will not harm the battery nor the trip unit but will defeat the function of the battery. Remember to press the reset button and check the battery icon in the display.

Figure 27. Replacing the battery.



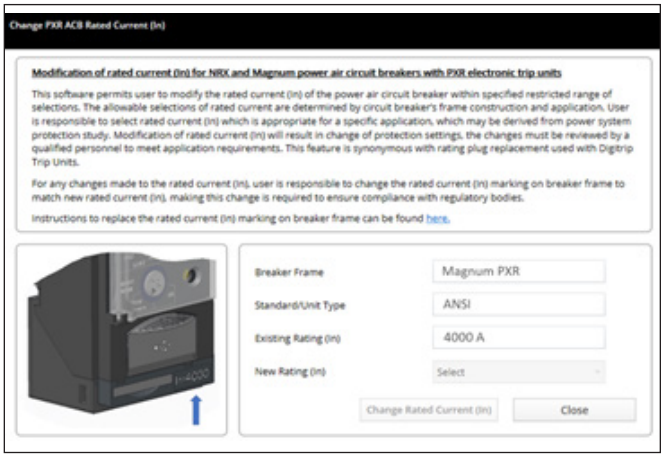
7.4 Changing the breaker rating

Magnum PXR and PD-SB circuit breakers are pre-configured at the factory for their I_n rating. The I_n rating is printed in the lower right corner of the frame module. A licensed feature of the PXPM software allows you to modify the rating within an allowed range based on the breaker construction and certification. You can purchase this online by navigating to www.eaton.com/PXPM and clicking Digital continuous current rating change. The labels are ordered using catalog number PXRIN-LABEL-KIT.

You are responsible to ensure that any rating change is appropriate for the application and settings specified by the engineer responsible for the installation. Protection settings are based on the I_n rating and affect the time-current response of the breaker. Modification of the rating requires re-evaluation of the circuit breaker's protection settings and the system coordination to ensure proper application.

To change the current rating (I_n) on the circuit breaker using PXPM, start the application and click on the Device Settings icon. Then click the Change Rated Current (I_n) icon and you will see the screen shown in Figure 27. Select an available rating from the drop down. After you apply the new rating, remember to apply the correct label from the kit.

Figure 28. Frame Rating Modification screen.



8. Documentation cross reference

Table 14 lists important documentation related to the Magnum PXR and PD-SB circuit breakers, and the available accessories. Please visit www.eaton.com/magnumpxr or www.eaton.com/powerdefensesb to access the most recent information about the family of circuit breakers, trip units and accessories.

Table 15. Documentation references.

Documentation number	Document name
IL0131154EN	Neutral current sensors Magnum PXR and Power Defense SB ACB trip units
IL0131158EN	Eaton Corporation EULA for Power Xpert Release (PXR 20/25/35)
IL0131132EN	Ethernet communication adapter module with Modbus TCP/IP
IL120009EN	Profibus (DP) communication adapter module
TD013172EN	Time current curves: Magnum PXR and Power Defense SB PXR 20/25/35 electronic trip units
MN013016EN	Magnum PXR and Power Defense SB low voltage power circuit breakers user manual (includes connection diagrams)
PA013006EN	Magnum PXR and Power Defense SB low voltage power circuit breakers Technical product guide
AP013002EN	Field testing instructions and application guide for ground fault and Maintenance Mode protection systems utilizing Eaton Magnum PXR and Power Defense SB circuit breakers
TD013168EN	Power Xpert Protection Manager Quick Start Guide
TD013198EN	Modbus Real Time Data Register Map
TD150066EN	IEC 61850 GOOSE Functionality for Magnum PXR 35

9. Modbus register map

A Modbus RTU communication port is integrated for certain trip unit styles. The trip unit is a server device on the network and will reply to messages from the client (master). The trip unit can support a maximum of 122 contiguous registers (244 data bytes) in a single Modbus transaction.

Table 16. Modbus function codes.

Supported function codes	Description
2	Read input status
3 or 4	Read holding or input register (interchangeable)
6	Write single register
16	Write multiple registers (contiguous block)
8	Diagnostics

Remember that the registers are numbered starting at 1 but addresses start at 0. In the following sections, register numbers and addresses (hex) are shown along with their definition and integer scaling if appropriate. Hex numbers are shown using the 0x0000 nomenclature.

9.1 Modbus configuration

9.1.1 Network parameters

Modbus connection parameters can be viewed and set from LCD display, via PXPM and Modbus communication. The settings are stored beginning at register number 4000. They may be read by using function code 3 or 4 and written using function code 6 (PXR 20/25/35) and 16 (PXR 35 only). Note that if you change these settings via Modbus, they may result in a loss of communication unless the corresponding change is done by the client.

Table 17. Modbus network parameters.

Function code	Register number	Address (hex)	Definition	Data range
3,4,6 (and 16 on PXR 35 only)	4000	0x0f9F	Address	1 to 247 (factory default = 1)
3,4,6 (and 16 on PXR 35 only)	4001	0x0fa0	Baud rate	0 = 9600 bits/s (factory default) 1 = 19200 bits/s 2 = 38400 bits/s 3 = 57600 bits/s 4 = 115200 bits/s
3,4,6 (and 16 on PXR 35 only)	4002	0x0fa1	Parity	0 = None 1 = Odd 2 = Even (factory default)
3,4,6 (and 16 on PXR 35 only)	4003	0x0fa2	Stop bit	0 = 1 bit (factory default) 1 = 2 bits

9.1.2 Exception codes

When there is an error in the request, an exception code will be returned for the following conditions.

Table 18. Modbus exception codes.

Returned value	Condition
0x01	Function or sub-function code in the query is not supported.
0x02	Requested data register/bit address is illegal.
0x03	Data in the query is illegal.
0x04	Trip unit does not support the query function.
0x05	Trip unit received the message and is processing (ACK).
0x06	Trip unit cannot perform the current request at this time (BUSY).
0x07	Trip unit cannot perform the requested action (NAK).
0x84	A partial register is used in the query.
0x87	Requested event entry does not exist.

9.1.3 Modbus data configuration

The following registers can be written to configure the trip unit response. They may also be read for verification.

Table 19. Modbus data configuration.

Function code	Register number	Address (hex)	Register definition
3,4,6	2001	0x07D0	**Invalid object access configuration:** If = Non-zero (default), any attempt to access a group of data objects that contain an invalid object will result in the 0x02 illegal data code. If = 0, the trip unit will respond with valid data in the valid objects and 0x0000 for invalid objects
3,4,6	2002	0x07D1	**Floating point data word order configuration:** If = Non-zero (default), the floating-point low order word is first. If = 0, the floating-point high order word is first.
3,4,6	2003	0x07D2	**32-bit fixed-point data word order configuration:** If = Non-zero (default), the fixed-point low order word is first. If = 0, the fixed-point high order word is first.

9.1.4 Diagnostics

The trip unit supports internal Modbus diagnostics to monitor port communication using function code 08 according to Table 19.

Table 20. Modbus diagnostics.

Sub-function code	Action
0	Echo query
1	Restart communication
4	Force listen
10	Clear counters
11	Modbus UART bus message count
12	Modbus UART CRC error count
13	Exception count
14	Slave message count
15	Slave no response count
16	Slave NAK count
17	PXR20/25: Slave busy count PXR35: Unused
18	Modbus UART over run error count
20	Clear Modbus UART counters
21	PXR 20/25: Unused PXR 35: Checksum error count
23	Modbus UART framing error count
24	PXR 20/25: Modbus UART noise error count PXR 35: Unused
25	Modbus UART parity error count
26	MCU1 firmware version
27	MCU1 firmware reversion
28	MCU1 firmware build
29	MCU2 firmware version
30	MCU2 firmware reversion
31	MCU2 firmware build
32	PXR 20/25: USB version PXR 35: MCU3 firmware version
33	PXR 20/25: USB reversion PXR 35: Unused
34	Reset block of registers

9.2 Real-time breaker status

Input status bits 1001 through 1032 are available using function code 2. The status definitions are defined in Table 20. The first 16 bits are the actual status state while the last 16 bits indicate if the corresponding status state is supported by the trip unit and valid.

Table 21. Modbus breaker status.

Register number	Address hex	Input status descriptions
1001	0x03E8	Breaker is in the closed position.
1002	0x03E9	Un-acknowledged trip condition.
1003	0x03EA	Un-acknowledged alarm.
1005	0x03EC	Maintenance mode is active.
1006	0x03ED	Test mode is active.
1009	0x03F0	Phase rotation is active
1010	0x03F1	Long delay pickup is active.
1011	0x03F2	Zone interlock is active.
1013	0x03F4	Ground is configured source ground.
1017	0x03F8	Breaker is in the closed position is valid.
1018	0x03F9	Un-acknowledged trip condition is valid.
1019	0x03FA	Un-acknowledged alarm is valid.
1021	0x03FC	Maintenance mode is active is valid.
1022	0x03FD	Test mode is active is valid.
1025	0x0400	Phase rotation is valid
1026	0x0401	Long delay pickup is active is valid.
1027	0x0402	Zone interlock is active is valid.
1029	0x0404	Ground is source ground is valid.

9.3 Real-time data registers

The data changing in real time, such as current, voltage, power, etc. is shown in the Table 21. Real-time data can be read either in 32-bit IEEE floating point or in 32-bit integer format. For integer data, the value is multiplied by a scale factor shown in the last column. Data objects are two registers (4 bytes) or four registers (8 bytes) in length. Since these objects have the capability to change in real-time, a complete data object must be read in a single transaction to avoid data tearing. See document TD013198EN for detailed register information for real time data.

9.3.1 Status within the real-time data register

Register Number 6145 and 6146 provide status information about the breaker according to Table 21.

Table 22. Modbus real-time data status decoding.

Primary status code register number 6145 (high)	Definition	Secondary status code register number 6145 (low)	Definition
0x01	Open	0x03	Test mode
0x02	Closed	0x07	Powered-up
0x03	Tripped	0x08	Alarm
0x04	Alarmed		
0x0D	Picked-up		

Cause of status register number 6146	Definition
0x0000	Unknown
0x0001	Normal
0x0002	Real Power
0x0003	Instantaneous
0x0004	Reactive Power
0x0005	Instant. neutral over current
0x0006	Apparent Power
0x0007	Under Power Factor
0x0008	Reverse Reactive Power
0x0009	Real Power Demand
0x0010	Under Frequency
0x0011	Current Unbalance
0x0012	Voltage Unbalance
0x0013	Apparent Power Factor
0x0014	Neutral
0x0016	Current THD
0x0017	Voltage THD
0x0018	Display Over Temperature
0x0019	Time to Trip
0x000A	Apparent Power Demand
0x000B	Over Voltage
0x000C	Under Voltage
0x000D	Goose Disconnect
0x000F	Over Frequency
0x000E	Aux-power under power
0x001A	Power Demand
0x001B	VA Demand
0x001C	Thermal Capacity
0x001D	Health Bus Communication Error
0x001E	THD
0x001F	Operations count
0x0020	Password Locked Out
0x0021	Control via communication
0x0022	Display MCU Internal Error
0x0023	Breaker Operations Count
0x0025	Coil Supervision
0x0026	Wrong Neutral Sensor Selected
0x0029	Battery low alarm
0x003D	Long delay
0x003E	Short delay
0x0041	Reverse Power
0x0044	Reverse sequence
0x0045	phase current loss
0x0049	High load alarm 1
0x0071	Calibration
0x0072	Breaker Config Error
0x004B	Making current release
0x004C	Fixed hardware instantaneous
0x004D	Setpoints error
0x004E	Over temperature
0x009B	Frame board version Fault
0x0050	Long delay neutral over current
0x0054	Ground fault
0x0055	Earth fault
0x0071	Calibration
0x0088	Real-time clock
0x0099	MM mode
0x009A	Breaker mechanism fault
0x009B	Frame board version Fault
0x009E	Short delay neutral over current
0x009F	Thermal memory alarm
0x00A0	Local test from LCD
0x00A1	High load alarm 2
0x00A2	Ground fault pre-alarm
0x00A3	High Temperature Alarm
0x00A6	Setpoint Mismatch between cassette/trip unit
0x00A7	Cassette UID Mismatch between cassette/ trip unit
0x00A8	GOOSE Configured but Ethernet is not connected
0x07FC	Digital Bypass
0x07FD	Non-volatile memory error
0x07FE	Watchdog Fault

9.3.2 Two and four register data formats

Format of two register (4 bytes) or four registers (8 bytes) data.

Table 23. Modbus multiple register data format.

2 register format Integer	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Register number	Low 16 bits																
Register number + 1	High 16 bits																
2 register format IEEE float	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Register number	Fraction																
Register number + 1	Sign	Exponent								Fraction							
4 register format Integer	**16**	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Register number	Byte 1 of mantissa								Byte 0 of mantissa								
Register number + 1	Byte 3 of mantissa								Byte 2 of mantissa								
Register number + 2	Byte 5 of mantissa								Byte 3 of mantissa								
Register number + 3	Engineering units								Mantissa multiplier								
Calculation:	Energy = $2^{\text{mantissa multiplier}}$ * 48-bit mantissa * $10^{\text{engineering units}}$																

9.3.3 Mapped register configuration

A block of registers can be established in trip unit to remap up to 100 data objects.

The remapping assignments are stored beginning at 1001; only the first data object address is used. For example, although data object IA occupies 0x1202 and 0x1203, only 0x1202 is loaded into the assignment register. It is important to account for data object length. For PXR 20/25, function code 16 is used to load the object assignments in a single write and all object assignments must be valid. PXR 35 allows function code 6 to write single object assignments or function code 16 to write one or more object assignments.

Data from the assigned objects is read from registers starting at 1201 and continuing for each object assigned. The number of objects and their placement order is dependent on the configuration of the assignment registers. The starting address must be aligned with the start of an object and the number of registers to obtain must align with the end of an object.

Table 24. Modbus mapped registers.

Func. code	Register number	Address (hex)	Length	R/W	Definition
3,4, 6,16	1001	0x03E8	100	R/W	Block of 100 registers for the mapping assignments.
3,4	1201	0x04B0	Maximum 200 (PXR 20/25) 800 (PXR 35)	R	Data that has been remapped (Actual length available is dependent on number of assigned objects and length of the individual objects.).

9.4 Setpoint registers

The trip unit's setpoints are organized into groups. Group 0 is the system group which provides information about the circuit breaker and trip unit. Group 1 is the protection group and Group 5 is the enhanced protection group. These three groups are common to the PXR 20, 25, and 35. The PXR 35 also has Group 2 for Modbus communications, Group 4 for frequency protection, Group 6 for power protection, group 7 for sync check and group 9 for assorted other setpoints.

Using groups, the trip units provide a method to read and change setpoints over Modbus. Certain setpoints are read only or may be restricted in enumerations or ranges based on trip unit. Each group can be considered an array of information which can be obtained through Modbus register access.

Register 3000 is used to write the 4-digit password used in PXR 20/25 and PXR 35 version 1.0.0, while registers 2999 and 3000 are used to write the 6-digit password used in PXR 35 version 2.0.0 and later. The password must be written before initiating a setpoint write (function code 6 or 16). If a correct password is written, you must write any new setpoints within ten minutes before the password must be entered again. If an incorrect password is written, any attempt to write a new setpoint will be rejected. If three incorrect passwords are written in a row, no additional password attempts will be accepted for 10 minutes. This 10-minute wait period will be repeated for all subsequent incorrect passwords. Note that any read of registers 2999 and 3000 will return a random value, not the password.

Register 3001 is used to select the specific setpoint group in which you are interested. The high byte selects the group number. The low byte is always 0xFF for the PXR 25, so writing 0x01FF selects group 1 for reading/writing. For PXR 35 setting the low byte to 0xFF selects the active protection set, whichever is currently being used, while 0-3 selects a specific set A-D respectively. In this case writing 0x01FF selects group 1 with the active set, while writing 0x0102 selects group 1, set C. Note that this setpoint set selection only applies to reading/writing setpoints via Modbus, it does not change the active protection set. The last value written to register 3001 applies until a new value is written and can be read at any time.

Each table of setpoint register tables include information on the register number, hex address, setpoint name, read/write support, and description. Addresses not listed do not have information available in them. The description should be

referenced to interpret the format of each register, and only the values in the description field are valid. Fractional values are entered with 10x or 100x multipliers. In these cases the actual value to be entered is shown in parentheses.

All registers can be read with function codes 3 or 4. For registers that can be written, PXR 20/25 only supports function code 6, write single register for setpoint writes. PXR 35 supports function codes 6 as well as function code 16, write multiple registers. With function code 16 it is possible to write the password, group/set, and setpoint values all with a single command.

9.4.1 Group 0 – system group

Table 25. Modbus system group data.

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition
3,4	3002	0x0BB9	Rating information	R	Values: 100, 200, 250, 300, 320, 400, 500, 600, 630, 800, 1000, 1200, 1250, 1600, 2000, 2500, 3000, 3200, 3600, 3800, 4000, 5000, 6000, 6300 (not all ratings are available for all circuit breaker frames)
3,4	3003	0x0BBA	Breaker frame	R	0 = NRX NF (PXR 20/25) 1 = NRX RF (PXR 20/25) 2 = Magnum standard 3 = Magnum narrow 4 = Magnum double wide 5 = Magnum double narrow
3,4	3004	0x0BBB	Style	R	0 = PXR20VN30LNNC 1 = PXR20VN30LNNM 2 = PXR20VN30LGNC 3 = PXR20VN30LGNM 4 = PXR20VN30LGAC 5 = PXR20VN30LGAM 6 = PXR25VN30LNNM 7 = PXR25VN30LGNM 8 = PXR25VN30LNAM 9 = PXR25VN30LGAM 10 = PXR20VN30LNAM 11 = PXR20VN30LNAC 20 = PXR35VN30LDAE 21 = PXR35VN30LDAB
	3005	0x0BBC	Trip unit style 2	R	The following bits will be selectable for the Digitization feature (PXR 35 only): bit 0 = 0 non-ground fault style bit 0 = 1 ground fault style bit 1 = 0 ground fault 1200A limit, bit 1 = 1 ground fault will use rating I_n without 1200A limit.
3,4,6	3006	0x0BBD	Maintenance mode: Enable	See note	The following bits will be selectable: bit 8 = MM current state (read only) bit 7 = MM rotary switch on/off (read only) bit 6 = MM rotary switch position valid or not (read only) bit 1 = MM remote control, 0=off, 1=on (read only) bit 0 = MM local control (read only)
3,4,6	3007	0x0BBE	Maintenance mode: Level	R/W	0 = 1.5*I_n 1 = 2.5*I_n 2 = 4.0*I_n 3 = 6.0*I_n 4 = 8.0*I_n 5 = 10.0*I_n
3,4	3008	0x0BBF	Frequency	R	PXR 20/25: 50 = 50 Hz 60 = 60 Hz 400 = 400 Hz PXR 35: 60 Hz only
3,4,6	3009	0x0BC0	Power feed	R/W	0 = forward 1 = reverse
3,4,6	3010	0x0BC1	Sign convention	R/W	0 = IEC 1 = IEEE 2 = IEEE alt
3,4,6	3011	0x0BC2	Demand window	R/W	0 = fixed 1 = sliding
3,4,6	3012	0x0BC3	Demand interval	R/W	0 = off 5 to 30 (minutes), steps of 1

Table 24. Modbus system group data, (continued).

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition
3,4,6	3013	0x0BC4	Language	R/W	0 = English 1 = Chinese (PXR 20/25) Spanish (PXR 35) 2 = German (PXR 20/25) 3 = Spanish (PXR 20/25) 4 = Portuguese (PXR 20/25) 5 = French (PXR 20/25)
3,4	3014	0x0BC5	UNUSED	R	UNUSED
3,4	3015	0x0BC6	UNUSED	R	UNUSED
3,4	3016	0x0BC7	UNUSED	R	UNUSED
3,4	3017	0x0BC8	UNUSED	R	UNUSED
3,4	3018	0x0BC9	UNUSED	R	UNUSED
3,4	3019	0x0BCA	UNUSED	R	UNUSED
3,4	3020	0x0BCB	UNUSED	R	UNUSED
3,4,6	3021	0x0BCC	Breaker health alarm level	R/W	0 - 50, step size 1 (PXR 20/25) 0 - 100, step size 1 (PXR 35)
3,4,6	3022	0x0BCD	Voltage level of breaker system	R/W	90-690, step size 1
3,4,6	3023	0x0BCE	Neutral sensor input	R/W	0 = Rogowski (required for PXR 35 in 4 or 8 pole circuit breakers) 1 = CT
3,4	3024	0x0BCF	Source ground sensor input	R	0 = Rogowski 21 = CT 10 : 1 51 = CT 25 : 1 101 = CT 50 : 1 201 = CT 100 : 1 401 = CT 200 : 1 801 = CT 400 : 1 1601 = CT 800 : 1 2001 = CT 1000 : 1 2401 = CT 1200 : 1
The following Group 0 setpoint registers are for PXR 35 only.					
3,4,6	3025	0x0BD0	Setpoint set selection method	R/W	0 = Remote Switch 1 = Communication
3,4	3026	0x0BD1	Active setpoint set	R	PXR 35 only: 0 = set A 1 = set B 2 = set C 3 = set D
3,4,6	3027	0x0BD2	UNUSED	R	UNUSED
3,4,6	3028	0x0BD3	UNUSED	R	UNUSED
3,4,6	3029	0x0BD4	Number of trip pre-trigger cycles for waveform capture	R/W	8 - 28, step size = 1
3,4,6	3030	0x0BD5	Number of alarm pre-trigger cycles for waveform capture	R/W	8 - 28, step size = 1
3,4,6	3031	0x0BD6	Number of extended-capture pre-trigger cycles for wf capture	R/W	8 - 28, step size = 1
3,4,6	3032	0x0BD7	IEC 61850 Communications configuration	R/W	0 = Disabled (no IEC 61850 comms) 1 = GOOSE
3,4,6	3033	0x0BD8	Time between log intervals in minutes	R/W	0 = Off 5 - 30, step size = 1
3,4,6	3034	0x0BD9	Voltage transformer configuration	R/W	0 = Wye 1 = Delta
3,4,6	3035	0x0BDA	Health bus enable	R/W	0 = Health bus disabled 1 = Health bus enabled
3,4	3036	0x0BDB	UNUSED	R	UNUSED

9.4.2 Group 1 – protection group

Table 26. Modbus protection group data.

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition	
3,4	3002	0x0BB9	Rating information	R	Values: 100, 200, 250, 300, 320, 400, 500, 600, 630, 800, 1000, 1200, 1250, 1600, 2000, 2500, 3000, 3200, 3600, 3800, 4000, 5000, 6000, 6300 (not all ratings are available for all circuit breaker frames)	
3,4	3003	0x0BBA	Breaker frame	R	0 = NRX NF (PXR 20/25) 1 = NRX RF (PXR 20/25) 2 = Magnum standard 3 = Magnum narrow 4 = Magnum double standard 5 = Magnum double narrow	
3,4	3004	0x0BBB	Style	R	0 = PXR20VN30LNNC 1 = PXR20VN30LNNM 2 = PXR20VN30LGNC 3 = PXR20VN30LGNM 4 = PXR20VN30LGAC 5 = PXR20VN30LGAM 6 = PXR25VN30LNNM 7 = PXR25VN30LGNM 8 = PXR25VN30LNAM 9 = PXR25VN30LGAM 10 = PXR20VN30LNAM 11 = PXR20VN30LNAC 20 = PXR35VN30LDAE 21 = PXR35VN30LDAB	
	3005	0x0BBC	Trip unit style 2	R	The following bits will be selectable for the Digitization feature (PXR 35 only): bit 0 = 0 non-ground fault style bit 0 = 1 ground fault style bit 1 = 0 ground fault 1200A limit, bit 1 = 1 ground fault will use rating I_n without 1200A limit.	
3,4,6	3006	0x0BBD	LD thermal memory	R/W	0 = disable 1 = enable	
3,4,6	3007	0x0BBE	ZSI	R/W	0 = disable 1 = enable hardwired only 2 = enable GOOSE only (PXR 35) 3 = enable GOOSE and hardwired (PXR 35)	
3,4,6	3008	0x0BBF	LD slope	See note	PXR 25 (read only) 0 = $I_{0.5}t$ 1 = It 2 = I^2t 3 = I_4t 4 = IEEE Moderately Inverse (PXR 35) 5 = IEEE Very Inverse (PXR 35) 6 = IEEE Extremely Inverse (PXR 35) 7 = IEC-A (PXR 35) 8 = IEC-B (PXR 35) 9 = IEC-C (PXR 35)	PXR 35 (read only) 0.40xI_n to 1.00xI_n in steps of 0.01 (40 - 100)
3,4,6	3009	0x0BC0	LD pickup (Ir)	See note	PXR 25 (read only) 40 = 0.4xI_n 50 = 0.5xI_n 60 = 0.6xI_n 70 = 0.7xI_n 75 = 0.75xIn 80 = 0.8xI_n 90 = 0.9xI_n 95 = 0.95xI_n 98 = 0.98xI_n 100 = 1.0xI_n	PXR 35 (read only) For $I^{0.5}t$ slope, 0.5s - 24.0s, step = 0.1s (50 - 2400) For It slope, 0.5s - 24.0s, step = 0.1s (50 - 2400) For I^2t slope, 0.5s - 24.0s, step = 0.1s (50 - 2400) For I^4t slope, 0.5s - 7.0s, step = 0.1s (50 - 700) For IEEE MOD slope, 0.1s - 5.0s, step = 0.1s (10 - 500) For IEEE VERY INV slope, 0.2s - 5.0s, step = 0.1s (20 - 500) For IEEE EXT INV slope, 0.2s - 5.0, step = 0.1 (20 - 500) For IEC A slope, 0.05 - 1.0s, step = 0.05s (5 - 100) For IEC B slope, 0.1 - 1.0s, step = 0.05s (10 - 100) For IEC C slope, 0.2 - 1.0s, step = 0.05s (20 - 100)

Table 25. Modbus protection group data, (continued).

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition	
3,4,6	3010	0x0BC1	LD time (tr)	See note	PXR 20 (read only) 5 = 0.5s 10 = 1s 20 = 2s 40 = 4s 70 = 7s 100 = 10s 120 = 12s 150 =15s 200 = 20s 240 = 24s	
3,4,6	3011	0x0BC2	High load alarm 1	R/W	Value is percentage x I_r Valid range: 50 to 120, must be < high load 2	
3,4,6	3012	0x0BC3	SD slope	R/W	0 = flat 1 = I^2T, For PXR 35 with IEEE or IEC Long Delay Slopes the Short Delay Slope is restricted to Flat.	
3,4,6	3013	0x0BC4	SD pickup (Isd)	See note	PXR 20 (read only) 15 = 1.5xI_r 20 = 2.0xI_r 25 = 2.5xI_r 40 = 4.0xI_r 50 = 5.0xI_r 60 = 6.0xI_r 70 = 7.0xI_r 80 = 8.0xI_r 100 = 10.0xI_r	PXR 35 1.5xI_r - 14.0xI_r in steps of 0.1 (15 - 140)
3,4,6	3014	0x0BC5	SD time (tsd)	See note	PXR 20 (read only) 5 = 50ms 10 = 100ms 20 = 200ms 30 = 300ms 40 = 400ms 50 = 500ms	PXR 35 50ms - 500ms in steps of 10ms Setpoint value is 100x actual value, so 5 = 50ms
3,4,6	3015	0x0BC6	Inst. pickup (Ii)	See note	PXR 20 (read only) 0 = OFF 20 = 2xI_n 40 = 4xI_n 50 = 5xI_n 60 = 6xI_n 70 =7xI_n 80 = 8xI_n 100 = 10xI_n 120 = 12xI_n 150 = 15xI_n	For PXR 35 0 = OFF 2.0xI_n - 15.0xI_n in steps of 0.1 (20 - 150)
3,4,6	3016	0x0BC7	Ground sensing	R/W	0 = residual 1 = source/zero sequence	
3,4,6	3017	0x0BC8	Ground action	R/W	0 = trip 1 = alarm 2 = OFF	
3,4,6	3018	0x0BC9	Ground slope	R/W	0 = flat 1 = I^2t	
3,4,6	3019	0x0BCA	Ground pickup (I_g)	See note	PXR 20 (read only) 0 = OFF 20 = 0.2xI_n 40 = 0.4xI_n 60 = 0.6xI_n 80 = 0.8xI_n 100 = 1.0xI_n	PXR 35 0 = OFF 0.20xI_n - 1.0xI_n in steps of 0.01 (20 - 100)
	3020	0x0BCB	Ground time(t_g)	See note	PXR 20 (read only) 10 = 0.1s 20 = 0.2s 30 = 0.3s 40 = 0.4s 50 = 0.5s	PXR 35 For flat slope, 0.05s - 1.00s in steps of 0.01s (5 - 100_ For I^2t slope, 0.10s - 1.00s in steps of 0.01s (10 - 100)
3,4,6	3021	0x0BCC	GF thermal memory	R/W	0 = disable 1 = enable	

Table 25. Modbus protection group data, (continued).

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition
3,4,6	3022	0x0BCD	Neutral protection ratio	R/W	0 = 0% 60 = 60% 100 = 100% 200 = 200% (PXR 35 for 3-pole circuit breakers with Long Delay Pickup <= 0.5xI_n)
3,4,6	3023	0x0BCE	High load alarm 2	R/W	Value is percentage x I_r. Valid range: 50 to 120, must be > high load 1.
3,4,6	3024	0x0BCF	GF pre-alarm	R/W	Value is percentage x I_r. Valid range: 50 to 100.
3,4,6	3025	0x0BD0	Thermal memory alarm	R/W	Value is percentage x I_r. Valid range: 50 to 100.
The following Group 1 setpoint registers are for PXR 35 only					
3,4,6	3026	0x0BD1	High load 1 time		Valid range: 1 - 60, steps of 1 0 = OFF (PXR 35)
3,4,6	3027	0X0BD2	GF ZSI		0 = disable 1 = enable hardwired only 2 = enable GOOSE only 3 = enable hardwired and GOOSE
3,4,6	3028	0x0BD3	Neutral alarm pickup		Valid range: 0 = OFF 10 to 100, steps of 5
3,4,6	3029	0x0BD4	Neutral alarm time		Valid range: 1 - 60, steps of 1
3,4,6	3030	0x0BD5	High load 2 time		Valid range: 1 - 60, steps of 1
3,4,6	3031	0x0BD6	High load 1 action		0 = Alarm 1 = Extended Capture (HL1 Action and HL2 action cannot both be set to 1 at the same time.) 2 = Off
3,4,6	3032	0x0BD7	High load 2 action		0 = Alarm 1 = Extended capture (HL1 Action and HL2 action cannot both be set to 1 at the same time.) 2 = Off
3,4,6	3033	0x0BD8	LDPU event action		0 = No event capture 1 = Summary and close call capture
3,4,6	3034	0x0BD9	Source ground CT ratio	R/W	Value = CT ratio 2401 = 1200:1 2001 = 1000:1 1601 = 800:1 801 = 400:1 (factory default) 401 = 200:1 201 = 100:1 101 = 50:1 51 = 25:1 21 = 10:1 0 = Rogowski
3,4,6	3035	0x0BDA	Short delay pick up event action	R	0 = No event capture 1 = Summary and close call capture

9.4.2 Group 2 - Modbus configuration group

Table 27. Modbus configuration - PXR35 only

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition
3,4,6	3002	0x0BB9	Modbus RTU address	R/W	1 - 247
3,4,6	3003	0x0BBA	Modbus RTU baud rate	R/W	0 = 9600 1 = 19200 2 = 38400 3 = 57600 4 = 115200
3,4,6	3004	0x0BBB	Modbus RTU parity	R/W	0 = None 1 = Odd 2 = Even

Table 26. Modbus configuration - PXR35 only (continued).

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition
3,4,6	3005	0x0BBC	Modbus RTU stop bits	R/W	1, 2
3,4,6	3006	0x0BBD	Modbus RTU invalid object Handling	R/W	0 = Return NaN or 0 1 = Return exception code
3,4,6	3007	0x0BBE	Modbus RTU floating point word Order	R/W	0 = high-order word is first 1 = low-order word is first
3,4,6	3008	0x0BBF	Modbus RTU fixed point word Order	R/W	0 = high-order word is first 1 = low-order word is first
3,4,6	3009	0x0BC0	Modbus RTU enable/disable	R/W	Bit 0 - Modbus RTU Communications Bit 1 - Open Breaker Command Bit 2 - Close Breaker Command Bit 3 - Maintenance mode Command Bit 4 - Setpoint Change for all bits, 0 = disable, 1 = enable
3,4,6	3010	0x0BC1	Modbus TCP invalid object Handling	R/W	0 = Return NaN or 0 1 = Return exception code
3,4,6	3011	0x0BC2	Modbus TCP floating point word Order	R/W	0 = high-order word is first 1 = low-order word is first
3,4,6	3012	0x0BC3	Modbus TCP fixed point word Order	R/W	0 = high-order word is first 1 = low-order word is first
3,4,6	3013	0x0BC4	UNUSED	R	UNUSED
3,4,6	3014	0x0BC5	UNUSED	R	UNUSED
3,4,6	3015	0x0BC6	Modbus TCP enable/disable	R/W	Bit 0 - Modbus RTU Communications Bit 1 - Open Breaker Command Bit 2 - Close Breaker Command Bit 3 - Maintenance mode Command Bit 4 - Setpoint Change for all bits, 0 = disable, 1 = enable

9.4.3 Group 4 frequency protection group

Table 28. Modbus frequency protection data - PXR35 only

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition
3,4,6	3002	0x0BB9	Freq protection enabled	R/W	0 = Disabled 1 = Enabled
3,4,6	3003	0x0BBA	Over freq trip action	R/W	0 = Disabled 1 = Enabled
3,4,6	3004	0x0BBB	Over freq trip pickup	R/W	Valid range: 100.1% - 110.0% 0.1% steps (1001 - 1100)
3,4,6	3005	0x0BBC	Over freq trip time	R/W	Valid range: 0.05s - 300s 0.05s step (5 - 30000, steps of 5)
3,4,6	3006	0x0BBD	Under freq trip action	R/W	0 = Disabled 1 = Enabled
3,4,6	3007	0x0BBE	Under freq trip pickup	R/W	Valid range: 90% - 99.9% 0.1% steps (900 - 999)
3,4,6	3008	0x0BBF	Under freq trip time	R/W	0.05s - 300s, 0.05s step (5 - 30000, steps of 5)
3,4,6	3009	0x0BC0	Over freq alarm action	R/W	0 = Disabled 1 = Enabled
3,4,6	3010	0x0BC1	Over freq alarm pickup	R/W	Valid range: 100.1% - 110.0% 0.1% steps (1001 - 1100)
3,4,6	3011	0x0BC2	Over freq alarm time	R/W	Valid range: 0.05s - 300s 0.05s step (5 - 30000, steps of 5)

Modbus configuration - PXR35 only, (continued).

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition
3,4,6	3012	0x0BC3	Under freq alarm action	R/W	0 = Disabled 1 = Enabled
3,4,6	3013	0x0BC4	Under freq alarm pickup	R/W	Valid range: 90% - 99.9% 0.1% steps (900 - 999)
3,4,6	3014	0x0BC5	Under freq alarm time	R/W	Valid range: 0.05s - 300s 0.05s step (5 - 30000, steps of 5)

9.4.4 Group 5 – Enhanced protection group

Table 29. Modbus enhanced protection group data - PXR 25.

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition (PXR 25/35 only)
3,4,6	3002	0x0BB9	Over voltage action	R/W	0 = trip, 1 = alarm, 2 = OFF
3,4,6	3003	0x0BBA	Over voltage pickup	R/W	Valid range: 180 to 720 (volts)
3,4,6	3004	0x0BBB	Over voltage time	R/W	Valid range: 1 to 300 (seconds)
3,4,6	3005	0x0BBC	Under voltage action	R/W	0 = trip, 1 = alarm, 2 = OFF
3,4,6	3006	0x0BBD	Under voltage pickup	R/W	Valid range: 60 to 670 (volts)
3,4,6	3007	0x0BBE	Under voltage time	R/W	Valid range: 1 to 300 (seconds)
3,4,6	3008	0x0BBF	Voltage unbalance action	R/W	0 = trip, 1 = alarm, 2 = OFF
3,4,6	3009	0x0BC0	Voltage unbalance pickup	R/W	Valid range: 5 to 25 (as % of phase voltage)
3,4,6	3010	0x0BC1	Voltage unbalance time	R/W	Valid range: 1 to 300 (seconds)
3,4,6	3011	0x0BC2	Current unbalance action	R/W	0 = trip, 1 = alarm, 2 = OFF
3,4,6	3012	0x0BC3	Current unbalance pickup	R/W	Valid range: 5 to 25 (as % of phase voltage)
3,4,6	3013	0x0BC4	Current unbalance time	R/W	Valid range: 1 to 300 (seconds)
3,4,6	3014	0x0BC5	Reverse power action	R/W	0 = trip, 1 = alarm, 2 = OFF
3,4,6	3015	0x0BC6	Reverse power pickup	R/W	Valid range: 1 to 65500 (kW)
3,4,6	3016	0x0BC7	Reverse power time	R/W	Valid range: 1 to 300 (seconds)

Table 30. Modbus enhanced protection group data - PXR 35.

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition (PXR 25/35 only)
3,4,6	3002	0x0BB9	Overvoltage trip action	R/W	0 = OFF, 1 = Enabled
3,4,6	3003	0x0BBA	Overvoltage trip pickup	R/W	Valid range: 102.0% - 150.0% of Vnominal 0.1% steps (1020 - 1500)
3,4,6	3004	0x0BBB	Overvoltage trip time	R/W	Valid range: 1 to 300 (seconds) 0.05s steps (100 - 30000, steps of 5)
3,4,6	3005	0x0BBC	Undervoltage trip action	R/W	0 = OFF, 1 = Enabled
3,4,6	3006	0x0BBD	Undervoltage trip pickup	R/W	Valid range: 50.0% - 98.0% of Vnominal 0.1% steps (500 - 980)
3,4,6	3007	0x0BBE	Undervoltage trip time	R/W	Valid range: 1 to 300 (seconds) 0.05s steps (100 - 30000, steps of 5)
3,4,6	3008	0x0BBF	Voltage unbalance trip action	R/W	0 = OFF, 1 = Enabled
3,4,6	3009	0x0BC0	Voltage unbalance trip pickup	R/W	Valid range: 2 - 90 (as % of phase voltage) 1% steps

Table 29 Modbus enhanced protection group data - PXR 35 (continued).

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition (PXR 25/35 only)
3,4,6	3010	0x0BC1	Voltage unbalance trip time	R/W	Valid range: 1 to 300 (seconds) 0.05s steps (100 - 30000, steps of 5)
3,4,6	3011	0x0BC2	Current unbalance trip action	R/W	0 = OFF, 1 = Enabled
3,4,6	3012	0x0BC3	Current unbalance trip pickup	R/W	Valid range: 2 - 90% 1% steps
3,4,6	3013	0x0BC4	Current unbalance trip time	R/W	Valid range: 1 to 300 (seconds) 0.05s steps (100 - 30000, steps of 5)
3,4,6	3014	0x0BC5	Reverse power action	R/W	0 = trip, 1 = alarm, 2 = OFF
3,4,6	3015	0x0BC6	Reverse power pickup	R/W	Valid range: 1 to 65500 (kW) 1 kW steps
3,4,6	3016	0x0BC7	Reverse power time	R/W	Valid range: 1 to 300 (seconds) 0.05s steps (100 - 30000, steps of 5)
3,4,6	3017	0x0BC8	Phase rotation type		0 = ABC 1 = CBA
3,4,6	3018	0x0BC9	Phase rotation action		0 = Trip 1 = Alarm 2 = Off
3,4,6	3019	0x0BCA	Phase loss action		0 = Trip 1 = Alarm 2 = Off
3,4,6	3020	0x0BCB	Phase loss time		1 - 240s, 1 second steps
3,4,6	3021	0x0BCC	Reserved		
3,4,6	3022	0x0BCD	Reserved		
3,4,6	3023	0x0BCE	Reserved		
3,4,6	3024	0x0BCF	Reserved		
3,4,6	3025	0x0BD0	Reserved		
3,4,6	3026	0x0BD1	Overvoltage alarm action		0 = Alarm 1 = Extended capture 2 = Off
3,4,6	3027	0x0BD2	Overvoltage alarm pickup		Valid range: 102.0% - 150.0% of Vnominal 0.1% steps (1020 - 1500)
3,4,6	3028	0x0BD3	Overvoltage alarm time		Valid range: 1 to 300 (seconds) 0.05s steps (100 - 30000, steps of 5)
3,4,6	3029	0x0BD4	Extended protection enable		0 = Disabled 1 = Enabled
3,4,6	3030	0x0BD5	Reverse reactive power action		0 = Trip 1 = Alarm 2 = Off
3,4,6	3031	0x0BD6	Reverse reactive power pickup		Valid range: 1 to 65500 (kVAR) 1 kVAR steps
3,4,6	3032	0x0BD7	Reverse reactive power time		Valid range: 1 to 300 (seconds) 0.05s steps (100 - 30000, steps of 5)
3,4,6	3033	0x0BD8	Reserved (phase rotation time)		Valid range: 0.05 to 300 (seconds) 0.05s steps (5 - 30000, steps of 5)
3,4,6	3034	0x0BD9	Overvoltage number of phases		1, 2, 3

Table 29 Modbus enhanced protection group data - PXR 35 (continued).

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition (PXR 25/35 only)
3,4,6	3035	0x0BDA	Undervoltage number of phases		1, 2, 3
3,4,6	3036	0x0BDB	Undervoltage alarm action		0 = Alarm 1 = Extended capture 2 = Off
3,4,6	3037	0x0BDC	Undervoltage alarm pickup		Valid range: 50.0% - 98.0% of Vnominal 0.1% steps (500 - 980)
3,4,6	3038	0x0BDD	Undervoltage alarm time		Valid range: (5 - 30000, steps of 5) 0.05s steps
3,4,6	3039	0x0BDE	Voltage unbalance alarm action		0 = Disabled 1 = Enabled
3,4,6	3040	0x0BDF	Voltage unbalance alarm pickup		Valid range: 2 - 90 5 1% steps
3,4,6	3041	0x0BE0	Voltage unbalance alarm time		Valid range: 1 to 300 (seconds) 0.05s steps (100 - 30000, steps of 5)
3,4,6	3042	0x0BE1	Current unbalance alarm function		0 = Disabled 1 = Enabled
3,4,6	3043	0x0BE2	Current unbalance alarm pickup		Valid range: 2 - 90% 1% steps
3,4,6	3044	0x0BE3	Current unbalance alarm time		Valid range: 1 to 300 (seconds) 0.05s steps (100 - 30000, steps of 5)

9.4.5 Group 6 power protection group

Table 31. Modbus power protection data - PXR 35 only

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition (PXR 25/35 only)
3,4,6	3002	0x0BB9	Power protection enabled	R/W	0 = Disabled 1 = Enabled
3,4,6	3003	0x0BBA	Forward real power action	R/W	0 = Trip 1 = Alarm 2 = Off
3,4,6	3004	0x0BBB	Forward real power pickup	R/W	Valid range: 1 - 65500 (kW) 1 kW steps
3,4,6	3005	0x0BBC	Forward real power time	R/W	Valid range: 1 - 300s 1 second steps
3,4,6	3006	0x0BBD	Forward reactive power action	R/W	0 = Trip 1 = Alarm 2 = Off
3,4,6	3007	0x0BBE	Forward reactive power pickup	R/W	Valid range: 1 - 65500 (kVAR) 1 kVAR steps
3,4,6	3008	0x0BBF	Forward reactive power time	R/W	Valid range: 1 - 300s 1 second steps
3,4,6	3009	0x0BC0	Apparent power action	R/W	0 = Trip 1 = Alarm 2 = Off
3,4,6	3010	0x0BC1	Apparent power pickup	R/W	Valid range: 1 - 65500 (kW) 1 kW steps

Table 30 Modbus power protection data - PXR 35 only (continued).

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition (PXR 25/35 only)
3,4,6	3011	0x0BC2	Apparent power time	R/W	Valid range: 1 - 300s 1 second steps
3,4,6	3012	0x0BC3	Power factor action	R/W	0 = Trip 1 = Alarm 2 = Off
3,4,6	3013	0x0BC4	Power factor pickup	R/W	Valid range: 0.20 - 0.95 0.05 steps (20 - 95, steps of 5)
3,4,6	3014	0x0BC5	Power factor time	R/W	Valid range: 1 - 60s 1 second steps
3,4,6	3015	0x0BC6	Real demand power action	R/W	0 = Trip (not available in V1.0.0) 1 = Alarm 2 = Off
3,4,6	3016	0x0BC7	Real demand power pickup	R/W	Valid range: 1 - 65500 (kW) 1 kW steps
3,4,6	3017	0x0BC8	Real demand power number of windows	R/W	Valid range: 1 - 10
3,4,6	3018	0x0BC9	Apparent demand power number of windows	R/W	Valid range: 1 - 10
3,4,6	3019	0x0BCA	Apparent demand power action	R/W	0 = Trip (not available in V1.0.0) 1 = Alarm 2 = Off
3,4,6	3020	0x0BCB	Apparent demand power pickup	R/W	Valid range: 1 - 65500 (kW) 1 kW steps
3,4,6	3021	0x0BCC	Demand mode and precision	R	Bit map: b15..b6: not used b5..b4: Precision. 00 = Precise, 01 = Normal, 10 = Relaxed b3..b1: not used b0: Mode. 0 = Internal time-based operation, 1 = External sync-based operation

9.4.6 Group 7 sync check group

Table 32. Modbus sync check data - PXR 35 only

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition (PXR 25/35 only)
3,4,6	3002	0x0BB9	Sync check action	R/W	0 = Disabled 1 = Enabled
3,4,6	3003	0x0BBA	Sync check min live voltage 1	R/W	Valid range: 10 - 130% $V_{nominal}$ 1% steps
3,4,6	3004	0x0BBB	Sync check max dead voltage 1	R/W	Valid range: 10 - 100% $V_{nominal}$ 1% steps
3,4,6	3005	0x0BBC	Sync check min live voltage 2	R/W	Valid range: 10 - 130% $V_{nominal}$ 1% steps
3,4,6	3006	0x0BBD	Sync check max dead voltage 2	R/W	Valid range: 1 - 100% $V_{nominal}$ 1% steps
3,4,6	3007	0x0BBE	Sync check max voltage difference	R/W	Valid range: 1 - 5% $V_{nominal}$ 1% steps
3,4,6	3008	0x0BBF	Sync check max slip frequency	R/W	Valid range: 0.01 - 2.0 Hz .01 Hz steps (1 - 200)
3,4,6	3009	0x0BC0	Sync check max angle difference	R/W	1 - 60, 1 steps
3,4,6	3010	0x0BC1	Sync check dead V1 - dead V2 enable	R/W	0 = Disabled 1 = Enabled
3,4,6	3011	0x0BC2	Sync check dead V1 - live V2 enable	R/W	0 = Disabled 1 = Enabled
3,4,6	3012	0x0BC3	Sync check live V1 - dead V2 enable	R/W	0 = Disabled 1 = Enabled
3,4,6	3013	0x0BC4	Sync check delay time	R/W	0 - 300s, 1 second steps

9.4.7 Group 9 other protection group

Table 33. Modbus other protection data - PXR 35 only

Func. code	Register number	Address (hex)	Setpoint name	R/W	Definition (PXR 25/35 only)
3,4,6	3002	0x0BB9	Over temperature alarm action	R/W	0 = Disabled 1 = Enabled
3,4,6	3003	0x0BBA	Over temperature alarm pickup	R/W	Valid range: 85 - 100C° 5C° steps
3,4,6	3004	0x0BBB	Current THD alarm pickup	R/W	0 = Off 10 - 30% 1% step
3,4,6	3005	0x0BBC	Current THD alarm time	R/W	Valid range: 1 - 60s 1 second step
3,4,6	3006	0x0BBD	Voltage THD alarm pickup	R/W	0 = Off 10 - 30% 1% step
3,4,6	3007	0x0BBE	Voltage THD alarm time	R/W	Valid range: 1 - 60s 1 second step
3,4,6	3008	0x0BBF	Operation count	R/W	0 = OFF 1 - 12500 steps of 1
3,4,6	3009	0x0BC0	UNUSED	R	UNUSED
3,4,6	3010	0x0BC1	Sneakers alarm	R/W	0 = Disabled 1 = Enabled
3,4,6	3011	0x0BC2	Internal error	R/W	0 = Disabled 1 = Enabled

9.5 Remote control commands

A set of three registers is reserved for sending commands to the trip unit. These registers should be written together using function code 16, with the data encoded as a 3-byte command and its 1's complement for a total of 6 bytes. If the three registers are valid, the trip unit will begin to execute the action. If the three registers are invalid, the trip unit returns exception code 03.

Once the command is successfully acknowledged by the trip unit, it returns a normal function code 16 response to the Modbus client. Since it may take some time for the trip unit to take the action, the client may further determine if the product completed the slave action function successfully after the normal response by reading status.

Table 34. Modbus remote control command encoding.

Register number	Address (hex)	Trip unit remote control - register definition	
2901	0x0B54	Command byte 1	Command byte 0
2902	0x0B55	1's complement of byte 0	Command byte 2
2903	0x0B56	1's complement of byte 2	1's complement of byte 1

Table 34 shows the list of commands available. The PW column indicates if the password must be written before issuing the command.

Table 35. Modbus remote control commands.

Register number		425088	425089	425090
Address (hex)		61FF	6200	6201
Definition	PW	Command (hex)		
Reset trip	No	0x0002	0xFD00	0xFFFF
Reset powered up indication	No	0x0003	0xFC00	0xFFFF
Reset peak demand watts	Yes	0x0004	0xFB00	0xFFFF
Reset energy	Yes	0x0008	0xF700	0xFFFF
Reset operations count	Yes	0x0102	0xFD00	0xFFFE
Reset runtime	Yes	0x0103	0xFC00	0xFFFE
Reset min/max currents	Yes	0x010D	0xF200	0xFFFE
Reset min/max line-to-neutral voltages	Yes	0x010F	0xF000	0xFFFE
Reset min/max line-to-line voltages	Yes	0x010E	0xF100	0xFFFE
Reset all min/max values	Yes	0x0104	0xFB00	0xFFFE
Reset trip count	Yes	0x0501	0xFE00	0xFFFA
Reset temperature	Yes	0x0502	0xFD00	0xFFFA
Reset all diagnostics information	Yes	0x0503	0xFC00	0xFFFA
Open breaker - activates the relay mapped to open breaker pulsed	Yes	0x0000	0xFF01	0xFEFF
Close breaker - activates the relay mapped to close breaker pulsed	Yes	0x0001	0xFE01	0xFEFF
Enable maintenance mode	No	0x0008	0xF701	0xFEFF
Disable maintenance mode	No	0x0009	0xF601	0xFEFF
Activate relay 1 (if relay 1 is mapped to remote control)	No	0x0101	0xFE04	0xFBFE
Deactivate relay 1 (if relay 1 is mapped to remote control)	No	0x0201	0xFE04	0xFBFD
Activate relay 2 (if relay 2 is mapped to remote control)	No	0x0102	0xFD04	0xFBFE
Deactivate relay 2 (if relay 2 is mapped to remote control)	No	0x0202	0xFD04	0xFBFD
Activate relay 3 (if relay 3 is mapped to remote control)	No	0x0103	0xFC04	0xFBFE
Deactivate relay 3 (if relay 3 is mapped to remote control)	No	0x0203	0xFC04	0xFBFD
Change active setpoint set to set A		0x0100	0xFF03	0xFCFE
Change active setpoint set to set B		0x0101	0xFE03	0xFCFE
Change active setpoint set to set C		0x0102	0xFD03	0xFCFE
Change active setpoint set to set D		0x0103	0xFC03	0xFCFE

9.6 Date and time

The date and time are available via registers. The Modbus client can synchronize the clock in the trip unit by writing the time at regular intervals. The entire block of registers should be written in a single command. A time change event is only generated when the time difference between old and new is greater than 60 seconds.

Table 36. Modbus date and time data.

Function code	Register number	Address (hex)	Definition	Value range
3, 4, 6, 16	2921	0x0B68	Month	1-12
3, 4, 6, 16	2922	0x0B69	Day	1-31
3, 4, 6, 16	2923	0x0B6A	Year	2015 - 2099
3, 4, 6, 16	2924	0xB6B	Day of week	1 = Sunday, 7 = Saturday
3, 4, 6, 16	2925	0x0B6C	Hour	0-23
3, 4, 6, 16	2926	0x0B6D	Minute	0-59
3, 4, 6, 16	2927	0x0B6E	Second	0-59
Alternate based on IEC 60870-5				
3, 4, 6, 16	2941	0x0B7C	Years after 2000	b15..7 = Reserved b6..0 = 0 - 127
3, 4, 6, 16	2942	0x0B7D	Month/Day	b15..12 = Reserved, b11..8 = 1 - 12 (Jan - Dec), b7..5 = Reserved, b4..0 = 1 - 31 (day)
3, 4, 6, 16	2943	0x0B7E	Hours/ Minutes	b15..13 = Reserved, b12..8 = 0 - 23 (hour), b6..7 = Reserved, b5..0 = 0 - 59 (minutes)
3, 4, 6, 16	2944	0x0B7F	Milliseconds	0 - 59,999

9.7 Event registers

The triggering of an event in the trip unit can provide historical data object values at the instant in time the event occurs or occurring around the time of the event. The trip unit categorizes the event information into summary, alarm, and trip event data. See Section 6.2 - Event recording and waveform capture for details. A maximum of 200 summary events, 10 alarm events, and 10 trip events are stored. A single event may place information into multiple objects. For example, the occurrence of an event triggered by a circuit breaker trip may provide both summary and trip information.

Access to event information is based on the selection of event type and event ID. When the event type selection is written in register 8193, the earliest and latest event ID can be obtained in registers 8194 and 8196 respectively to determine the range of events saved for the selected event type.

Register 8198 is used to select the requested event ID. If the request event exists in the trip unit, registers 8200 and 8202 provide the Previous Event ID and Next Event ID respectively. If the requested event does not exist in the trip unit, exception code 0x87 is returned.

The date and time when the requested event happened is read in registers 8204 through 8211. This value corresponds to the time of occurrence of the historical event.

Register 8212 provides an indication of the selected event type's data content. This is a constant value for the three event types Modbus port supports.

Event data also provides a validity bit for each data object, starting from register 8213. Bit 0 set to be 1 indicates that the first data object is valid, bit 1 is used to indicate the second data object is valid, and so on.

The triggering of an event in the trip unit can provide historical data object values at the instant in time the event occurs or occurring around the time of the event. The trip unit categorizes the event information into the following events:

1. Summary events
2. Trip events
3. Alarm events
4. Extended capture events [PXR 35 only]
5. Close call events [PXR 35 only]
6. Time adjustment events [PXR 35 only]
7. Demand log events [PXR 35 only]

Access to event information is based on the selection of event type and event ID. When the event type selection is written in register 8193 (types defined in the table), the earliest and latest event ID can be obtained in registers 8194 and 8196 respectively to determine the range of events saved for the selected event type.

Register 8198-99 is used to select the requested event ID. If the requested event exists in the trip unit, registers 8200-01 and 8202-03 provide the Previous Event ID and Next Event ID. If the requested event does not exist in the trip unit, exception code 0x87 is returned.

The date and time when the requested event happened is read in registers 8204 through 8207 [PXR 35] or 8211 [PXR 20/25]. This value corresponds to the time of occurrence of the event. Register 8207 [PXR 35] or 8212 [PXR 20/25] provides an indication of the selected event type's data content. This is a constant value for all the event types Modbus supports.

Table 37. Modbus event registers for PXR 20/25.

Summary events

Function code	Register number	Address (hex)	Format	Definition
3, 4, 6, 16	8193	0x2000	Encoded	Event type: event = 0x8EFF
3, 4	8194	0x2001	Unsigned32	Earliest event ID
3, 4	8196	0x2003	Unsigned32	Latest event ID
3, 4, 16	8198	0x2005	Unsigned32	Requested event ID
3, 4	8200	0x2007	Unsigned32	Previous event ID
3, 4	8202	0x2009	Unsigned32	Next event ID
3, 4	8204	0x200B	Date/Time	Date/time

Table 36. Modbus event registers for PXR 20/25 (continued).

Trip or major alarms

Function code	Register number	Address (hex)	Format	Definition
3, 4	8212	0x2013	Encoded	Format of data: 0x0000, 0x0001, 0x0004, 0x0005, 0x0006
3, 4	8213	0x2014	B0	Object validity bits
3, 4	8214	0x2015	Encoded	Event cause: 00 = power up – time OK 01 = setpoints download 02 = time adjusted 03 = trip 04 = alarm 05 = enter test mode 06 = exit test mode 08 = power up - no time 09 = test completed 10 = maintenance mode active 11 = maintenance mode inactive 12 = opened by communications 13 = closed by communications

Minor alarm

Function code	Register number	Address (hex)	Format	Definition
3, 4, 6, 16	8193	0x2000	Encoded	Event type: alarm = 0x81FF
3, 4	8194	0x2001	Unsigned32	Earliest event ID
3, 4	8196	0x2003	Unsigned32	Latest event ID
3, 4, 16	8198	0x2005	Unsigned32	Requested event ID
3, 4	8200	0x2007	Unsigned32	Previous event ID
3, 4	8202	0x2009	Unsigned32	Next event ID
3, 4	8204	0x200B	Date/time	Date/time
3, 4	8212	0x2013	Encoded	Format of data: minor alarm 0x0006
3, 4	8213	0x2014	Bits	Object validity bits
3, 4	8214	0x2015	Encoded	Status cause (primary, secondary, cause)
3, 4, 6, 16	8193	0x2000	Encoded	Event type: trip = 0x80FF, alarm = 0x81FF
3, 4	8194	0x2001	Unsigned32	Earliest event ID
3, 4	8196	0x2003	Unsigned32	Latest event ID
3, 4, 16	8198	0x2005	Unsigned32	Requested event ID
3, 4	8200	0x2007	Unsigned32	Previous event ID
3, 4	8202	0x2009	Unsigned32	Next event ID
3, 4	8204	0x200B	Date/time	Date/time
3, 4	8212	0x2013	Encoded	Format of data: trip = 0x0004, major alarm 0x0005
3, 4	8213	0x2014	Bits	Object validity bits
3, 4	8214	0x2015	Bits	Object validity bits
3, 4	8215	0x2016	Encoded	Status cause (primary, secondary, cause)
3, 4	8217	0x2018	Unsigned32	IA
3, 4	8219	0x201A	Unsigned32	IB
3, 4	8221	0x201C	Unsigned32	IC
3, 4	8223	0x201E	Unsigned32	IN
3, 4	8225	0x2020	Unsigned32	IG source
3, 4	8227	0x2022	Unsigned32	IG residual
3, 4	8229	0x2024	Unsigned16	VAB
3, 4	8230	0x2025	Unsigned16	VBC
3, 4	8231	0x2026	Unsigned16	VCA
3, 4	8232	0x2027	Unsigned16	VAN
3, 4	8233	0x2028	Unsigned16	VBN
3, 4	8234	0x2029	Unsigned16	VCN
3, 4	8235	0x202A	Signed32	Real 3-phase power (W)
3, 4	8237	0x202C	Signed32	Reactive 3-phase power (VAR)

Table 36. Modbus event registers for PXR 20/25 (continued).

Trip or major alarms

Function code	Register number	Address (hex)	Format	Definition
3, 4	8239	0x202E	Unsigned32	Apparent 3-phase power (VA)
3, 4	8241	0x2030	Signed32	Real 3-phase power demand (W)
3, 4	8243	0x2032	Signed32	Reactive 3-phase power demand (VAR)
3, 4	8245	0x2034	Unsigned32	Apparent 3-phase power demand (VA)
3, 4	8247	0x2036	Singed16	Device temperature (1/10 °C)
3, 4	8248	0x2037	Unsigned16	Frequency (1/10 Hz)
3, 4	8249	0x2038	Singed16	Apparent power factor (1/100 pf)
3, 4	8250	0x2039	Unsigned16	Operations count
3, 4	8251	0x203A	Bits	Binary status with validity bits

Table 38. Modbus event registers for PXR 35.

Demand/energy logs				
Function code	**Register number**	**Address (hex)**	**Format**	**Definition**
3, 4, 6, 16	8193	0x2000	Encoded	Event type: Trip = 0x80FF Alarm = 0x81FF Extended capture = 0x82FF Close call = 0x83FF Time adjustment = 0x84FF Demand = 0x85FF Summary = 0x8EFF
3, 4	8194	0x2001	Unsigned32	Earliest event ID
3, 4	8196	0x2003	Unsigned32	Latest event ID
3, 4, 16	8198	0x2005	Unsigned32	Requested event ID
3, 4	8200	0x2007	Unsigned32	Previous event ID
3, 4	8202	0x2009	Unsigned32	Next event ID
3, 4	8204	0x200B	Date/Time	Date/time
3, 4	8208	0x2013	Encoded	Format of data: Trip = 0x0004 Alarm = 0x0005 Extended capture = 0x0006 Close call = 0x0007 Time adjustment = 0x0008 Demand = 0x0009 Summary = 0x0010
Summary				
Function code	**Register number**	**Address (hex)**	**Format**	**Definition**
3, 4	8209	0x2010	Unsigned32	Summary code: See Table 38 for summary codes
Tips, alarms, extended captures				
Function code	**Register number**	**Address (hex)**	**Format**	**Definition**
3, 4	8209	0x2010	Encoded	Primary & secondary causes: See Table 38 for codes
3, 4	8210	0x2011	Encoded	Cause: See Table 38 for codes
3, 4	8211	0x2012	Encoded	Summary code: See Table 38 for summary codes
3, 4	8217	0x2018	Float	IA
3, 4	8219	0x201A	Float	IB
3, 4	8221	0x201C	Float	IC
3, 4	8223	0x201E	Float	IN
3, 4	8225	0x2020	Float	IG source
3, 4	8227	0x2022	Float	VAN 1
3, 4	8229	0x2024	Float	VBN 1
3, 4	8231	0x2026	Float	VCN 1
3, 4	8233	0x2028	Float	VAB 1
3, 4	8235	0x202A	Float	VBC 1
3, 4	8237	0x202C	Float	VCA 1
3, 4	8239	0x202E	Float	VAN 2
3, 4	8241	0x2030	Float	VBN 2
3, 4	8243	0x2032	Float	VCN 2
3, 4	8245	0x2034	Float	VAB 2
3, 4	8247	0x2036	Float	VBC 2
3, 4	8249	0x2038	Float	VCA 2
3, 4	8251	0x203A	Float	Real power
3, 4	8253	0x203C	Float	Reactive power
3, 4	8255	0x203E	Float	Apparent power

Table 37. Modbus event registers for PXR 35 (continued)

Trips, alarms, extended captures				
Function code	**Register number**	**Address (hex)**	**Format**	**Definition**
3, 4	8257	0x2040	Float	Demand IA
3, 4	8259	0x2042	Float	Demand IB
3, 4	8261	0x2044	Float	Demand IC
3, 4	8263	0x2046	Float	Demand IN
3, 4	8265	0x2048	Float	Demand W
3, 4	8267	0x204A	Float	Demand Var
3, 4	8269	0x204C	Float	Demand Var
3, 4	8271	0x204E	Float	Device temperature (1/10 °C)
3, 4	8273	0x2050	Float	Frequency 1
3, 4	8275	0x2052	Float	Frequency 2
3, 4	8277	0x2054	Float	Apparent power factor phase A
3, 4	8279	0x2056	Float	Apparent power factor phase B
3, 4	8281	0x2058	Float	Apparent power factor phase C
3, 4	8283	0x205A	Float	Apparent power factor total
3, 4	8285	0x205C	Float	THD IA
3, 4	8287	0x205E	Float	THD IB
3, 4	8289	0x2060	Float	THD IC
3, 4	8291	0x2062	Float	THD IN
3, 4	8293	0x2064	Float	THD VAN 1
3, 4	8295	0x2066	Float	THD VBN 1
3, 4	8297	0x2068	Float	THD VCN 1
3, 4	8299	0x206A	Float	THD VAB 1
3, 4	8301	0x206C	Float	THD VBC 1
3, 4	8303	0x206E	Float	THD VCA 1
3, 4	8305	0x2070	Float	CF IA
3, 4	8307	0x2072	Float	CF IB
3, 4	8309	0x2074	Float	CF IC
3, 4	8311	0x2076	Float	CF IN
3, 4	8313	0x2078	Float	Operations count
Close call				
Function code	**Register number**	**Address (hex)**	**Format**	**Definition**
3, 4	8217	0x2018	Float	Duration
3, 4	8219	0x201A	Unsigned32	Reserved & critical value code
3, 4	8221	0x201C	Float	Critical value
3, 4	8223	0x201E	Float	Average value
3, 4	8225	0x2020	Float	Min. thermal value
3, 4	8227	0x2022	Unsigned32	Trigger event ID
Time adjustment				
Function code	**Register number**	**Address (hex)**	**Format**	**Definition**
3,4	8217	0x2018	Date/Time	New time

9.8 Demand/Energy log registers (PXR 35)

The PXR 35 trip unit logs up to 12992 Demand/Energy data entries, enough for 45 days at 5-minute intervals. Register 10241 is used to select the requested log entry, with 0 being the most recent log. If the requested log exists in the trip unit, registers 10241 – 10272 provide the data. If the requested log does not exist in the trip unit, the previous requested entry is used.

Table 39. Demand/Energy logs

Function code	Register number	Address (hex)	Format	Definition
3,4,6,16	10241	0x2800	Unsigned16	Requested log index
3,4	10242	0x2801	Unsigned16	Number of stored logs
3, 4	10243	0x2802	Date/Time	Log Timestamp
3, 4	10247	0x2806	Float	IA Demand
3, 4	10249	0x2808	Float	IB Demand
3, 4	10251	0x280A	Float	IC Demand
3, 4	10253	0x280C	Float	IN Demand
3, 4	10255	0x280E	Float	W Demand
3, 4	10257	0x2810	Float	VAR Demand
3, 4	10259	0x2812	Float	VA Demand
3, 4	10261	0x2814	Unsigned 32	Interval
3, 4	10263	0x2816	Unsigned 32	Wh forward
3, 4	10265	0x2818	Unsigned 32	Wh reverse
3, 4	10267	0x281A	Unsigned 32	VARh lagging
3, 4	10269	0x281C	Unsigned 32	VARh leading
3, 4	10271	0x281E	Unsigned 32	VAh leading

Table 40. Summary codes

Event code	Definition
No event	0x0000
Power up – Time OK	0x0001
Power up – Time bad	0x0002
Setpoint download – Modbus RTU	0x0003
Setpoint download – CAM	0x0004
Setpoint download – Modbus TCP	0x0005
Setpoint download – Display	0x0006
Setpoint download – USB	0x0007
Setpoint download - Bluetooth	0x0008
Setpoint set change – CAM	0x0009
Setpoint set Change – Display	0x000A
Setpoint set Change – Modbus RTU	0x000B
Setpoint set Change – Modbus TCP	0x000C
Setpoint set Change – USB	0x000D
Setpoint set Change – GOOSE	0x000E
Setpoint set Change – Bluetooth	0x000F
Setpoints frame mismatch	0x0010
Setpoints error	0x0011
Enter maintenance mode	0x0012
Exit Maintenance Mode	0x0013
Opened by Modbus RTU	0x0014
Opened by CAM	0x0015
Opened by display	0x0016
Opened by Modbus TCP	0x0017
Opened by USB	0x0018
Opened by Bluetooth	0x0019
Opened by GOOSE	0x001A
Closed by Modbus RTU	0x001B
Closed by CAM	0x001C
Closed by display	0x001D
Closed by Modbus TCP	0x001E
Closed by USB	0x001F
Closed by Bluetooth	0x0020
Closed by GOOSE	0x0021
Enter test mode	0x0022
Exit test mode – Abort	0x0023
Exit test mode – Completed	0x0024
Admin password changed	0x0025
Admin password entered	0x0026
User password changed	0x0027
User password entered	0x0028

Table 39. Summary codes (continued).

Event code	Definition
Incorrect password entered	0x0029
TA coil failure	0x002A
Watchdog restart	0x002B
Calibration sonstants error	0x002C
Setpoints fault	0x002D
Frame Module fault	0x002E
NV memory error 1	0x002F
NV memory error 2	0x0030
NV memory error 3	0x0031
Breaker health	0x0032
Aux power failure	0x0033
Battery failure	0x0034
RTC error	0x0035
Wrong sensor	0x0036
Digital bypass	0x0037
Operations count	0x0038
Mechanical 2	0x0039
Display overtemperature	0x003A
System change	0x003B
Sensor error	0x003C
ZSI input active	0x003D
ZSI output active	0x003E
LDPU entry	0x003F
LDPU neutral entry	0x0040
SDPU entry	0x0041
Trip actuator voltage low	0x0042
Alarm overvoltage entry test	0x0043
Alarm undervoltage entry test	0x0044
Alarm high load 1 entry test	0x0045
Alarm high load 2 entry test	0x0046
Alarm overtemperature test	0x0047
Alarm voltage unbalance test	0x0048
Alarm current unbalance test	0x0049
Alarm reverse kW entry test	0x004A
Alarm reverse kVar entry test	0x004B
Alarm phase loss entry test	0x004C
Alarm overfrequency entry test	0x004D
Alarm underfrequency entry test	0x004E
Alarm over kW entry test	0x004F
Alarm over kVar entry test	0x0050
Alarm over kVA entry test	0x0051

Table 39. Summary codes (continued).

Event code	Definition
Alarm under PF entry test	0x0052
Alarm ground fault pre test	0x0053
UnderVoltage pickup	0x0054
OverVoltage pickup	0x0055
Voltage unbalance pickup	0x0056
Current unbalance pickup	0x0057
Reverse W pickup	0x0058
Reverse var pickup	0x0059
Phase loss pickup	0x005A
Over frequency pickup	0x005B
Under frequency pickup	0x005C
Over W pickup	0x005D
Over Var pickup	0x005E
Over VA pickup	0x005F
Under PF pickup	0x0060
Ground fault pickup	0x0061
GOOSE disconnect	0x0062
Health bus error	0x0063
Internal display error	0x0064
BT idle	0x0065
BT pairing	0x0066
BT connected	0x0067
BT polling	0x0068
BT fault	0x0069
BT attention	0x006A
Password locked out	0x006B
User password created	0x006C
Health bus cassette error	0x006D
Health bus IO block error	0x006E
Health bus unsupported device connected	0x006F
Setpoint set IO block switch	0x0070
Cassette setpoints mismatch	0x0071
Cassette Unit ID mismatch	0x0072
Protection CAN0 FIFO full	0x0073
Protection CAN1 FIFO full	0x0074
Protection CAN bus initialization error	0x0075
I/O Block communications timeout	0x0076
I/O Block invalid input	0x0077
Overfrequency exit	0x0078
Underfrequency exit	0x0079

Table 39. Summary codes (continued).

Event code	Definition
Bluetooth error	0x007A
Touchscreen error	0x007B
Web user interface error	0x007C
GOOSE - reconnected	0x007D
GOOSE - no Ethernet	0x007E
GOOSE Ethernet restored	0x007F
Display processor rebooted	0x0080
Trip – Long delay	0x0200
Trip – Short delay	0x0201
Trip – Instantaneous	0x0202
Trip – Override	0x0203
Trip – Long delay neutral	0x0204
Trip – Short delay neutral	0x0205
Trip – Instantaneous neutral	0x0206
Trip – Mechanical 1	0x0207
Trip – Overtemperature	0x0208
Trip – Other	0x0209
Trip – Overvoltage	0x020A
Trip – Undervoltage	0x020B
Trip – Voltage unbalance	0x020C
Trip – Current unbalance	0x020D
Trip – Reverse real power (kW)	0x020E
Trip – Reverse reactive power (kVar)	0x020F
Trip – Reverse sequence	0x0210
Trip – Phase current loss	0x0211
Trip – Overfrequency	0x0212
Trip – Underfrequency	0x0213
Trip – GOOSE	0x0214
Trip – Maintenance mode / ARMS	0x0215
Trip – Overpower real (kW)	0x0216
Trip – Overpower reactive (kVar)	0x0217
Trip – Overpower apparent (kVA)	0x0218
Trip – Under PF	0x0219
Trip – Over demand power real (demand kW)	0x021A
Trip – Over demand power reactive (demand kVar)	0x021B
Trip – Over demand power apparent (demand kVA)	0x021C
Trip – Ground fault	0x021D
Trip – MCR	0x021E
Alarm – Mechanical 1	0x0300
Alarm – Ground fault pre-alarm	0x0301

Table 39. Summary codes (continued).

Event code	Definition
Alarm – Ground fault	0x0302
Alarm – Neutral current	0x0303
Alarm – THD current	0x0304
Alarm – THD voltage	0x0305
Alarm – Over demand power real (demand kW)	0x0306
Alarm – Over demand power reactive (demand kVar)	0x0307
Alarm – Over demand power apparent (demand kVA)	0x0308
Alarm – Thermal memory	0x0309
Alarm – Overtemperature entry	0x030A
Alarm – Overvoltage	0x030B
Alarm – Undervoltage	0x030C
Alarm – Voltage Uunbalance	0x030D
Alarm – Current unbalance	0x030E
Alarm – Reverse real power (kW)	0x030F
Alarm – Reverse reactive power (kVar)	0x0310
Alarm – Reverse sequence	0x0311
Alarm – Phase current loss	0x0312
Alarm – Overfrequency	0x0313
Alarm – Underfrequency	0x0314
Alarm – Overpower real (kW)	0x0315
Alarm – Overpower reactive (kVar)	0x0316
Alarm – Overpower apparent (kVA)	0x0317
Alarm – Under PF	0x0318
Alarm – High load 1	0x0319
Alarm – High load 2	0x031A
Alarm – GOOSE capture entry	0x031B
Extended capture – GOOSE capture entry	0x0400
Extended capture – Overvoltage entry	0x0401
Ext capture – Undervoltage entry	0x0402
Extended capture – High Load 1 entry	0x0403
Ext capture – High Load 2 entry	0x0404
Ext capture – GOOSE capture entry – No snapshots	0x0405
Ext cap – OV entry – No snapshots	0x0406
Ext cap – UV entry – No snapshots	0x0407
Ext cap – HL1 entry – No snapshots	0x0408
Ext cap – HL2 entry – No snapshots	0x0409
LDPU exit	0x0500
High temperature exit	0x0501
Overvoltage exit	0x0502
Undervoltage exit	0x0503

Table 39. Summary codes (continued).

Event code	Definition
Voltage unbalance exit	0x0504
Current unbalance exit	0x0505
Reverse real power (kW) exit	0x0506
Reverse reactive power (kVar) exit	0x0507
Phase current loss exit	0x0508
Overfrequency exit	0x0509
Underfrequency exit	0x050A
Overpower real (kW) exit	0x050B
Overpower reactive (kVar) exit	0x050C
Overpower apparent (kVA) exit	0x050D
Under PF exit	0x050E
High load 1 exit	0x050F
High load 2 exit	0x0510
Ground fault exit	0x0511
GOOSE capture exit	0x0512
SDPU exit	0x0513
PRIMARY	
Open	0x01
Closed	0x02
Tripped	0x03
Alarmed	0x04
In pickup	0x0D
SECONDARY	
Not applicabled	0x01
Test mode	0x03
Power since last trip/alarm reset	0x07
Alarm	0x08
CAUSE	
Unknown	0x00
Normal	0x01
Real power	0x02
Instantaneous	0x03
Reactive power	0x04
Instantaneous neutral over current	0x05

Table 41. Modbus waveform download for events for PXR 35 Modbus TCP Only

Function code	Register number	Address (hex)	Format	Definition
3, 4, 6	10001	0x2710	Encoded	**WAVEFORM REQUEST** **Higher byte meaning:** 0x01 = Waveform event ID overview for trips 0x02 = Waveform event ID overview for alarms 0x03 = Waveform event ID overview for extended capture 0x11 = Waveform for Trip 0x12 = Waveform for alarms 0x13 = Waveform for extended capture 0x14 = Waveform for extended capture 6s 0x15 = Waveform for extended capture 60s **Lower byte meaning:** Value range 0 - 10, select which current or voltage waveform you want to download: Ia = 0, Ib = 1 Ic = 2 In = 3 Ig = 4 Van1 = 5 Vbn1 = 6 Vcn1 = 7 Van2 = 8 Vbn2 = 9 Vcn2 = 10 For waveform event ID overviews (Higher byte is 0x01,0x02,0x03), the lower byte is not needed.
3, 4, 6	10002	0x2711	Unsigned16	**WAVEFORM CYCLE** Specifies which cycle of waveform data the user wants. For trips, alarms, and extended capture, **the value range is 0 – 35.** For 6s and 60s extended capture, **the value range is 0 - 5.** **Not needed for waveform event ID overview.**
3, 4, 16	10003	0x2712	Unsigned32	**WAVEFORM EVENT ID** Specify the event ID for which you want to download the waveform
Data to download for event ID overview request				
3, 4	10005	0x2714	Unsigned16	Number of log entries = N (max 20 events)
3, 4	10006	0x2715	Unsigned32	Event ID 0 (earliest waveform entry)
3, 4	10008	0x2717	Unsigned32	Event ID 1
3, 4	10010	0x2719	Unsigned32	Event ID 2
3, 4	…		Unsigned32	…
3, 4	10042	0x2739	Unsigned32	Event ID 18
3, 4	10044	0x273B	Unsigned32	Event ID 19 (max)
Data to download for Current request				
3,4	10005	0x2714	Unsigned32	Event ID
3,4	10007	0x2716	Unsigned16	Cycle (0-35)
3,4	10008	0x2717	Unsigned64	Timestamp of the first sample
3,4	10012	0x271B	Unsigned16	Number of samples in the cycle (80 max)
3,4	10013	0x271C	Float	Current sample 0 (first, earliest sample)
3,4	10015	0x271E	Float	Current sample 1
3,4	10017	0x2720	Float	Current sample 2
3,4	…		Float	…
3,4	10169	0x27B8	Float	Current sample 78
3,4	10171	0x27BA	Float	Current sample 79 (max)
Data to download for Voltage request				
3,4	10005	0x2714	Unsigned32	Event ID
3,4	10007	0x2716	Unsigned16	Cycle (0-35)
3,4	10008	0x2717	Unsigned64	Timestamp of the first sample
3,4	10012	0x271B	Unsigned16	Number of samples in the cycle (80 max)
3,4	10013	0x271C	Signed16	Voltage sample 0 (first, earliest sample)
3,4	10014	0x271D	Signed16	Voltage sample 1
3,4	10015	0x271E	Signed16	Voltage sample 2
3,4	…		Signed16	…

Table 40. Modbus waveform download for events for PXR 35 Modbus TCP Only (continued).

Function code	Register number	Address (hex)	Format	Definition
3,4	10091	0x276A	Signed16	Voltage sample 78
3,4	10092	0x276B	Signed16	Voltage sample 79 (max)

10. Records and forms

You may use these forms to create a hard-copy record of settings and trip events.

Table 42. Records - trip function settings.

Trip function settings			
Circuit no./address:		**Breaker shop order reference:**	
Rating amperes (I_n)		I_r = continuous ampere rating = long delay setting x I_n	
Trip function	Pickup setting	x Ampere equivalent	Time delay setting
Instantaneous		I_n A.	N.A.
Long delay (I_r)		I_n A.	Sec.
Short delay		I_r A.	Sec.
Ground fault		I_n A.	Sec.
Date:		**By:**	

Table 43. Records - trip operations.

Automatic trip operation record				
Circuit/address:	**Breaker shop order reference:**			
Trip function	**Settings reference**			
Instantaneous				
Long delay setting				
Long delay time				
Short setting				
Short time				
Ground fault setting				
Ground fault time				
Date of trip	**Cause of trip**	**Settings**	**Setting changed**	**Investigated by:**

Table 44. Records - ground fault test.

Ground fault test record form **Ground fault test record should be retained by those in charge of the building's electrical installation in order to be available to the authority having jurisdiction.**			
Test date:	**Breaker number:**	**Results:**	**Tested by:**

PXPM can print (pdf) the complete configuration of the PXR trip unit. Use Setpoint Configuration -> Connect to Unit and click on "PDF Report" in the upper right of the window. Alternately this form can be used to record settings.

Table 45. Records - PXR 20 factory defaults and settings.

Factory defaults and settings record			
Circuit no./address:		**Breaker shop order reference:**	
Rating amperes (I_n)		**I_r = continuous current rating = LD x I_n**	
Parameter:	**Factory default:**	**Settings:**	**Notes:**
Long delay slope	I^2t		
Long delay pickup	100%		
Long delay time	2 s		
High load alarm 1 pickup	85%		
High load alarm 2 pickup	105%		
Thermal memory	Enabled		
Short delay slope	Flat		
Short delay pickup	250%		
Short delay time	100 ms		
Instantaneous pickup	200%		
Ground fault type	Residual		
Ground fault slope	Flat		
Ground fault pickup	40%		
Ground fault delay time	100 ms		
Ground fault pre-alarm	85%		
External ground ratio	N/A		
Zone Selective Interlocking	Disabled		
Neutral protection	100%		
Neutral sensor	Iron core CT		
System frequency	60 Hz		
Modbus address	1		
Modbus baudrate	9600 bps		
Modbus parity	Even		
Modbus stop bit	1 bit		
CAM	None		
Maintenance Mode Pickup	250%		
Health alarm setting	25%		

Table 46. Records - PXR 25 factory defaults and settings.

Factory defaults and settings record			
Circuit no./address:		**Breaker shop order reference:**	
Rating amperes (I_n)		**I_r = continuous current rating = LD x I_n**	
Parameter:	**Factory default:**	**Settings:**	**Notes:**
Long delay slope	I^2t		
Long delay pickup	100%		
Long delay time	2 s		
High load alarm 1 pickup	85%		
High load alarm 2 pickup	105%		
Thermal memory	Enabled		
Short delay slope	Flat		
Short delay pickup	250%		
Short delay time	100 ms		
Instantaneous pickup	200%		
Ground fault type	Residual		
Ground fault slope	Flat		
Ground fault pickup	40%		
Ground fault delay time	100 ms		
Ground fault pre-alarm	85%		
External ground ratio	N/A		
Zone Selective Interlocking	Disabled		
Neutral protection	100%		
Neutral sensor	Iron core CT		
System frequency	60 Hz		
Reverse feed breaker	Top to bottom		
Power sign convention	IEEE		
Current demand window	Fixed window		
Current demand interval	5 minutes		
Power demand window	Fixed window		
Power demand interval	15 minutes		
Energy log window	Off		
Current unbalance pickup	Off		
Current unbalance time	1 s		
Phase loss pickup	Off		
Phase loss time	1 s		
Under voltage pickup	Off		
Under voltage time	1 s		
Over voltage pickup	Off		
Over voltage time	1 s		
Voltage unbalance pickup	Off		
Voltage unbalance time	1 s		
Reverse power pickup	Off		
Reverse power time	30 s		
Modbus address	1		
Modbus baudrate	9600 bps		
Modbus parity	Even		
Modbus stop bit	1 bit		
CAM	None		
Maintenance mode pickup	250%		
Health alarm setting	25%		

Table 47. Records - PXR 35 factory defaults and settings.

Factory defaults and settings record			
Circuit no./address:		**Breaker shop order reference:**	
Rating amperes (I_n)		**I_r = continuous current rating = LD x I_n**	
Parameter:	**Factory default:**	**Settings:**	**Notes:**
Long delay slope	I^2t		
Long delay pickup	100%		
Long delay time	2 s		
High load alarm 1 pickup	85%		
High load alarm 2 pickup	105%		
Thermal memory	Enabled		
Short delay slope	Flat		
Short delay pickup	250%		
Short delay time	100 ms		
Instantaneous pickup	200%		
Ground fault type	Residual		
Ground fault slope	Flat		
Ground fault pickup	40%		
Ground fault delay time	100 ms		
Ground fault pre-alarm	85%		
External ground ratio	N/A		
Zone Selective Interlocking	Disabled		
Neutral protection	100%		
Neutral sensor	Iron core CT		
System frequency	60 Hz		
Reverse feed breaker	Top to bottom		
Power sign convention	IEEE		
Current demand window	Fixed window		
Current demand interval	5 minutes		
Power demand window	Fixed window		
Power demand interval	15 minutes		
Energy log window	Off		
Motor protection	Off		
Under voltage pickup	Off		
Under voltage time	60 s		
Over voltage pickup	Off		
Over voltage time	60 s		
Voltage unbalance pickup	Off		
Voltage unbalance time	30 s		
Current unbalance pickup	Off		
Current unbalance time	10 s		
Phase loss pickup	Off		
Phase loss time	1 s		
Phase rotation (sequence) pickup	Off		
Reverse power pickup	Off		
Reverse power time	30 s		
Reverse reactive power pickup	Off		
Reverse reactive power time	30 s		
Power protection	Off		
Real power pickup	Off		
Real power time	30 s		
Apparent power pickup	Off		
Apparent power time	30 s		

Table 45. Records - PXR 35 factory defaults and settings (continued).

Factory defaults and settings record			
Circuit no./address:		**Breaker shop order reference:**	
Rating amperes (I_n)		**I_r = continuous current rating = LD x I_n**	
Parameter:	**Factory default:**	**Settings:**	**Notes:**
Reactive power pickup	Off		
Reactive power time	30 s		
Real power demand pickup	Off		
Apparent power demand pickup	Off		
Reactive power demand pickup	Off		
Power factor pickup	Off		
Power factor time	30 s		
Frequency protection	Off		
Over frequency pickup	Off		
Over frequency time	60 s		
Under frequency pickup	Off		
Under frequency time	60 s		
THD current alarm pickup	Off		
THD current alarm time	1 s		
THD voltage alarm pickup	Off		
THD voltage alarm time	1 s		
Sync check action	Off		
Min live line voltage	65% Vnom		
Max dead line voltage	3% Vnom		
Min live load voltage	65% Vnom		
Max dead load voltage	3% Vnom		
Max voltage difference	5% Vnom		
Max slip frequency	0.20 Hz		
Max angle difference	10 deg		
Dead line dead load	Off		
Dead line live load	Off		
Live line dead load	Off		
Voltage dead time	50 s		
Modbus address	1		
Modbus baudrate	9600 bps		
Modbus parity	Even		
Modbus stop bit	1 bit		
CAM	None		
Maintenance mode pickup	250%		
Health alarm setting	25%		

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